

The WORKHOUSE Program

Exact $SU(N)$ Flux-Band Spectra, the Fixed-Spacing Wilson Construction,
and a Verified Register of What the Continuum Limit Still Requires

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Abstract

This is the master reference edition of a four-year program on the strong-coupling spectral structure of Hamiltonian $SU(N)$ lattice gauge theory and on the obligations that separate that structure from a continuum Yang–Mills mass gap. It is organized so that its two halves can be read, and used, independently.

Part I is a closed body of exact mathematics. A rank-one cellular subspace — the kernel of the plaquette boundary map — carries the strong-coupling band, and its hopping amplitude is the rational function

$$t_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)},$$

positive for $N \geq 3$, vanishing identically at $N = 2$. The four shared-link channel weights that produce it are order-two Weingarten values rather than an isotropy assumption, so the second-order theory rests on representation theory alone. At fourth order the effective operator splits into Hodge terms and one geometric term coupling the carrier to its complement, with the off-axis coefficient decided by derivation across three independent implementations. The rank grading is exact: every even order of the one-plaquette splitting carries the same single power of N , with coefficients $c_2 = -4$, $c_4 = -32$, $c_6 = -1748/3$, $c_8 = -123332/9$, $c_{10} = -49593808/135$, and a four-channel cancellation that sums to zero at every even order from four to ten; while the odd orders are determinant families — for even N every odd order vanishes identically, and for odd N the first surviving odd order is $N - 2$, with a closed-form single vertex. At sixth order, a rational band shape is proved to survive every local direct term. Of the statements in Part I, none is conditional on an open estimate.

Part II states the construction at fixed lattice spacing: an actual infinite-volume Wilson transfer limit, a complete physical odd band isomorphic to $\ell^2(\mathbb{Z}^3) \otimes \mathbb{C}^3$, and a literal plaquette-source frame that is onto that entire band, on an explicit interval of the coupling.

Part III is the part written for whoever works on this next. It states six obstructions between the fixed-spacing construction and the continuum limit — five of them with explicit finite counterexamples, four of them having killed arguments this program had written down as complete — of which the decisive one is that the interval where the construction is proved and the trajectory along which the continuum is taken are disjoint sets of couplings. It then gives the refutation register: 4 falsified claims with what falsified each, and the mechanisms this program advanced and withdrew. It closes with an obligation register of eight numbered problems and the repository’s full route record — 111 routes, of which 64 are complete, 20 live, 12 untried and 15 dead, each dead one carrying the reason it died.

No claim is made here about the existence of continuum Yang–Mills theory or about a continuum mass gap. §15 is the argument that no such claim currently follows, and §17 is the list of what would have to change.

Every count quoted above is generated from the repository’s ledgers rather than typed, and every statement carries its machine-verification tier and the command that re-establishes it. Currently: 439 Lean 4 theorems with 0 occurrences of `sorry`, 143 registered analytic results, and 54 verified evidence edges to published work across 34 obtained papers.

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1 Introduction

1.1 What is actually here

Strong-coupling expansions of Hamiltonian lattice gauge theory are old. What is not old is the object this program organizes them around.

Write the cellular chain complex of the cubic lattice as $C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1$, with C_2 spanned by plaquettes. The subspace

$$Z_2 = \ker \partial_2 \subset C_2 \tag{1.1}$$

— the closed plaquette chains — is the carrier of the strong-coupling band. It is a homological object, fixed before any perturbation theory is done, and the entire content of Parts [I](#) and [II](#) is a description of what the Kogut–Susskind Hamiltonian does to it, order by order and rank by rank.

Three things follow from taking that object seriously, and they are the three claims of this document.

First: the coefficients are exact rational functions of the rank. Not asymptotic in N , not fitted, not computed rank by rank and extrapolated. The second-order hopping amplitude is

$$t_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)}, \quad t_2 = 0, \quad t_3 = \frac{5}{612}, \tag{1.2}$$

and the four channel weights behind it are order-two Weingarten values, so [\(1.2\)](#) imports nothing from any expansion. The factor $N^2 - 4$ is not a coincidence of small rank: it is why $SU(2)$ has no band at this order at all. The same discipline runs through fourth order, through the rank grading of [§8](#), and into a symbolic engine that works over $\mathbb{Q}(N)$ with N an indeterminate rather than a number.

Second: the rank grading has exact structure, and it comes from cancellation. For the C -parity splitting of the one-plaquette diagonal, every even order carries the same single power of N once the coupling is written in the planar variable, and the leading coefficients of the four contributing intermediate channels are in the ratio $(-4, +2, +1, +1)$ and *sum to zero*. The grading is a cancellation theorem, not a power count. Complementarily, the odd orders vanish identically for even N , and for odd N the first surviving one is order $N - 2$, with a closed-form vertex. Neither statement is visible from any single rank.

Third: the construction at fixed lattice spacing is complete, and it does not reach the continuum. [§13](#) states an actual infinite-volume Wilson transfer limit with a complete physical odd band and a literal plaquette-source frame that is onto it. [§15](#) states why that is not a continuum mass gap, in six independent ways, five of them with finite counterexamples. The most important is structural rather than technical: the theorem holds on $|u| \leq u_*/(10\,022\,400\,000N)$ and the continuum trajectory runs to $u \rightarrow \infty$, so the two are disjoint and no sharpening of the constant changes that.

1.2 Why the document is shaped this way

Alex, the maintainer of this program, asked for a paper that would serve as the base for further work on the mass gap rather than as a report of work completed. That request has consequences for the form.

It means that failure is content. §15 and §16 exist because a route this program has already closed is more valuable to the next attempt than a route it has already opened, and because at least four arguments were written down here as complete and then found not to be. It means the route register in §17 reproduces 15 dead routes at length, including their measured costs, rather than listing only the 20 live ones. And it means every count in the prose is a macro generated from the repository’s ledgers, so that a stale number is a build failure rather than a quiet misstatement.

It also means the document distinguishes, everywhere and visibly, three things that are routinely conflated: what a statement mathematically establishes, what artefact backs it, and what a machine checks. A result can be fully proved analytically and carry the weakest machine badge. A Lean theorem can be checked by the kernel and prove something trivial. This program has made both errors and the notation of §2 is the correction.

1.3 Relation to the literature

The external comparisons are recorded as an evidence map rather than a citation list: 108 indexed papers, 152 recorded relations, of which 54 are verified against a named claim here. A published paper is not authority in this program — it is unchecked until something checks it, exactly like an internal document. What makes an external agreement evidence is that it was produced without knowledge of this program, and §12 reports the comparisons that hold and, separately, the two that contradict.

1.4 What is claimed, and what is not

Claimed: the contents of Part I, unconditionally, at the tiers recorded; and the contents of Part II on the stated interval and under the stated hypotheses.

Not claimed: the existence of continuum four-dimensional Yang–Mills theory; a continuum mass gap; Poincaré covariance; restoration of rotational invariance from the hypercubic lattice; or the construction of a continuum measure. None of these has an argument in this document, and §15 explains for each of the six obstructions why the present machinery does not supply one.

2 How to use this document

This document has two audiences and it is worth saying which paragraph is for whom.

A reader who wants the mathematics should read Part I, which is self-contained and compiles on its own as a separate paper (`paper/master/workhouse_band_paper.tex`). Nothing in Part I depends on anything later, and nothing in it is conditional on an open estimate.

A reader who intends to *work* on the continuum problem should read §15, §16 and §17 first, in that order, and only then come back for the mathematics. The reason is economic. This program has spent four years discovering that most of the obvious routes to the continuum limit are closed, and the record of *which* routes are closed and *why* is worth more than the record of what is open. Four sessions of one earlier effort were spent on a single route that a static argument had already

shown could not decide the question it was aimed at; that experience is why the route register in §17 exists and why dead routes are reproduced in it at full length rather than deleted.

2.1 What the badges mean

Every statement in this document carries a machine-verification badge.

- **[T0 Lean]** — Lean 4 compiles the statement from explicit definitions with no **sorry**, using only **propext**, **Classical.choice** and **Quot.sound**. Nothing written in prose can weaken a T0 badge.
- **[T1 exact]** — the statement is re-derived symbolically, in exact arithmetic, from stated definitions.
- **[T2 numeric]** — floating-point agreement within a tolerance that the check itself prints.
- **[T3 analytic]** — no single repository check covers the whole statement.

The badge is *independent* of mathematical status. A theorem with a complete analytic proof and no formalization carries [T3 analytic]: the badge says what the machine checks, not what the mathematics establishes. This is the one convention a reader must internalize to read the document correctly, because the opposite reading — T3 as a synonym for “unproved” — makes most of Part III look weaker than it is, and the reading of T0 as a synonym for “important” makes it look stronger. Both errors have been made in this program’s own internal documents and corrected.

Where a statement’s scope is narrower than its headline suggests, a **Does not assert** paragraph follows it. Those paragraphs are not hedging. They are the part of each result that took longest to get right, and several of them exist because an earlier version of the claim was broader and was refuted.

2.2 If you are the next agent on this program

The following is an operating procedure, not advice. It is written for the case — now the normal case — in which the next piece of work on this program is done by an automated agent with a fresh context window.

1. **Do not trust this document over the ledgers.** Every number quoted in the prose here is generated from `ledger/*.yaml` by `paper/master/generate_apparatus.py`. If the ledger and this document disagree, the ledger is right and this document is stale; regenerate it. The counts in §17, §12 and Appendix B are macros, not typed literals, precisely so that this failure mode is visible rather than silent.
2. **Establish the state before choosing a target.** Run `workhouse brief --startup`, then `workhouse brief <gap> --json` for the gap you mean to work on, and retain the snapshot. A saved graph in a fresh clone reports freshness **unknown** until local generation provenance exists; saved briefing mode never silently rebuilds it.
3. **Read the dead routes for your gap before starting.** `workhouse why <gap>` prints them. §17 reproduces them here. Of 111 recorded routes across all gaps, 15 are dead, and several died only after substantial expenditure with a measured cost recorded in the ledger. A route marked **cannot_decide** is not merely unpromising: it is structurally incapable of resolving the claim it was aimed at, and re-running it will produce output that answers a different question.

4. **Do not aim at the continuum limit directly.** §15 states six independent obstructions, five of them with explicit finite counterexamples. Each is a constraint on what a correct argument can look like, and an argument that does not explicitly clear all six is not a candidate. In particular, Break 5 (§15.5) shows that the regime in which this program’s strongest construction holds and the regime in which the continuum limit lives are disjoint sets of couplings. That is a statement about the shape of any possible proof, not a gap in the present one.
5. **Pick a numbered Problem from §17,** not a topic. Each Problem states its available hypotheses, what it unblocks downstream, and the specific next computation. Problems are ordered by the ledger’s research judgement, which is not a claim that earlier ones are prerequisites for later ones.
6. **Record the attempt, including its failure.** A route that dies is a result. Set its state in `ledger/gaps.yaml`, name what closed it in `closed_by`, and if the instrument could not have reached the claim, say so in `cannot_decide`. The working agreement in `CLAUDE.md` is the binding version of this instruction; this paragraph only explains why it is there.

2.3 What would falsify the contents of this document

Three things, in decreasing order of how much they would cost.

First, an error in the Haar or Weingarten input to §6 would propagate to every order and to both the fourth-order band and its rank grading. That input is now representation theory rather than assumption (§6), and it is the load-bearing algebraic fact of Part I.

Second, a demonstration that the plaquette source family of §13 is not total in the physical complement would empty the fixed-spacing band theorem of physical content without making it false — this is exactly the failure mode that Break 4 (§15.4) exhibits in a three-dimensional model, and totality is therefore asserted in that theorem as a proved conclusion rather than assumed.

Third, and least costly, a published result reproducing any of the exact coefficients of §8 or §7 would remove this program’s claim to have derived them first without touching their correctness. §12 records the searches already performed against 108 indexed papers.

3 The source layers

Every derivation in this program can be traced to bytes. This section says how, because the tracing is not uniform: there are three source systems, they certify different things, and reading a statement out of the wrong one is the most common way to get this corpus wrong.

An agent that internalizes only one thing from this document should internalize the table below.

Layer	What it holds	What it certifies
Theory graph	332 derivation statements in 40 documents; 143 results; 439 theorems; 111 routes; 32503 edges in 0 relations	That a statement was <i>recorded</i> with a status, hypotheses, a source locator, and a digest of the bytes it was read from. It does not certify the mathematics; the tier does that separately.
Published work	108 papers in the evidence map, 54 edges verified; 260 acquired papers in the reading library; 777 extracted pages across 28 works	That an external result exists and, for a verified edge, that it was read against a named claim here. Obtained is not read; read is not checked.
Local archive	Four years of manuscripts, runs, notes and Lean, with 248 aliased documents, 17 note archives and 313 recorded reviews	That the bytes are preserved and pinned. Nothing about whether the content is current, or correct.

3.1 Layer 1: the theory graph

The graph is the index of record. Its node identifiers are prefixed, and the prefix tells you what kind of object you are holding and therefore what it can support.

CITE:

A source document, pinned by path and SHA-256. 40 of them. A CITE: record is bytes, not a claim.

DERIV:

A statement located *inside* a CITE: document, by heading and line range. 332 of them, every one with a locator. This is the finest granularity the corpus tracks, and it is the right unit to work at.

RESULT:

A first-class analytic claim, independent of any one document. 143 of them, each with status, evidence, hypotheses, scope, **depends_on** and **bears_on**.

LEAN:

A registered theorem. 439 of them. **formalizes** links it to a catalogue record; **promotes** — which only 49 carry — means it proves an entire named check and therefore lifts that check's tier.

CHK:

A registered invariant check, carrying its tier and the numbers it establishes. A FINDING: check is one that asserts a *discrepancy* rather than an agreement.

RUN:

A pinned computational run, with its own manifest.

ADR:

A numbered decision record: what was decided, on what evidence, and what it retracts or amends.

CONST:, SYM:

A named constant and its symbol registry entry — canonical name, corpus spellings, code names,

and exact values. Appendix I.

Gn, Cn, Un, Rn

Gaps, contradictions, unifying candidates and registry claims.

The edges matter as much as the nodes. 32503 of them across 0 relations, and three deserve separate mention because they are routinely confused. **depends_on** is a mathematical input: if it falls, the dependent falls. **supported_by** is narrower — a control that bears on part of a statement, and a result with **supported_by**: `[]` has no registered machine support at all, which is a fact worth checking before quoting a tier. **bears_on** is looser still: a topical connection, and it establishes nothing.

3.2 Layer 2: published work

There are two published-work systems and they answer different questions. Using the wrong one produces either a false novelty claim or a false authority claim.

The evidence map (`literature/index.yaml`) is small, curated and adversarial. 108 papers, 152 recorded relations, 54 of them verified against a named claim of this program. An entry earns its place by naming a specific claim. Query it with `workhouse lit --for G19`. This is the only layer from which an agreement or a contradiction may be quoted.

The reading library (`literature/library/paper_manifest.csv` and `volume2/`) is large and inclusive: 260 acquired papers — 99 and 161 — each with tier, category, journal reference, DOI, arXiv identifier, INSPIRE citation count, local file, page count, SHA-256, abstract, and a curated one-line **workhouse_connection**. Appendix G reproduces it. Its purpose is coverage: to answer “has anyone done this?” before an agent spends a week. Its connection field is a reading note and carries no evidential weight.

The gap between the two is the useful measurement. 34 papers have been read against a claim; 260 have been acquired. Most of the library has never been checked against anything here, and a novelty claim asserted against the library is therefore weaker than one asserted against the evidence map — which is itself weaker than priority, for the reasons in §12.

3.3 Layer 2b: the page-level extraction index

This is the piece an agent is most likely to miss, and it is the one that saves the most time.

777 pages of local PDFs have been extracted and indexed at `literature/yangmills/extraction/pages.jsonl`, covering 28 works. Each record carries the paper identifier, the PDF page number, the SHA-256 of the source *and* of the extracted text, the path of the reading copy, and the line positions of every hit against 7 topic keys. Appendix H tabulates it.

The workflow it supports:

1. Search `pages.jsonl` for a paper identifier and a topic key (`ground_state`, `spectral_gap`, `curvature`, `reconstruction`, `uniform_limits`, `constructive`, `gauge_geometry`).
2. Follow the page and line locator into the reading copy under `literature/inbox/<ID>/extracted/`.
3. Verify any equation you intend to use against the pinned PDF.

Three cautions the index states about itself. A topic hit may be a citation rather than a result, or an extraction error. Candidate theorem labels are recorded on many pages, but a label can sit

inside a citation or carry an OCR fault — the index is a search aid and asserts nothing. And five pages extracted fewer than eighty characters and are explicitly flagged as unreliable; their contents must not be inferred from the text index. Exact equations from legacy scans must be checked in the PDF.

3.4 Layer 3: the local archive

The archive is preserved rather than curated, and its governing instruction is that nothing is erased. Failed attempts, superseded drafts, duplicates and uncommitted work are all kept, because deleting a refuted claim destroys the evidence that it was tried.

Consequently the field that matters when reading an archive document is its *standing*. 248 documents carry a registered alias and a standing, tabulated in Appendix J. A superseded document is not marked in its own text and reads exactly like a current one; the register is the only thing that distinguishes them. Anything under **theory/superseded/** is retained for audit and must never be read as current.

The maintainer's own research notes enter through a separate discipline: 17 declared archives with 313 recorded reviews, each carrying a content digest, a verdict from a closed vocabulary, and a mandatory reason. An **import** is byte-verified against its digest; an **extract** must name the claims it entered through; a **set-aside** is a recorded judgement and never a deletion. No verdict promotes anything past the default machine tier.

3.5 Finding things: the corpus joins on values, not concepts

The single most important retrieval fact about this corpus: *its join keys are exact rationals and spellings, not concepts*. No semantic query retrieves 109151/249696, and the same quantity appears in a manuscript, in code, and in a claim record under three different names.

workhouse search exists for this. It matches by *value* rather than spelling — searching $-10/96$ finds $-5/48$ — across the prose index, the code and certificate index, the claim catalogue, and the curated alias table of Appendix I. It also carries two warnings no text search can: names the register *forbids* because using them regenerates a dissolved contradiction, and names coined inside this program that its own source corpus never uses.

Two indexes sit underneath it, and the second exists because the first could not see the sealed core: values such as $5/48$, $5/12$, $5/612$, $11/306$ and $7/102$ can be absent from the prose index while their computations are present in code.

3.6 The queries worth knowing

<code>workhouse why <id></code>	everything recorded about one claim:
	both sides of a dispute, every check and
	its verdict, routes tried and which died
<code>workhouse search '5/612'</code>	search by exact VALUE, not by name
<code>workhouse branches C2</code>	both sides of a dispute, side by side
<code>workhouse derive C2 G3</code>	the evidence chain, registered edges only
<code>workhouse lit --for G19</code>	published work bearing on a claim
<code>workhouse notes --queue</code>	the next unreviewed notes, by signal
<code>workhouse triage <path></code>	survey an unpinned collection
<code>workhouse ask "<question>"</code>	candidate finder over prose; ends in 'why'

`workhouse brief <gap> --json target snapshot, with freshness`

`why` is the front door. It resolves a claim identifier, a constant, a gap, an ADR, or a corpus file path, and it prints the routes with their states — which is how a dead route announces itself before an agent re-runs it.

3.7 What each layer cannot do

- The graph cannot tell you a statement is *true*. It tells you what was recorded, by whom, from which bytes, under which hypotheses. Status, evidence and machine tier are three independent fields precisely so that this stays visible.
- The library cannot establish novelty. It establishes that a paper was acquired.
- The extraction index cannot establish what a paper says. It tells you which page to read.
- The archive cannot tell you what is current. Standing does.
- A digest cannot tell you an anchor resolves. Check both: a correct SHA-256 over a fabricated locator has occurred here.

3.8 Operational rules live elsewhere, deliberately

How to add an invariant, register a Lean theorem, record a route, order the regeneration passes, and land work: those are specified in `CLAUDE.md` (the working agreement), `AGENTS.md` (the research posture) and `CONTRIBUTING.md`. They are not restated here.

That is a decision, not an omission. Those files are binding and they change; a copy in a document that is rebuilt on its own schedule would drift, and an agent following a stale copy of a rule is worse off than one following no copy. This document carries the *mathematical* working knowledge — which sources support what, how the derivations connect, and where the traps are. For procedure, read the source of record.

Two of its rules are worth naming here anyway, because they explain the shape of everything above. A claim is established by its proof or computation under explicit hypotheses, and its artefact must not claim more than that argument establishes. And a disputed value closes by derivation or not at all: both sides stay recorded, and no code path may pick one, average them, or prefer whichever is an exact rational because exactness reads as authority.

Part I

Exact spectral structure at strong coupling

4 Setting and conventions

4.1 The Hamiltonian

Work on the cubic lattice $\mathbb{Z}^3$ with gauge group $G = \mathrm{SU}(N)$, one group element U_ℓ per oriented link. The Kogut–Susskind Hamiltonian is

$$H = H_0 + V, \quad H_0 = \frac{1}{2} \sum_{\ell} E_{\ell}^2, \quad V = -u W, \quad (4.1)$$

where E_ℓ^2 is the quadratic Casimir of the left-invariant electric field on link ℓ , and W is the sum of plaquette characters in both orientations,

$$W = \sum_p (\chi_p + \bar{\chi}_p), \quad \chi_p = \text{Tr} \prod_{\ell \in \partial p} U_\ell. \quad (4.2)$$

Strong-coupling perturbation theory is the expansion in u around the electric vacuum, organized as degenerate perturbation theory — a des Cloizeaux effective operator — on the sector of interest.

4.2 The coupling, and one erratum that matters

The canonical coupling variable throughout is

$$u = \frac{\beta_N}{2N}, \quad (4.3)$$

whose $\text{SU}(3)$ specialization is $u = \beta_3/6$.

A warning, because it has corrupted transcriptions of this material before. An archived line reading $Y = 2\beta/3 = 4u$ is a definition-label erratum, not a change of variable. The coefficients printed alongside it were *already* in the canonical coordinate (4.3). Applying a 4^r rescaling at order r to “correct” them corrupts every order, and the repository carries an explicit check that the printed towers reproduce their canonical values verbatim.

For rank-uniform statements the natural variable is the planar one,

$$\tau = \frac{2u}{N^2} = \frac{\beta_N}{N^3}, \quad (4.4)$$

in which the grading theorems of §8 take their simplest form: every even order of the one-plaquette splitting then carries the same single power of N .

4.3 Two projections that must not be conflated

Two distinct projections appear throughout, and several contradictions in this program’s own register were naming collisions between them.

The *band* projection is onto the degenerate strong-coupling subspace carrying the excitation of interest — for the flux band, the cellular subspace of §5. Coefficients anchored to it are band-kernel quantities.

The *physical* projection is onto the gauge-invariant, vacuum-subtracted spectrum at a definite momentum. Coefficients anchored to it are physical quantities.

These are related by translation-local scalar shifts, and such a shift changes neither the centered operator, nor its eigenvectors, nor the bandwidth. It does change the scalar. Consequently the band-kernel anchor and the physical Γ -point coefficient are two differently anchored coordinates, not two competing estimates of one quantity, and they are written $q_{\text{band}}^{(4)}$ and $m_\Gamma^{(4)}$ respectively. Writing either as “ m_4 ” regenerates a contradiction that does not exist, and the repository’s search tooling refuses the name for that reason.

4.4 Exactness

Every coefficient displayed in Part I is an exact rational, or an exact rational function of N . Where a quantity is known here only numerically it carries an explicit numerical marker in the source

registry and is never displayed as though exact. The single most dangerous defect available in this material is a float that reads as exact, and the boundary between the two is machine-guarded rather than maintained by care.

5 The homological carrier

5.1 Where the flat band comes from

Project the Kogut–Susskind Hamiltonian (4.1) onto the fundamental, charge-odd, one-plaquette manifold. In three spatial dimensions a fundamental plaquette excitation carries one of three face orientations, and at second order neighbouring faces communicate through a shared link. The projected problem is therefore a three-orbital Bloch Hamiltonian, and it looks at first like a tight-binding model.

It is not an engineered one. The signs of the matrix elements are fixed by the orientation of the plaquette boundary and by charge conjugation, and non-abelian group integration generates, exactly, the oriented edge–face boundary Gram operator

$$B(k)B(k)^\dagger, \quad B = \partial_2^\dagger, \quad (5.1)$$

with the coefficient of §6 in front of it. The structure of the result is a clean separation:

$$\boxed{\text{colour dynamics} \longrightarrow t_N, \quad \text{cellular geometry} \longrightarrow BB^\dagger.} \quad (5.2)$$

Representation theory fixes the amplitude; the cubical chain complex fixes the kernel and the spectrum. Neither borrows from the other.

5.2 The projected spectrum

Theorem 5.1 (Second-order projected spectrum). *The projected second-order spectrum consists of one momentum-independent level together with a doubly degenerate branch shifted by $t_N u^2 q(k)$, where*

$$q(k) = 4 \sum_{j=1}^3 \sin^2(k_j/2). \quad (5.3)$$

The flat level is exact and it is exact for a reason: the flat carrier is $\ker \partial_2$.

Theorem 5.2 (The carrier and its multiplicity). *Elementary cube boundaries are compact localized states of the carrier; three wrapping sheets complete it on the torus. For $L \geq 3$ the exact multiplicity is*

$$(L^3 - 1) + 3 = L^3 + 2. \quad (5.4)$$

Graph id: RESULT:UNIVERSAL_CELLULAR_HODGE_SPECTRUM

The two contributions are structurally different and the distinction matters later. The $L^3 - 1$ cube boundaries are local and finitely supported. The three sheets are not: they wrap the torus, and they are the entire zero-momentum content of the carrier. §13 returns to the consequence — that no finitely supported exactly-flat local operator can see the Γ point at all.

5.3 Third order, and the coefficients that survive

For $SU(3)$, exact third-order perturbation theory stays inside the same boundary-factorized algebra. It does not generate a new operator structure; it renormalizes the coefficient in front of the same $BB^\dagger$:

$$H_{\text{eff},-} = E_{\text{flat}}(u) I + \left(\frac{5}{612} u^2 + \frac{1975}{124848} u^3 \right) BB^\dagger + O(u^4), \quad (5.5)$$

with

$$E_{\text{flat}}(u) = \frac{8}{3} + u + \frac{11}{306} u^2 - \frac{109151}{249696} u^3. \quad (5.6)$$

That (5.5) remains boundary-factorized through third order is the non-trivial part. It is what makes the fourth order the first place where the carrier can couple to its complement, and §7 is about exactly that coupling.

5.4 What is and is not novel here

This program does not claim that incidence flatness, cellular compact localized states, or homological counting of zero modes are new. All three have a substantial prior literature — line-graph constructions, compact localized states, the necessity of non-contractible states on periodic systems, rectangular factorizations and index arguments, and closely related closed-surface structures in two-form spin liquids — and a recent independent construction treats face-graph and higher-cell flat bands directly.

What is claimed is that the flatness here is *dynamical*: it is produced by exact strong-coupling matrix elements of non-abelian $SU(N)$ Hamiltonian lattice gauge theory, with the amplitude fixed by group representation theory and the kernel fixed by the cubical chain complex, as in (5.2). The flat band is not put in; it comes out.

Scope 5.3. Theorem 5.1 and (5.5) are finite-volume, finite-order statements about a projected flux sector. They are not a continuum glueball theorem and not a mass-gap theorem, and the source manuscript says so in its own abstract.

6 Second order at every rank

This section is the algebraic base of the whole program. Everything in it is proof-checked, and its conclusion — that the band exists for $N \geq 3$ and not for $N = 2$, with an exact amplitude — follows from representation theory with no input from any expansion.

6.1 The shared-link weights are Weingarten values

Two plaquettes contributing at second order either share a link or do not. The amplitude for the shared-link case was for a long time taken in this program with an isotropy premise: that the four contributing representation channels carry weights in a fixed ratio. That premise is not needed. It is a consequence of the order-two Weingarten values.

Proposition 6.1 (The shared-link weights, from representation theory alone). *Let two plaquettes share exactly one link. Then:*

1. *The six non-shared links collapse. Each plaquette contributes a product of three independent Haar links, and a product of independent Haar matrices is Haar. So the two-plaquette amplitude reduces to $\text{Tr}(AU) \text{Tr}(BU^{\pm 1})$ with A and B independent Haar, and integrating them out leaves a pure degree-(2, 2) moment of the shared link U .*
2. *Two such moments settle both families. With $M_{\text{direct}} = N^2$ and $M_{\text{cross}} = N$, the like family splits as $(N + 1)/(2N)$ and $(N - 1)/(2N)$; and the singlet component of $U_{ij}\overline{U}_{lk}$ is $\delta_{il}\delta_{jk}/N$, of squared norm 1, giving $1/N^2$.*

All four weights are exactly d_R/N^2 , where d_R is the dimension of the channel.

Graph id: theshared-linkweightsareWeingarten,notanisotropyassumption Reproduce: workhouse verify -only 'the shared-link weights are Weingarten, not an isotropy assumption'

[T1 exact]. The relevant Weingarten pair is the inverse of the S_2 Gram matrix and the index sums are explicit, so the check imports nothing from the corpus. Registered as the discharge of the gap that had carried this premise as the one unproved input of the second-order chain; it was opened and closed on the same day, 28 August 2026, once the question was asked in the right form.

6.2 The four channels in closed form

Write $C_F = (N^2 - 1)/(2N)$ for the fundamental Casimir. Each channel R contributes a resolvent weight $-(d_R/N^2)/(C_F + \frac{1}{2}\Delta C_2(R))$, and all four evaluate in closed form.

Theorem 6.2 (Per-channel resolvent weights).

$$w_{\mathbf{1}} = -\frac{1/N^2}{C_F} = -\frac{2}{N(N^2 - 1)}, \quad (6.1)$$

$$w_{\text{Adj}} = -\frac{(N^2 - 1)/N^2}{C_F + N/2} = -\frac{2(N^2 - 1)}{N(2N^2 - 1)}, \quad (6.2)$$

$$w_{\text{Sym}} = -\frac{N(N + 1)/2N^2}{C_F + \frac{(N-1)(N+2)}{2N}} = -\frac{N + 1}{(N - 1)(2N + 3)}, \quad (6.3)$$

$$w_{\Lambda} = -\frac{N(N - 1)/2N^2}{C_F + \frac{(N+1)(N-2)}{2N}} = -\frac{N - 1}{(N + 1)(2N - 3)}. \quad (6.4)$$

Graph id: LEAN:resolventWeight_singlet,LEAN:resolventWeight_adjoint,LEAN:resolventWeight_sym,LEAN:resolventWeight_antisym

[T0 Lean] — all four, at lean/Workhouse/Basic.lean:72,80,89,104. Reproduce with make lean.

6.3 The two family sums

The channels group into a *mixed* family $\{\mathbf{1}, \text{Adj}\}$ and a *like* family $\{\Lambda^2 F, \text{Sym}^2 F\}$, and these are the antiparallel and parallel channel sums.

Theorem 6.3 (Family sums).

$$A_N = w_{\mathbf{1}} + w_{\text{Adj}} = -\frac{2N^3}{(N^2 - 1)(2N^2 - 1)}, \quad B_N = w_{\text{Sym}} + w_{\Lambda} = -\frac{4N(N^2 - 2)}{(N^2 - 1)(4N^2 - 9)}. \quad (6.5)$$

Graph id: LEAN:mixed_family_sum,LEAN:like_family_sum

[T0 Lean], at `lean/Workhouse/Basic.lean:118,129`. The mixed sum is obtained from the dimension and Casimir table rather than by transcription from any source document, which is the point: the family sums are re-derived, not quoted.

6.4 The hopping amplitude

Theorem 6.4 (The all-rank second-order hopping amplitude).

$$t_N = B_N - A_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)}. \quad (6.6)$$

It is strictly positive for every $N \geq 3$, and

$$t_2 = 0, \quad t_3 = \frac{5}{612}. \quad (6.7)$$

Graph id: CONST:t_N Reproduce: workhouse why t_N

[T0 Lean] for the rank law and for both special values (`lean/Workhouse/Basic.lean:rank_law_numerator`, `hopping_two`, `hopping_three`). Clearing denominators, the channel difference has numerator exactly $2N(N^2 - 4)$.

Two consequences deserve to be stated separately, because both are structural rather than numerical.

Corollary 6.5 ($SU(2)$ has no band at this order). *The factor $N^2 - 4$ in (6.6) vanishes at $N = 2$. The exclusion of $SU(2)$ is therefore not a smallness but an identity: the parallel and antiparallel channel sums coincide exactly at rank two.*

Graph id: LEAN:hopping_two

Corollary 6.6 (The planar approach is from below). *With $D = (N^2 - 1)(2N^2 - 1)(4N^2 - 9)$,*

$$D - 4N^3 \cdot 2N(N^2 - 4) = 2N^4 + 31N^2 - 9 > 0 \quad (N \geq 1), \quad (6.8)$$

so $4N^3 t_N = 1 - (2N^4 + 31N^2 - 9)/D < 1$. Hence $t_N < 1/(4N^3)$ for every N , and $t_N \sim 1/(4N^3)$: the N^{-3} cancellation is exact and what it leaves is a strictly positive deficit.

Graph id: LEAN:hopping_deficit_numerator

[T0 Lean], at `lean/Workhouse/Basic.lean:48`. Corollary 6.6 is the first appearance of the rank grading that §8 develops: the leading power of N is fixed by a cancellation, and the surviving term is a polynomial one can write down.

6.5 Independent corroboration of the channel weights

The weights of Proposition 6.1 have an external provenance, from work written without knowledge of this program.

An efficient-truncation study of $SU(N_c)$ lattice gauge theory for quantum simulation [21] gives, in its Appendix D, the exact finite-rank plaquette matrix element

$$|M_\rho| = \sqrt{\frac{\dim \rho}{\dim A \dim R}}. \quad (6.9)$$

Taking both factors fundamental gives $|M_\rho|^2 = d_\rho/N^2$, which is *identically* the numerator of the weight formula used here. The four shared-link channel weights, and with them A_N , B_N and $t_N = B_N - A_N$, therefore rest on a matrix element an independent group wrote down for an unrelated purpose. That truncation is also the first published one retaining all four shared-link irreducible representations.

The agreement is evidence rather than bookkeeping precisely because neither side knew of the other. It does not make t_N published: the rank law (6.6), the vanishing at $N = 2$, and the deficit of Corollary 6.6 are assembled here.

A caution attaches in the other direction. The leading-large- N_c qutrit plaquette Hamiltonian [20] explicitly decouples the charge-odd combination, so the sector in which t_N lives is inert in that model — effectively $t_3 = 0$ at leading order in large N_c , against the finite- N channel-complete value 5/612. That paper is recorded in this program’s index as not yet obtained, and the statement here is taken from imported bridge documents rather than checked against the paper itself.

6.6 What this section does not establish

Does not assert. Convergence of the strong-coupling series; anything about the spectrum at finite u beyond second order; and nothing about $SU(2)$ at higher orders, where Corollary 6.5 does not apply. Theorem 6.4 is a statement about one coefficient of a formal expansion, established exactly.

7 Fourth order, and how a disputed coefficient was decided

Fourth order is where the carrier first couples to its complement, and it is where this program spent the largest single block of its effort. The mathematics is one theorem and one number. The number was disputed for months between two irreconcilable values, and the way it was settled is, in this author’s view, the more instructive half of the story.

7.1 The structure

The fourth-order effective operator splits into Hodge terms and one geometric term coupling the carrier to its complement. Restricted to range-one operator structures, the shape content collapses: each such structure contributes a fixed shape coefficient, and every one of them gives $B = D = 0$ identically. The tier collapse is simply what range-one looks like.

Theorem 7.1 (Fourth-order support, at $SU(3)$). *The recorded fourth-order kernel has support $\{I, U, S, S^2, R\}$, and the shape data closes in that span. Solved over the whole Brillouin zone in exact rationals rather than fitted, the shape table is determined by three signed amplitudes:*

$$C_{\text{shp}} = -\frac{5}{96} - u - \frac{1}{2}(\rho + \pi), \quad u_2 = 2u, \quad \nu = -\left(\frac{5}{48} + 4u\right), \quad (7.1)$$

where u is the two-hop chain cumulant.

Graph id: RESULT:TIER_COLLAPSE_ACTUAL_H4_SUPPORT, RESULT:TIER_COLLAPSE_IS_R_DEGREE

Scope 7.2. “The recorded kernel” means the 189-record historical kernel, which is $N = 3$. The operator splitting is an $SU(3)$ statement; it is not established at general rank, and framing it as an $SU(N)$ identity would be an overreach. The rank-uniform statements of this program are the coefficient laws of §6 and §8, not this splitting.

Clearing the $1/q$ makes the shape ansatz a linear system over Laurent polynomials in e^{ik_j} , which is why the whole zone can be solved exactly instead of sampled.

There is a second, equivalent presentation. In the Hodge form the two-hop weight is absorbed, $\tilde{\pi} = \pi + 2u$, and

$$C_{\text{shp}} = -\frac{5}{96} - \frac{1}{2}(\rho + \tilde{\pi}), \quad (7.2)$$

so that u drops out of C entirely and exactly two shared-link amplitudes remain. Equations (7.1) and (7.2) are the same statement. They must not be combined — writing u and $\tilde{\pi}$ in the same expression double-counts the two-hop weight, and that error has been made.

7.2 The dispute

Two independently recorded kernels — a historical pipeline from June 2026 and a later cold dump — gave different values for the off-axis shape coefficient C_{shp} . Both were exact rationals. Neither could be promoted, because the repository’s governing rule is that a dispute closes by derivation or not at all: code may not pick a side, average them, or prefer whichever rational looks more authoritative.

Comparing the two kernels record by record localized the disagreement rather than resolving it. Both have the same six weight orbits and the same normalized shape table; both satisfy the same two relations above. What differs is the skeleton scale and the *signs* of two amplitudes. One scale factor explains 144 of the 189 records to 2×10^{-12} : the rotation bulk is structurally shared, and the dispute is three amplitudes, not two kernels.

7.3 How it was settled

By building a third implementation that shares no primitive with either, and by recovering the historical pipeline’s own intermediate ledger.

The third implementation is a Wilson-loop calculus written from scratch: the free Hamiltonian by the Fierz identity as a rewiring of index ports — a like pair swaps wires, an unlike pair is cut and crossed, which is unitarity; Haar integrals by the $U(N)$ Weingarten function as the Gram pseudoinverse on S_n , with $\det U = 1$ inserted on virtual slots for the determinant families; and the resolvent by per-link irreducible projectors, so that every component is an exact eigenvector of H_0 with a known energy. It reproduces the four second-order constants, the two-hop weight with the correct sign on three chain types, and the single-contact and shared-link dressings, in seconds — and then gives the corner cumulant to the digit.

Separately, a survey of the corpus located the historical pipeline’s own Stage-3I ledger: 4,221 ordered words whose hash matches the input the kernel copies quote. Its rooted assembly, ported verbatim, reproduces the pinned 189 records exactly. Grouping each word’s rooted images by the set of plaquettes it touches then yields the historical kernel’s own connected cumulants — a reading the original pipeline never performed on its own data.

Theorem 7.3 (Adjudication of the off-axis coefficient). *The historical kernel and the independent assembly agree on every cluster of the rotation record — pair, fourteen chains, two fans, two corners — with identical rationals in both C -parity sectors, except one. The exception is the adjacent-face cube completion. The historical ledger holds 8 of the 24 insertion orderings, each with exactly the weight the multi-loop model assigns it, namely $(2, 1, 4, 2, 8, 8, 2, 4)$ in units of $1/1536$, totalling $31/1536$; the sixteen it lacks sum to $-25/512$. The opposite-face completion in the same ledger has all 24.*

Consequently, in the kernel's basis,

$$C_{\text{shp}} = C_{\text{historical}} + \frac{25}{1024} = -\frac{13035490122347}{550663802582400} = -0.0236723\dots \quad (7.3)$$

Substituted into the 24 records and solved over the whole zone, this gives $A = 5/48$ and $B = D = 0$ with no residual.

Graph id: C2,ADR:0024 Reproduce: workhouse why C2

[T1 exact], suite “the third implementation and the historical ledger”: nine exact checks, three of them registered findings.

The shortfall has a closed form at every rank. The adjacent-face completion is

$$c_4^{\text{adj}}(N) = \frac{-106}{N(N^2 - 1)^3}, \quad c_4^{\text{adj}}(3) = -\frac{53}{768}, \quad (7.4)$$

in ratio 53/80 to the opposite-face completion $-160/(N(N^2 - 1)^3)$; the -106 is the primitive -88 plus ten multi-loop orderings at -18 .

7.4 Three things this episode establishes

The disagreement was an enumeration shortfall, not a disagreement about mathematics. A pipeline dropped sixteen of twenty-four orderings on one cluster and nowhere else. That is a diagnosable defect, and it was diagnosed ordering by ordering rather than adjudicated by preference.

A basis error can survive validation. An earlier assembly of the same amplitude was correct and was read in the conjugate basis, giving a third value. Every C -odd component of the historical ledger is the negative of the assembly's value in one traversal while every C -even one is identical — the signature of conjugating a single face. The assembly was right; its basis was not. The decision record amends it rather than withdrawing it.

The right answer arrived by a forbidden route first. The value (7.3) is the one an all-rank continuation formula produces at $N = 3$ — and that derivation was retracted, because the substitution it used is not permitted. The retraction stands. That the forbidden route happened to reach the number the legitimate assembly later confirmed is not a vindication of the route, and the register keeps both facts.

Scope 7.4. Both recorded kernels remain in the registry with their verdicts: the historical superseded, the cold dump falsified, beside the proven assembled value. Nothing was deleted. One follow-up remains untried — the pair cluster from the third engine, pure-six family included — which can sharpen this result but not overturn it.

8 Rank grading: cancellation and determinant families

The strong-coupling coefficients of §6 and §7 are exact rational functions of N . That makes a question available which is invisible at any single rank: how does each order of the series scale with the rank, and why?

The answer has two halves. At even orders the scaling is fixed by an exact cancellation among four representation channels. At odd orders the coefficients vanish outright for even N , and for odd N they first appear at order $N - 2$ with a closed-form vertex. Neither statement can be seen from numerics at $N = 3$.

8.1 The one-face grading law

Let $S_m(N)$ be the C -parity splitting of the one-plaquette strong-coupling diagonal at order m ,

$$S_m(N) = (\text{odd})_m - (\text{even})_m = -2 K_m[0][1]. \quad (8.1)$$

Theorem 8.1 (Single-face planar grading through tenth order). *At even orders $m = 2, 4, 6, 8, 10$ the exact rational function of N satisfies*

$$S_m(N) = c_m N^{-(2m-1)}(1 + O(N^{-2})), \quad (8.2)$$

with

$$c_2 = -4, \quad c_4 = -32, \quad c_6 = -\frac{1748}{3}, \quad c_8 = -\frac{123332}{9}, \quad c_{10} = -\frac{49593808}{135}. \quad (8.3)$$

Writing $u = N^2\tau/2$ as in (4.4), the leading part of the order- m contribution is $(c_m/2^m) N\tau^m$. Every even order therefore carries the same single power of N , with

$$b_2 = -1, \quad b_4 = -2, \quad b_6 = -\frac{437}{48}, \quad b_8 = -\frac{30833}{576}, \quad b_{10} = -\frac{3099613}{8640}, \quad b_m = c_m/2^m. \quad (8.4)$$

Graph id: RESULT:SINGLE_FACE_PLANAR_GRADING

Recorded proven with output-certified evidence. The exponent and the coefficients are exact limits of exact rational functions of N , not numerical extrapolations. An independent integer-rank engine reproduces c_2, c_4, c_6, c_8 by extrapolation to within 1.4×10^{-5} relative — a cross-check by a different method, not the source of the values.

The word *leading* in Theorem 8.1 is doing work and should not be dropped. Each S_m is an exact rational function of N with relative corrections in N^{-2} , so the reduction to τ is a planar-limit statement, not an identity at finite rank. Explicitly, at second order

$$S_2(N) = -4N^{-3} - 4N^{-7}, \quad \text{so} \quad \frac{S_2 u^2}{N} = -\tau^2 - \frac{\tau^2}{N^4}. \quad (8.5)$$

The single-face band per unit N is therefore a function of τ alone as $N \rightarrow \infty$ at fixed τ , and not at finite N . A statement of the second form would be false, and an earlier internal summary of this result made it.

8.2 The grading is a cancellation theorem

Equation (8.2) is a much smaller number than its constituents, and the reason is exact.

Theorem 8.2 (Four-channel cancellation). *Decomposing S_m by the intermediate state of the perturbed vector gives exactly four contributing channels at every even m from 4 to 10: the singlet $((), ()),$ the adjoint $((1), (1)),$ and the two-box states $((2), ())$ and $((1, 1), ())$. Each is of order $N^{-(m-1)}$, with leading coefficients*

$$(-4, +2, +1, +1) \times 2^{m/2-2}, \quad (8.6)$$

which sum to zero. At $m = 2$ only the singlet and the adjoint contribute, at -2 and $+2$, and these also cancel.

Graph id: RESULT:SINGLE_FACE_PLANAR_GRADING

So the individual channels are of order $N^{-(m-1)}$ and their leading behaviour annihilates, leaving (8.2) at $N^{-(2m-1)}$ — two further orders down in the N^{-2} expansion. At $m = 4$ this is the statement that channels of order N^{-3} conspire to produce a band of order N^{-7} .

This is why the grading is stated as a theorem about a mechanism rather than as an observed power. A power law read off from fitted data at several ranks would carry no information about *why* the leading terms are absent, and would give no reason to expect (8.2) to persist. The cancellation does.

Scope 8.3. Theorem 8.1 concerns the *one-plaquette* diagonal splitting at even orders $m \leq 10$. It is not the cluster cumulant of the multi-face conjecture, which is not discharged by it: the one-face case at sixth order was already established separately, and the shared-link pair route carries the multi-face statement. The general- m law remains a conjecture. All statements are about coefficients at fixed order, not about convergence.

8.3 Odd orders are determinant families

Theorem 8.4 (Centre parity). *For even N , every odd-order matrix element of the des Cloizeaux effective operator of the strong-coupling band vanishes identically — on every cluster, in both C -parity sectors, at every order.*

For odd N , an odd order m vanishes unless $m \geq N - 2$. At $m = N - 2$ the only surviving element is the one-plaquette vertex

$$H_{N-2}(\bar{F}, F) = - \frac{(N/(N+1))^{N-3}}{((N-3)!)^2}, \quad (8.7)$$

which is -1 at $N = 3$ and $-\frac{25}{144}$ at $N = 5$. Hops and leakages vanish at that order.

Graph id: RESULT:ODD_ORDER_CENTRE_PARITY

The mechanism is the centre. The $SU(N)$ Haar integral of a word vanishes unless every link's net flux is $0 \bmod N$; on a single plaquette the four links are private, so the intermediate states are characters with energy $2C_2(R)$, and the flux condition at odd order can be met only once enough boxes have accumulated to close modulo N . That threshold is $N - 2$, and it is why the first odd order is a function of the rank rather than a fixed number.

Scope 8.5. Statements about coefficients of the strong-coupling series at fixed order, not about convergence. On a periodic lattice the link-set lemma requires even extents. The even orders and the overlap theorem are not addressed here.

8.4 Why this refuted a claim

The register records a claim that centre-only circuits are dynamically dark — that configurations carrying only centre flux cannot contribute to the dynamics. Theorem 8.4 and its fifth-order companion refute it: the exact fifth-order determinant correction $\delta c_{5,\det}$ is nonzero. The claim was falsified by an exact computation rather than by an estimate, which is the only way a claim of that shape can be settled.

9 The symbolic-rank engine

Every closed form in §6–§8 is a rational function of N computed as such, not interpolated through ranks. This section explains how, because the method is the most transferable thing in the program and because getting it wrong is the default.

9.1 The problem with sweeping ranks

The natural procedure is to run an exact engine at $N = 3, 4, 5, \dots$ and fit a rational function. That is what was done first. An assembled cluster cumulant β_N agreed with a candidate closed form $P_{17}(N^2)/(N R_{20}(N^2))$ at every rank from 4 to 70: sixty-seven exact agreements between exact rationals.

Sixty-seven exact agreements do not prove the identity. Rational interpolation is unique only once the degrees are bounded, and without a degree bound the agreements constrain nothing about $N = 71$. The review of that work made exactly this objection — no “for all N ” without a degree bound — and it was correct.

A degree bound could have been argued: Weingarten denominators, energy denominators, and a count of vertices together bound the degree, and the sixty-seven agreements would then prove the identity by uniqueness. That argument was not made. What was done instead makes it unnecessary, and is better.

9.2 Running the engine over $\mathbb{Q}(N)$

Inspect where the rank actually enters the exact engine. It enters in exactly two places that are not rational arithmetic:

- the charge predicate, $\text{charge} \equiv 0 \pmod{N}$;
- the flux-difference predicate, $|\text{diff}| \leq N$.

Everywhere else the rank appears only through rational expressions in $C_F = (N^2 - 1)/(2N)$, in $1/(2N)$, and in N^{closed} .

So make the two predicates overridable and promote everything else to the rational function field:

- scalars become elements of $\mathbb{Q}(N)$, carried as a numerator and a monic denominator in exact polynomial arithmetic;
- the Weingarten function becomes the symbolic inverse of the Gram matrix N^{cycles} on S_n ;
- the single-link spectra become the $\text{SU}(N)$ quadratic Casimirs of the irreducible representations that link’s flux content can carry.

A cumulant then emerges as *one rational function of N* , computed rather than fitted. The per-rank engine is otherwise untouched and its own test suite still holds — which is the check that the symbolic run is the same calculation and not a different one.

Theorem 9.1 (Closed forms over $\mathbb{Q}(N)$). *The fourth-order cluster cumulants, the two-hop weight in both C -parity sectors, and the assembled β_N are exact elements of $\mathbb{Q}(N)$ produced by the symbolic engine. The identity $\beta_N = P_{17}(N^2)/(N R_{20}(N^2))$ holds as an identity of rational functions.*

The final polynomial identity is promoted to [T0 Lean], and it is worth saying exactly which identity that is. The corpus *states* $\beta_N = P_{17}(N^2)/(NR_{20}(N^2))$ and never derives it. What Lean checks is that the closed forms the symbolic engine returns for the three cumulants and the adjacent-face cube completion *sum to* the corpus's coefficient — a polynomial identity between the engine's output and the corpus's formula. It is not a Lean derivation of the cumulants, and the Lean source says so in its own docstring. This document repeats the qualification because an internal summary of the same fact dropped it.

Every eigencomponent of every resolvent is verified to be an exact eigenvector of every single-link H_0 , symbolically — a check that has no per-rank analogue, because at a fixed rank a coincidence of numbers is indistinguishable from an identity.

9.3 Why this is the right shape of argument

The methodological content generalizes beyond this program.

An identity between rational functions can be established in two ways: verify it at enough points and bound the degrees, or compute in the function field. The first requires an argument about the size of the objects, which is a second piece of mathematics and is where the difficulty hides. The second requires only that the algorithm's dependence on the parameter be rational — which is a property one can *inspect*, predicate by predicate, rather than estimate.

The title of the decision record is the summary: the degree bound is proved by removing it.

Scope 9.2. The symbolic engine is valid at ranks where the generic partition labelling applies. Exceptional low ranks, where representations degenerate, are handled separately and are the reason $SU(2)$ and in places $SU(4)$ carry their own entries in the register.

This degree bound must not be confused with the *other* degree bound in this program's history — the proposed kinematic mechanism for the fourth-order tier collapse, which was retracted (§7, §16). That one was withdrawn because it was wrong. This one was removed because it became unnecessary.

10 Sixth order: a shape that cannot be removed

The fourth-order shape data closes in a five-element span. The question this section answers is whether the sixth order stays inside it. It does not, and the proof is a divisibility argument that needs no knowledge of the sixth-order dynamics at all.

10.1 Conventions

In the Bloch variables z_j write

$$a_j = 2 - z_j - z_j^{-1}, \quad q = \sum_j a_j, \quad e_2 = \sum_{i < j} a_i a_j, \quad e_3 = a_1 a_2 a_3, \quad (10.1)$$

and let $U = \psi\psi^*$ with $\psi^*\psi = q$, $P_c = U/q$, $Q_c = I - P_c$, $S = L_\downarrow - 4I$, and R the cross-plane part of S .

Two warnings, both of which have caused errors here. The invariants e_2, e_3 are Bloch invariants, *not* representation Casimirs; and U is the up-Laplacian, not its normalized projector, so its kernel is the complementary plane rather than the carrier line. Cleared identities hold at $q = 0$; normalized carrier formulas require $q > 0$.

10.2 The complete folded operator

Let P_E be the isolated electric model-space projector, $D = Q_E/(E_0 - H_0)$, and V_0 the perturbation. If $P_E V_0 P_E = a P_E$, set $V = V_0 - a I$ on the whole Hilbert space: shifting the complementary block as well is essential. With $\chi_0 = P_E$, intermediate normalization gives the exact recursions

$$K_n = P_E V \chi_{n-1}, \quad \chi_n = D V \chi_{n-1} - D \sum_{j=1}^{n-1} \chi_{n-j} K_j, \quad (10.2)$$

and with $M = P_E + \chi^* \chi$ the canonical Hermitian effective operator is $M^{1/2} K M^{-1/2}$. Writing

$$A_m = P_E V (D V)^{m-1} P_E, \quad C_2 = P_E V D^3 V P_E, \quad (10.3)$$

and B_m for the sum of A_m with exactly one of its $m - 1$ slots replaced by D^2 :

Theorem 10.1 (The eighteen-word sixth-order fold).

$$H_6 = A_6 - \frac{1}{2}\{A_4, B_2\} - \frac{1}{2}\{A_3, B_3\} - \frac{1}{2}\{A_2, B_4\} + \frac{1}{2}\{A_2^2, C_2\} + \frac{3}{8}\{A_2, B_2^2\} + \frac{1}{4}B_2 A_2 B_2. \quad (10.4)$$

Expanded, this is exactly eighteen words in P_E, V, D . The same recursion reproduces the established $H_4 = A_4 - \frac{1}{2}\{A_2, B_2\}$.

Graph id: RESULT:G9_HERMITIAN_FOLDED_SUPPORT

The $\{A_3, B_3\}$ term is present and it matters: dropping odd electric histories gives an incomplete formula. No odd order is dropped anywhere in the recursion. Independent rational matrix checks verify both the invariant-subspace equation and the normalization through degree six, with a rank-two P_E and non-commuting effective coefficients.

Does not assert. Equation (10.4) is the universal folded operator support in this convention. It does not evaluate Wilson plaquette matrix elements, establish a specific support, or carry out connected multi-plaquette rooted subtraction. Those require the actual electric and Haar implementation on each support.

10.3 Exact word symbols

In a polynomial carrier gauge, exact symmetrization gives each word its cleared symbol in $\mathbb{Q}[q, e_2, e_3]$. The table below is computed, not inferred, and all forty words of length zero through three agree coefficient by coefficient with independently implemented spatial Laurent operators.

Word	$\sigma(W)$	Word	$\sigma(W)$
I	q	RR	$q e_2 + 3 e_3$
U	q^2	RUR	$4 e_2^2$
S	$-4q$	RSR	$(q - 4)(q e_2 + 3 e_3) - 4 e_2^2$
S^2	$16q$	RRR	$-2 e_2^2 - 2 q e_3$
R	$-2 e_2$	URR, RRU	$q^2 e_2 + 3 q e_3$
UR, RU	$-2 q e_2$		

A word census must retain coefficients and combine them before testing shape membership: $\sigma(RUR + 2 RRR) = -4 q e_3$, although both individual words contain e_2^2 . Support is not unique modulo operator identities, and the invariant object being tested is the complete carrier polynomial. Missing direct input is represented as unknown, not as zero.

10.4 The band coefficient at sixth order

At fixed nonzero momentum, with scalar anchors removed,

$$H_{\text{eff}}(u) = u^2 t_3 q Q_c + u^3 H_3 + u^4 H_4 + u^5 H_5 + u^6 H_6 + O(u^7). \quad (10.5)$$

The recorded $SU(3)$ third-order factorization gives $Q_c H_3 P_c = 0$, so the off-diagonal block first appears at order u^2 in the scaled operator, from H_4 . Expanding the Schur complement for the scalar carrier branch,

$$\lambda_6 = \frac{\sigma(H_6)}{q} - \frac{q \sigma(H_4^2) - \sigma(H_4)^2}{t_3 q^3}. \quad (10.6)$$

This is not an assumption that H_5 vanishes. H_5 first couples to H_4 at order u^7 in H_{eff} , as does the uH_3 correction to the complementary inverse, and unknown off-diagonal H_6 contributes later still. The decoupling of H_3 is an explicit recorded input rather than an assumption, and any scalar vacuum subtraction is part of H_6 and changes nothing.

Evaluating all twenty-five ordered pairs from the actual H_4 support $\{I, U, S, S^2, R\}$, including their projected subtraction, gives zero for twenty-four of them. Only RR survives, in agreement with direct composition of the full assembled spatial kernel. With

$$\kappa = \frac{4C_{\text{shp}}^2}{t_3} = \frac{169924002729806205028788409}{619343593697933825385600000} > 0, \quad t_3 = \frac{5}{612}, \quad (10.7)$$

the known dynamical mixing support is $-(\kappa/q)RR + (\kappa/q^2)RUR$, with normalized expectation

$$\lambda_6^{\text{mix}} = -\kappa \left(\frac{e_2}{q} + \frac{3e_3}{q^2} - \frac{4e_2^2}{q^3} \right). \quad (10.8)$$

Does not assert. The RUR here arises from inserting $P_c = U/q$ in the projected subtraction. It is not evidence that a direct six-insertion electric history carries the local word RUR , and the ratio $1 : 3 : -4$ belongs to this mixing term rather than to the full coefficient.

10.5 Non-cancellation

Theorem 10.2 (An extra rational shape survives every local direct term). *Assume the order-six electric-shell effective operator is a finite-range, translation-covariant stencil with Laurent-polynomial entries in this basis, and write $F = \sigma(H_6)$. Then (10.6) gives*

$$q^3 \lambda_6 = q^2 F - \kappa(q^2 e_2 + 3q e_3 - 4e_2^2), \quad (10.9)$$

and q does not divide the right-hand side, for any Laurent polynomial F .

Proof. Modulo q , the right-hand side of (10.9) is $4\kappa e_2^2$. Evaluate at

$$z_1 = z_2 = -1, \quad z_3 = 5 + 2\sqrt{6}, \quad (10.10)$$

all nonzero, so the point lies in the Laurent ring. Since $z_3^{-1} = 5 - 2\sqrt{6}$, we get $a = (4, 4, -8)$, hence $q = 0$ and $e_2 = -48$. The value there is $4\kappa \cdot 48^2 = 9216\kappa \neq 0$, and a Laurent polynomial divisible by q would vanish at that point. $\square$

Graph id: RESULT:G9_LOCAL_SIXTH_NONCANCELLATION

[T1 exact]. Consequently λ_6 cannot lie in the fourth-order shape span $\{1, q, e_2, 4e_2/q, e_3/q\}$, and no local direct RUR term removes the extra rational shape: a direct $hRUR$ contributes $4he_2^2/q$, which cannot cancel an e_2^2/q^3 component. For symmetric F the remainder modulo q^2 is exactly $4\kappa e_2^2 - 3\kappa q e_3$, so both e_3/q^2 and e_2^2/q^3 are present. The e_2/q term alone *can* be cancelled, by $F = \kappa e_2$ — for instance a local $H_6 = -(\kappa/2)R$ — so the $1 : 3 : -4$ ratio is not a constraint on all sixth-order shapes.

Note what the proof does not use: any knowledge of F . The extra shape is forced by the fourth-order data alone, through the Schur complement, whatever the sixth-order dynamics turns out to be.

10.6 A withdrawn derivation

An earlier version of this work enumerated closed rooted walks — 90 four-step unit-edge walks, of which 66 retrace immediately and 24 are planar squares, together with 192 non-retracing six-step walks using all three axes — and read a sixth-order amplitude off that census.

The derivation is withdrawn. Edge walks are not ordered plaquette insertions with electric states, Haar weights, projectors and folds, and no map between them was supplied; the census therefore determines no net coefficient of any of the candidate words. The old amplitude formula was assigned rather than derived. The old carrier function used $\sigma(R) = 0$ and $\sigma(U) = q$, where the actual unnormalized symbols are $-2e_2$ and q^2 ; it returned a nonzero expression for a defect it separately reported as zero. The implementation now computes the symbols and tests the defect, and the old interface raises an explicit missing-dynamics error rather than returning a number.

What is withdrawn is a claimed derivation, not the possibility of a physical amplitude of that shape. Theorem 10.2 is independent of the census entirely. Computing the direct term on a connected multi-plane support is Problem 8, and it is the cheapest decisive computation currently open in this program.

11 A rank-uniform local shell construction

This section is a separate result in its own regime, included because it is exact, rank-uniform, and independent of everything above — not because it continues it. The object is the formal local Wilson oscillator expansion around a single plaquette, and the point is that its spectrum admits a finite exact recursion at every perturbative order with coefficients in $\mathbb{Q}[N, N^{-1}]$, needing no Gram-matrix inversion.

Conflating this construction with the flux band of §5–§10, or with the Wilson transfer operator of §13, would be an error. They are four different objects — the physical $SU(N)$ transfer band, the $SU(3)$ internal Hamiltonian carrier, the conditional block, and this local oscillator — and no cross-regime continuation between them is asserted anywhere in this document.

11.1 The exact algebra

Let X be a traceless Hermitian $N \times N$ matrix with density proportional to $e^{-\text{Tr}(X^2)}$, write $P_k = \text{Tr}(X^k)$ with $P_0 = N$ and $P_1 = 0$, and use a traceless Hermitian basis normalized by $\text{Tr}(T_a T_b) = \delta_{ab}$. The covariance is

$$\mathbb{E}[X_{ab} X_{cd}] = \frac{1}{2} \left(\delta_{ad} \delta_{bc} - \frac{1}{N} \delta_{ab} \delta_{cd} \right). \quad (11.1)$$

With $D = \frac{1}{2}\Delta$ and E the polynomial Euler operator, the completeness identity for this basis gives the exact relations

$$DP_k = \frac{k}{2} \sum_{l=0}^{k-2} P_l P_{k-2-l} - \frac{k(k-1)}{2N} P_{k-2}, \quad (11.2)$$

$$D(fg) = (Df)g + f(Dg) + \nabla f \cdot \nabla g, \quad (11.3)$$

$$\nabla P_k \cdot \nabla P_l = kl \left(P_{k+l-2} - \frac{1}{N} P_{k-1} P_{l-1} \right). \quad (11.4)$$

These are exact at every integer rank, including ranks at which some traces become dependent — which is the property that makes the construction rank-uniform rather than rank-by-rank. For homogeneous F of positive degree d , Gaussian integration by parts gives $\mathbb{E}_\mu[F] = \mathbb{E}_\mu[DF]/d$, and with $\mathbb{E}_\mu[1] = 1$ and vanishing odd moments this is a finite exact moment recursion.

11.2 Shell projectors without a Gram inverse

Proposition 11.1 (Terminating Wick conjugation). *Put $L = E - D$. Since $[E, D] = -2D$, the terminating exponential $W = \exp(-D/2)$ satisfies*

$$LW = WE. \quad (11.5)$$

Hence, with H_d the extraction of the homogeneous degree- d part,

$$\Pi_d = \exp(-D/2) H_d \exp(D/2) \quad (11.6)$$

is the exact oscillator-shell projector, and on each finite polynomial space $R_s = \sum_{d \neq s} \Pi_d / (s - d)$ is the exact shell resolvent.

The content of Proposition 11.1 is that L and E are conjugate by a *terminating* exponential, so the shell decomposition of the interacting operator is the shell decomposition of the Euler operator, transported. No Gram matrix is formed and none is inverted, which is precisely the step that makes rank-uniformity delicate elsewhere in this program.

11.3 The coefficients

Theorem 11.2 (All-rank local coefficients). *The terminating conjugation gives exact shell projectors and a unique all-order simple-branch formal recursion in $\mathbb{Q}[N, N^{-1}]$, with state degree at most $s + 4r$. The two formal local gaps have exact all-rank coefficients through β^{-2} , including c_3^- and $c_4^\pm$. The five corrections $c_0, \dots, c_4$ are strictly negative at every allowed rank, by positive shifted numerator polynomials.*

Graph id: RESULT:LOCAL_CLASS_WICK_RECURRENCE, RESULT:LOCAL_CLASS_WICK_COEFFICIENTS, RESULT:LOCAL_CLASSES_WICK_SIGNS

The construction applies to the vacuum, the first charge-even shell for $N \geq 2$, and the first charge-odd shell for $N \geq 3$. It recovers the previously archived c_0, c_1, c_2 formulas in both sectors and the archived all-rank c_3^+ , and additionally derives c_3^- and both all-rank c_4 formulas. The certificate carries full polynomial eigen-equation residuals rather than only agreement with the archived constants — a distinction worth stating, because agreement with a previous value is a weaker check than satisfying the equation.

The recovery is by an independent route: the archive engine uses full-GUE cut-and-join and then subtracts the trace, while this one applies D directly in traceless coordinates.

11.4 A companion algebraic identity

Theorem 11.3 (Shifted-power carrier symbols). *For the cubic Laurent operators with $S = L_\downarrow - 4I$,*

$$\sigma(RS^m R) = (q - 4)^m (qe_2 + 3e_3) - 4\Pi_m(q) e_2^2 \quad (11.7)$$

for every natural m , with Π defined polynomially also at $q = 0$.

Graph id: RESULT:RSM_R_CARRIER_SYMBOL_MASTER_THEOREM

This closes the word table of §10 at all powers, and is formalized in Lean as an all-power operator statement.

Scope 11.4. Theorem 11.2 concerns the *formal* local class oscillator only. No error bound for the compact-group eigenvalues is claimed, and local spectral coefficients do not by themselves identify the interacting Wilson spectrum, an interacting-volume estimate, or a continuum source measure. Theorem 11.3 is an algebra identity and carries no sixth-order dynamical population claim.

12 External comparison

Published work enters this program as evidence rather than as authority. A paper is unchecked until something checks it, exactly like an internal document; the reason an external agreement counts for more is not that it is published but that it was produced without knowledge of this program, so agreement is not bookkeeping.

The map holds 108 indexed papers and 152 recorded relations to named claims here. 54 of those edges are verified — the paper was obtained, read, and compared against the claim — across 34 distinct papers. The remainder are recorded targets for papers not yet obtained, which is itself useful: a paper heavily cited within this web and still unobtained is automatically the next acquisition target.

An entry earns its place by naming a specific claim and saying what relationship it has to it. `workhouse lit -for G19` answers the query in the other direction, and validation rejects an entry whose target does not resolve.

12.1 A 1989 table, and rationals computed cold

The sharpest external check available to this program is a Hamiltonian strong-coupling series for SU(3) glueball masses published in 1989 [51], whose Table 1 is pinned here by digest. Under the normalization bridge $m_n = 2^{n-1}a_n$, its coefficients are compared against the Γ -point coefficients this program derives as exact rationals.

In the C -odd (1^{+-}) channel — the sector of the carrier, whose series is (5.5) — the agreement runs through third order:

Order	Published	This program	Gap (bound)
$n = 0$	$\frac{16}{3}/2$	$\frac{8}{3} = E_{\text{flat}}(0)$	exact
$n = 1$	$a_1 = +1$	the u coefficient +1	exact
$n = 2$	$2a_2 = 0.0359477124184$	$\frac{11}{306}$	$9.94 \times 10^{-14} \ (2 \times 10^{-13})$
$n = 3$	$4a_3 = -0.437135556836$	$-\frac{109151}{249696}$	$1.11 \times 10^{-12} \ (3 \times 10^{-12})$

In the C -even (0^{++}) channel, $2a_2 = -0.21274509804$ against $-\frac{217}{1020}$ with gap 7.84×10^{-13} , and $4a_3 = -0.1039004229144$ against $-\frac{54049}{520200}$ with gap 1.36×10^{-13} . The sign structure matches as well: $a_1 = -1$ in the scalar channel, where the carrier's is $+1$.

Each bound above is the printed half-ulp of the published table scaled with margin, so these are agreements to the precision at which the 1989 table was printed — not approximate corroborations. The rationals $11/306$ and $109151/249696$ are the third-order coefficients of E_{flat} in (5.5), computed here from Haar integration and the cellular complex with no knowledge of the table.

At fourth order, $8a_4 = -0.7751458630184$ against $m_{\Gamma}^{(4)} = -0.7751458630189173$, gap 5.17×10^{-13} against a bound of 5.3×10^{-13} .

Scope 12.1. These are [T2 numeric] checks: float agreement within a stated tolerance, to the precision of a printed table. They corroborate the coefficients; they do not prove them, and the exact statements of §6–§8 do not rest on them. Note also that the pinned revision of the source compares only $2a_2$ and $4a_3$ in its own text — the fourth-order agreement is registered here and not leaned on there.

12.2 The two contradicting edges

Two edges in the map are recorded as *contradicts*, both from a 1976 strong-coupling series, and both against this program: one against the third-order C -odd coefficient, one against the Γ -point fourth-order value. They originate in a disagreement the 1989 paper itself reports at orders x^3 and x^4 with the series it supersedes.

Both are recorded as not yet obtained. Until that primary source is read, this program has two unresolved external contradictions against coefficients it otherwise regards as settled, and says so rather than resolving them by the age of the sources. Obtaining that paper is listed with the acquisition targets below.

12.3 How far a novelty claim here can be trusted

Not as far as the size of the index suggests, and the limitation is worth stating precisely rather than as a disclaimer.

Only 54 of 152 recorded edges are verified. The unverified remainder is not a random sample: it includes several of the *nearest* neighbours to this program's own results. The main Nuclear Physics B paper of the 1981 strong-coupling glueball series is recorded as not obtained — only its errata were — and the 1974 and 1979 large- N papers that bear most directly on the rank grading of §8 are likewise unobtained, with the ledger noting for one of them that its definitions were used from memory and only their sign consequences checked.

Consequently: where this document says a result is novel, that means *no verified entry in this index contains it*, which is a weaker statement than priority. Two of the strongest results are affected. The rank-grading cancellation of §8 sits closest to large- N counting arguments in papers this program has not read; and the fixed-spacing transfer construction of §13 sits alongside the classical positivity and self-adjoint transfer-matrix results of Osterwalder and Seiler, which the index does register.

Acquiring the unobtained near neighbours is cheap relative to everything else in §17, and it is the one item on that list that could *subtract* from the program rather than add to it. That is a reason to do it, not a reason to defer it.

Table 1: Verified external-evidence edges. Each row is a published result that was obtained, read, and checked against a named claim of this program. Relation is the ledger’s own classification; *contradicts* and *confusable* are listed first because they are the rows that cost the program something.

Paper	Ref.	Year	Bears on	Relation
HSB_2000	[53]	2000	HAMER_A4_NUM	confusable with
HAMER_1989	[51]	1989	BAND_EVEN_BOTTOM	corroborates
HAMER_1989	[51]	1989	C1	corroborates
LLL_2006	[69]	2006	C1	corroborates
HAMER_1989	[51]	1989	D_3	corroborates
SCHIERHOLZ_1988	[91]	1988	G18	corroborates
HAMER_1989	[51]	1989	M3_EVEN_K0	corroborates
KPS_1981	[64]	1981	R14	corroborates
CBB_2026	[21]	2026	R2	corroborates
KPS_1981	[64]	1981	G7	supplies a value
HAMER_1989	[51]	1989	HAMER_A4_NUM	supplies a value
MUNSTER_1985_TM	[79]	1985	C2	supplies a comparison
CAO_2023	[16]	2023	C7	supplies a comparison
LEMOINE_2026	[68]	2026	C7	supplies a comparison
HAZRA_2026	[54]	2026	G14	supplies a comparison
AT_2020_SU3	[2]	2020	G18	supplies a comparison
BESIII_2026	[22]	2026	G18	supplies a comparison
CHEN_2006	[18]	2006	G18	supplies a comparison
LLL_2006	[69]	2006	G18	supplies a comparison
MORNINGSTAR_2025	[76]	2025	G18	supplies a comparison
MP_1999	[77]	1999	G18	supplies a comparison
AT_2021_SUN	[3]	2021	G19	supplies a comparison
BB_1983	[10]	1983	G19	supplies a comparison
FLPS_2026	[27]	2026	G19	supplies a comparison
URZUA_2019	[102]	2019	G19	supplies a comparison
BORGA_2024	[11]	2024	G5	supplies a comparison
CAO_2023	[16]	2023	G5	supplies a comparison
OBZ_1985	[82]	1985	G5	supplies a comparison
AT_2021_SUN	[3]	2021	G6	supplies a comparison
CM_2003	[17]	2003	G6	supplies a comparison
BALAJI_2026	[8]	2026	R2	supplies a comparison
CKS_2021	[19]	2021	U1	supplies a comparison
DRS_2021	[25]	2021	U1	supplies a comparison
KRS_2023	[62]	2023	U1	supplies a comparison
SZH_1997	[92]	1997	U1	supplies a comparison
CS_2006	[23]	2006	C7	supplies a method
JW_2006	[61]	2006	G17	supplies a method
UELTSCI_2004_CLUSTER	[101]	2004	G17	supplies a method
YAROTSKY_2004_GS	[107]	2004	G17	supplies a method
BB_1983	[10]	1983	G18	supplies a method

Paper	Ref.	Year	Bears on	Relation
JW_2006	[61]	2006	G18	supplies a method
NSY_2019	[80]	2019	G18	supplies a method
YAROTSKY_2004_QP	[108]	2004	G18	supplies a method
JW_2006	[61]	2006	G19	supplies a method
KLEIN_ROSENBERGER_2020_WKB	[63]	2020	G19	supplies a method
SCHIERHOLZ_1988	[91]	1988	G19	supplies a method
UELTSCHI_2004_CLUSTER	[101]	2004	G19	supplies a method
YAROTSKY_2004_GS	[107]	2004	G19	supplies a method
YAROTSKY_2004_QP	[108]	2004	G19	supplies a method
JW_2006	[61]	2006	G20	supplies a method
JW_2006	[61]	2006	G22	supplies a method
JW_2006	[61]	2006	G23	supplies a method
MUNSTER_1985_TM	[79]	1985	G3	supplies a method
KRS_2023	[62]	2023	R2	supplies a method

Part II

The construction at fixed lattice spacing

13 The construction at fixed lattice spacing

This is the strongest constructive statement in the program. It is also the one most often misread, so it is stated here with its hypotheses, its interval, and an explicit list of what it does not assert.

Throughout this section ε denotes the temporal mesh. It is *not* the planar variable τ of (4.4); the two are unrelated and the symbols are kept distinct deliberately.

13.1 The theorem

Theorem 13.1 (Infinite-volume Wilson band and literal source frame). *Under the hypotheses listed below, the Perron-normalized Wilson transfer operator has a common strong limit along open-box and centered-periodic exhaustions, in the transported product representation. The limit is a positive contraction fixing the vacuum; it differs from the free product in norm by at most $1/998$, and its norm on the non-vacuum subspace is at most $\frac{4}{5} + \frac{1}{998}$.*

On the coupling interval

$$|u| \leq \frac{u_*}{10\,022\,400\,000\,N}, \quad (13.1)$$

the complete infinite-volume physical odd band is isomorphic to $\ell^2(\mathbb{Z}^3) \otimes \mathbb{C}^3$. The projected literal plaquette-source synthesis is onto that entire band, with Gram bounds

$$\frac{9}{16} I < \mathcal{G} \leq \frac{81}{64} I. \quad (13.2)$$

The Euclidean Osterwalder–Schrader band and the quantum GNS band coincide, intertwining the physical transfer and the sources.

Graph id: RESULT:WILSON_INFINITY_PHYSICAL_BAND

Recorded **proven** with **analytic** evidence; [T3 analytic] for whole-statement machine certification, with its component estimates checked across several suites.

13.2 Hypotheses

1. All calibrated kinetic, positive Wilson-kernel, incidence and common-disk hypotheses of the weighted-activity and cardinality-chart theorems.
2. Fixed $SU(N)$ with $N \geq 3$, and fixed *spatial* lattice spacing. Compatible induced open boxes, or centered periodic exhaustions, carrying the same local data.
3. Temporal mesh $0 < \varepsilon \leq \varepsilon_0 \leq \log(5/4)/(4\gamma)$, with $m = \lceil \log(5/4)/(\gamma\varepsilon) \rceil$, and the full neutral physical odd free-window premise.
4. $\gamma = C_F/2 = (N^2-1)/(4N)$, the *exact* calibrated fundamental-link energy. This is a hypothesis about an identity, not a gap lower bound one is free to weaken: substituting a smaller number invalidates the conclusion rather than weakening it.
5. Literal odd sources $\mathcal{O}_p = (\chi_p - \bar{\chi}_p)/\sqrt{2}$, with free orthonormal plaquette vectors of norm at most $\sqrt{2}N$.
6. The calibrated representation-uniform kinetic-window mesh premise, whose N -dependent admissible temporal threshold is the previously established one. Every estimate is uniform in volume and in meshes satisfying these premises, and the common coupling interval depends on fixed N and is nonzero.

The interval (13.1) unpacks as $u_0 = u_*/1252800000$ and $u_1 = u_0/(8N)$, with the accompanying constants $\eta_{\text{act}} = 1/2500$, $\epsilon = 1/998$, $g_* = 1024/15625$ and $q_0 = \frac{4}{5} + \frac{1}{998} < 1$. The bound $\eta_{\text{act}} \leq 1/2500$ on the integrated generator norm at weight $1 + \log 2$ is a *hypothesis* of the source specialization, not a conclusion of the theorem; it is listed here to prevent the reading in which the theorem is taken to establish it.

13.3 Why totality is the substance

The clause that carries the physical content of Theorem 13.1 is that the source synthesis is *onto* the band, not merely bounded into it.

Break 15.4 is the reason. A local probe family can decay at one rate while the true spectrum contains a much slower state the probe cannot see; the three-dimensional model of Proposition 15.4 exhibits a factor of twenty-one between the observed rate and the actual gap. A construction that produced a band and a decay estimate, without totality, would prove nothing about the spectrum.

Totality here is a conclusion. The tagged source norm controls the entire synthesis operator; close projections and a Neumann inverse establish surjectivity in infinite dimension; and reflection-positive history completion contains the whole band by selective source cyclicity. Fine time steps are the positive m -th root of the block.

13.4 What Theorem 13.1 does not assert

Does not assert. Global equality of the Osterwalder–Schrader and full quantum high-energy spaces. Global transfer injectivity. Any spatial continuum limit. Extension beyond $SU(N)$, $N \geq 3$. The positive energy gap is in electric-time units, and its conversion to physical units is not part of the statement.

Most importantly, and separately from the list above: the interval (13.1) is a neighbourhood of $u = 0$, and the continuum limit is taken along a trajectory on which $u \rightarrow \infty$. Theorem 13.1 therefore supplies no control whatever on the continuum limit. This is Break 15.5, it is structural, and no improvement of the constant 10 022 400 000 addresses it. A version of exactly that inference was made in this program’s own drafts and retracted; see §16.

13.5 The relation to the Hamiltonian carrier

Theorem 13.1 is a statement about the Wilson transfer operator. The Hamiltonian carrier of Part I is a different object, spectrally identified with the band only on momentum patches away from zero.

The obstruction at zero momentum is proved, not merely encountered, and it is a statement about *chains*:

1. every vector in the image of the cube boundary map has exactly zero zone-average component;
2. the zero-momentum carrier content is therefore carried entirely by the three wrapping sheets of §5;
3. a sheet’s normalized overlap with any fixed local observable is exactly $1/L$;
4. compact two-cycles on open windows are exactly cube boundaries — checked directly for small windows, and Alexander duality in general.

The repository’s own prose summarizes (i)–(iv) as “every finitely supported exactly-flat operator is Γ -blind”. That phrasing is a convenient shorthand and this document does not adopt it: the four statements are about chain vectors, and read literally as a claim about arbitrary operators the shorthand is false. What follows from (i)–(iv) is the thing actually needed — that an interpolator for the zero-momentum protected level must leave the exactly-flat local algebra, coupling either through in-sector directions that are flat-degenerate at Γ anyway, or through out-of-sector multi-plaquette content.

13.6 A prediction, sharper than the measurement it explains

The published few-percent overlap of a local operator with the lightest state is not a constant: it degrades toward the continuum. For a local lattice source whose lowest-dimension continuum operator has dimension d , the overlap law is

$$Z_0(a) \sim \frac{|f_d|^2 a^{2d-3}}{2M}, \quad (13.3)$$

which at $d = 4$ gives a^5 and so reproduces the published measurement — the agreement that makes (13.3) worth recording at all.

But a^5 is the wrong exponent for *this* carrier. The C-odd carrier’s leading continuum operator is $d^{abc} F^a F^b F^c$, of dimension six, so the applicable exponent is $a^{2 \cdot 6 - 3} = a^9$. The extrapolation is brutal: $Z_0 \approx 1.2 \times 10^{-6}$ at $a\sqrt{\sigma} = 0.10$ and 2.4×10^{-9} at 0.05. Still a power, so formally survivable, and a serious practical obstacle to measuring this state with a local operator.

The same law is the argument for smearing, quantitatively: at fixed physical radius R , $Z_0 = |f_R|^2 R^{2d-3}/(2M)$ is a -independent. A smeared basis is structural here, not a convenience.

Scope 13.2. Equation (13.3) is asserted, not proved here. What is checked is the dimension arithmetic and the agreement of its $d = 4$ case with the published a^5 . An earlier version of this program’s notes quoted a^5 for the carrier itself; that is corrected above.

Scope 13.3. The out-of-sector dressing, the transfer-matrix identification at zero momentum, and everything non-perturbative remain open. A particle statement lives at fixed nonzero momentum, where no collapse occurs; the L^{-2} statement in the literature is a minimal-grid-momentum statement governing the approach to Γ , not a fixed-momentum one.

Part III

The continuum limit: what is required, and what is ruled out

14 The continuum problem

Part II constructs an infinite-volume theory with a positive gap at fixed lattice spacing. This part is about the distance between that and a continuum quantum Yang–Mills theory with a mass gap, and it is written to be useful to someone trying to close that distance.

14.1 The target, and what has no argument here

The statement in question is the existence of a non-trivial quantum field theory on $\mathbb{R}^{1,3}$ for $G = \text{SU}(N)$, $N \geq 2$, satisfying the Wightman axioms, with

$$\text{spec}(H)|_{\Omega^\perp} \subset [m, \infty), \quad m > 0. \quad (14.1)$$

Four of its ingredients have no argument anywhere in this program and are stated here so that no reader has to infer their absence.

1. **Poincaré covariance.** No construction of a unitary representation of the covering group of $\mathcal{P}_+^\uparrow$ is given.
2. **Restoration of rotational invariance.** The hypercubic lattice breaks it; nothing here restores it in the limit.
3. **The continuum measure.** It is never constructed.
4. **Reconstruction.** Even granting a limiting kernel, the passage to a Wightman theory is not carried out.

What *is* carried out is a set of estimates on the lattice side, and this part states them, states why the six obstructions of §15 block their assembly, and states in §17 what would have to be proved.

14.2 The three routes that are live

The estimates in this program group into three approaches, which are alternative rather than sequential. The ledger records them as independent routes and explicitly notes that no chain of necessary conditions runs between them.

The conditional-score route. Control the conditional law of a link given its complement, in the small-coupling limit, by dominating the score of the conditional measure. The available inputs

are strong: exact heat-bath conditional laws, synchronized radial dilation following the conditional minimum curve, the antipodal magnetic normal geometry, exponential suppression outside the complete minimizing set, and a uniform Agmon action gap on the complement. The obligation is Problem 1, and what stands between the inputs and the conclusion is polynomial amplitude control and the tube moments, not a missing structural idea.

The transport route. Compare the fine and coarse theories by transporting the vacuum and the sources along a common energy domain against a physical clock. Here the recorded failure is instructive: an earlier closure used a vacuum-only skew generator, and the source generator transports the ground state rather than annihilating it. The corrected obligation is Problem 2, and its companion, Problem 3, is the requirement that the comparison’s constants not degrade with the number of interacting blocks.

The moving-time route. Extract a limiting gap from approximate sources and one growing observation time per cutoff. The implication from its hypotheses to a gap is proved, and sharp: pairwise inequalities give the optimal normalized rate, the horizon penalty and a witnessing time, and one-atom examples show the rate cannot be improved. Its hypotheses are the difficulty — see Break 15.3, which shows the theorem returns exactly zero when the background error rate fails to exceed twice the probe tilt rate, an inequality no calculation certifies for four-dimensional Yang–Mills.

14.3 Why none of the three currently closes

Each route reaches a point where a summability or uniformity statement is required, and at each such point the natural estimate is one power short.

That is not a coincidence of technique. Break 15.1 exhibits the arithmetic: on an asymptotically free trajectory the running coupling gives increments of size $O(g_k^2) = O(1/k)$, the normalization of the scale derivative carries no small factor, and $\sum 1/k$ diverges. Break 15.2 exhibits the same shortfall in the polymer variable, where a contracting activity and a scale-independent physical mass cannot be read off one schedule. Problem 3 is the same shortfall again in the block count.

Underneath all three sits Break 15.5: the estimates are proved near $u = 0$ and the continuum lives at $u \rightarrow \infty$. A summable increment would connect scales *within* a regime; it would still not move the construction between regimes. That is why Problem 6 is marked load-bearing and the others are not.

15 Six obstructions, with counterexamples

This section states the obstructions that any argument from the fixed-spacing construction of §13 to a continuum mass gap must clear. They were produced by a deliberate adversarial audit of this program’s own drafts, and four of them killed arguments that had been written down here as complete.

They are stated as mathematics, not as caveats. Five carry explicit finite counterexamples, and a finite counterexample to a proof strategy is a permanent fact about that strategy. A reader who wants to know where the remaining difficulty of the problem actually sits should read this section and §17 and can safely skip everything else in Part III.

The counterexamples are not rhetorical. Those of Break 15.4 are registered checks in this repository’s own test suite — `tests/test_os_kernel_gap.py`, six checks, passing — and the two collapse conditions of Break 15.3 are checked in `tests/test_moving_time_gap.py`. An obstruction

that a repository asserts against itself, and then runs in continuous integration, is harder to forget than one recorded in prose.

15.1 Break 1: an $O(g_k^2)$ increment bound does not give convergence

A renormalization-group argument for the continuum limit produces a sequence of effective actions S_k^{eff} and observable expectations $m_k(\mathcal{O})$ indexed by scale, and controls the one-step increment by the running coupling:

$$|m_{k+1}(\mathcal{O}) - m_k(\mathcal{O})| \leq C_{\mathcal{O}} g_k^2 \leq \frac{C'_{\mathcal{O}}}{k + k_0}, \quad (15.1)$$

the second inequality being asymptotic freedom, $g_k^2 = \kappa/(k + k_0)$. The step that must not be taken is to conclude from (15.1) that (m_k) is Cauchy. Telescoping gives only

$$|m_n(\mathcal{O}) - m_m(\mathcal{O})| \leq C'_{\mathcal{O}} \sum_{k=m}^{n-1} \frac{1}{k + k_0} \asymp C'_{\mathcal{O}} \log\left(\frac{n}{m}\right), \quad (15.2)$$

which does not tend to zero as $m, n \rightarrow \infty$.

Proposition 15.1 (Counterexample). *Let $x_k = \sin(\log(k + 1))$. Then $x_k \in [-1, 1]$ and, by the mean value theorem, $|x_{k+1} - x_k| \leq 1/(k + 1) \rightarrow 0$, yet the set of subsequential limits of (x_k) is the whole interval $[-1, 1]$.*

So the increment bound (15.1) is consistent with the observable expectations oscillating forever across a fixed interval. What is needed is one more power: $|m_{k+1} - m_k| = O(g_k^4) = O(k^{-2})$.

The obstruction to obtaining it is not technical. On the asymptotically free trajectory $u_k = (\alpha + \beta k)^{-1}$,

$$\Delta u_k = -\frac{\beta}{(\alpha + \beta k)(\alpha + \beta(k + 1))}, \quad \frac{|\Delta u_k|}{4u_k u_{k+1}} = \frac{\beta}{4} = O(1), \quad (15.3)$$

and differentiating an expectation along the trajectory,

$$\frac{d}{ds} \mu_s(\mathcal{O}_s) = \mu_s(\partial_s \mathcal{O}_s) - \text{Cov}_{\mu_s}(\mathcal{O}_s, \partial_s S_s), \quad (15.4)$$

the normalization carries no small factor. The natural size of the variation is therefore $O(g_k^2)$, and a bounded-covariance estimate cannot improve it. Converting $1/k$ into $1/k^2$ requires an exact cancellation of the leading partition-function covariance in (15.4). No such cancellation is proved here.

Scope 15.2. This obstruction applies to the multiscale route recorded as

DERIV:YANGMILLS_CONTINUUM_BALABAN_MULTISCALE_PROOF:THEOREM_7_1,

whose Cauchy input is exactly what (15.1) fails to supply. It does not bear on any statement in Part I or II, which are at fixed scale.

15.2 Break 2: the polymer activity and the mass schedule are algebraically incompatible

A multiscale polymer cluster expansion carries an activity parameter κ_k that must contract for the expansion to converge, say

$$\kappa_{k+1} = \frac{3}{4} \kappa_k, \quad \kappa_k = \kappa_0 \left(\frac{3}{4}\right)^k \longrightarrow 0, \quad (15.5)$$

while a physical mass is extracted at each scale by

$$\frac{\kappa_k - \log C_4}{a_k} = c_0 \Lambda > 0 \quad \text{for all } k \geq 0, \quad a_k = a_0 L^{-k}, \quad L = 3, \quad (15.6)$$

with $C_4 > 1$ the polymer coordination constant.

Proposition 15.3. *The two displays above are incompatible. Since $\kappa_k \rightarrow 0$ and $\log C_4 > 0$, there is a finite*

$$k_* = \left\lceil \frac{\log(\kappa_0 / \log C_4)}{\log(4/3)} \right\rceil$$

with $\kappa_k - \log C_4 < 0$ for every $k > k_*$; beyond that scale the cluster sum $\sum_n e^{-(\kappa_k - \log C_4)n}$ diverges and the expansion does not exist. Setting $C_4 = 1$, which discards polymer entropy, does not repair it: then $\kappa_k / a_k = (\kappa_0 / a_0)(3L/4)^k = (\kappa_0 / a_0)(9/4)^k \rightarrow \infty$, so (15.6) cannot equal a finite constant either.

The content of Proposition 15.3 is that a contracting activity and a scale-independent physical mass cannot both be read off the same schedule. Any repair must decouple them — the mass must come from a quantity that is not the activity divided by the spacing.

15.3 Break 3: the moving-time gap can be exactly zero

The moving-time theorem of §14 extracts a limiting gap from approximate sources and one growing observation time per cutoff. Under the error bound $K_n(t) \leq A a_n^{-p} e^{-mt} + B a_n^r$ and probe tilt $|\alpha_n| \geq c_\alpha a_n^s$, it gives

$$M_* = \frac{m(r-2s)}{p+r} \quad \text{at} \quad t_n = \frac{p+r}{m} \log \frac{1}{a_n}. \quad (15.7)$$

Two ways this returns nothing.

Zero margin. If $r \leq 2s$ then $M_* \leq 0$: the theorem is sharp and it yields exactly zero. Here r is the convergence rate of the coarse-graining background and s the rate at which probes must shrink to stay in the quadratic basin of the Wilson action. No calculation in this program, or to this author's knowledge anywhere, certifies $r > 2s$ for four-dimensional Yang–Mills. The inequality $r > 2s$ is therefore a hypothesis with no current route, not a technical condition.

Finite horizon. If the observation time is capped by the box, $t_n \leq c_{\max} \log(1/a_n)$, the optimum becomes

$$M_{\text{opt}} = \min\left(m - \frac{p+2s}{c_{\text{opt}}}, \frac{r-2s}{c_{\text{opt}}}\right), \quad c_{\text{opt}} = \min\left(c_{\max}, \frac{p+r}{m}\right), \quad (15.8)$$

and $c_{\max} \leq (p+2s)/m$ forces $M_{\text{opt}} \leq 0$. Both collapses are registered checks rather than remarks.

15.4 Break 4: a local probe can be blind to the lightest state

Suppose one measures a plaquette correlator and observes

$$\langle \mathcal{O}_{\text{plaq}}, e^{-tH} \mathcal{O}_{\text{plaq}} \rangle \leq C e^{-M_{\text{obs}} t}, \quad M_{\text{obs}} = \log 4 \approx 1.386. \quad (15.9)$$

This does not bound the spectrum.

Proposition 15.4 (Dark sector). *On $\mathbb{C}^3$ let*

$$T = \text{diag}\left(1, \frac{15}{16}, \frac{1}{4}\right), \quad \Omega = e_1, \quad v_{\text{obs}} = e_3.$$

Then $\langle v_{\text{obs}}, T^t v_{\text{obs}} \rangle = 4^{-t} = e^{-t \log 4}$, while the true first excited energy is $-\log(15/16) \approx 0.0645$, smaller by a factor above 21. The probe has zero overlap with the state realizing the gap.

Consequently a decay rate for local plaquettes is a statement about the gap only after the plaquette family is proved *total* in $\Omega^\perp$. This is why totality is a conclusion, not a hypothesis, in the fixed-spacing band theorem (§13); and it is why a continuum argument may not inherit totality from the fixed-spacing statement without proving it again in the limit.

A second, subtler version of the same failure concerns positivity rather than overlap.

Proposition 15.5 (Osterwalder–Schrader null states). *Let $(x, y) \in \{-1, 1\}^2$ be uniform and let time reflection act by $\Theta(x, y) = (y, x)$. The centered observable $F(x, y) = y$ has Euclidean variance $\mathbb{E}[F^2] = 1 > 0$ but physical norm $\|F\|_{\text{OS}}^2 = \langle \Theta F, F \rangle = \mathbb{E}[xy] = 0$.*

So positive Euclidean variance does not certify that a probe exists in the physical Hilbert space at all. An algorithm that selects probes by Euclidean variance can select states that vanish after reconstruction.

A third version concerns the generator rather than the states: with $H_n = \text{diag}(0, n)$ one has $e^{-tH_n} \rightarrow \text{diag}(1, 0)$ for every $t > 0$, so the limiting semigroup is not strongly continuous at $t = 0$ and no self-adjoint generator exists on the second coordinate. Convergence of Gram matrices alone therefore does not produce a limiting Hamiltonian.

15.5 Break 5: the proved regime and the continuum regime are disjoint

This is the structural obstruction, and the one that most changes what a correct argument must look like.

The fixed-spacing band theorem (§13) holds on

$$|u| \leq \frac{u_*}{10\,022\,400\,000\,N}, \quad (15.10)$$

and the spatial-closure estimates hold for $0 \leq \lambda \leq \lambda_c$ with $\lambda_c \in \left(\frac{1}{73}, \frac{1}{72}\right)$. Both are neighbourhoods of zero in the strong-coupling variable.

The continuum limit lives elsewhere. With $g^{-2}(a) = \beta_0 \log(1/a\Lambda)$ and $u = 2/(g^2(a)N)$,

$$u(a) \rightarrow \infty, \quad \lambda(a) \sim g^{-4}(a) \rightarrow \infty \quad (a \rightarrow 0). \quad (15.11)$$

So the interval on which the theorem is proved and the trajectory along which the continuum is taken have empty intersection:

$$[0, \lambda_c] \cap [\lambda(a), \infty) = \emptyset \quad \text{for all sufficiently small } a. \quad (15.12)$$

Scope 15.6. The fixed-spacing results establish a strictly positive gap at fixed lattice spacing in a neighbourhood of strong coupling. They supply no control on the weak-coupling trajectory. An inference from (15.10) to a continuum mass gap is a regime error, and this program has made and retracted a version of it.

What Break 5 demands of a future argument is a *controlled trajectory*: an estimate that survives continuation in the coupling, not a better constant at $u \approx 0$. Improving the constant 10 022 400 000 in (15.10) by any finite factor does not move the boundary of the difficulty. This is why Problem 6 in §17, and not a sharper fixed-spacing estimate, is the load-bearing open problem.

15.6 Break 6: diffusion time is not Euclidean time

A recurring route derives a spectral gap from curvature: positive Bakry–Émery–Ricci curvature on the configuration manifold gives a logarithmic Sobolev inequality, which gives a spectral gap $\lambda_{\text{diff}} > 0$ for the Langevin generator. The step that fails is the identification of λ_{diff} with a physical mass.

Langevin time is a fictitious parabolic parameter; Euclidean time is a spacetime coordinate of the reconstructed theory. In free models the two are related by $\Delta_{\text{phys}} = \sqrt{\lambda_{\text{diff}}}$, but that relation is a consequence of Gaussian structure and not a general isometry. For non-abelian gauge theory no such correspondence is available here.

The companion obstruction concerns the coarse-graining itself. A scale map built from a gauge-covariant Markov kernel that preserves both reflection positivity and locality is not available: any such kernel either forces the holonomies to commute — reducing the theory to an abelian one — or destroys reflection positivity, so that the reconstructed inner product is indefinite. This is why the scale route actually used in §14 is a *deterministic* pushforward rather than a Markov kernel, and why the two-spin counterexample $\mathbb{E}[s_- s_+] = -1$ is recorded as a proposition rather than as an anecdote.

Scope 15.7. Break 6 does not refute the curvature computations themselves. The Haar Ricci constant and the Wilson Hessian bound are correct statements about the configuration manifold. What is refuted is their use as a route to Δ_{phys} , and the curvature results are consequently recorded in this program as inputs to coercivity estimates rather than as a mass-gap mechanism.

15.7 What survives

1. The exact strong-coupling spectral theory of Part I, in its entirety. None of the six obstructions touches a fixed-order statement.
2. The fixed-spacing infinite-volume Wilson band and its literal source frame (§13), on the interval (15.10), with the scope stated there.
3. The moving-time spectral exclusion as an abstract interface: the implication from its hypotheses to a limiting gap is proved and sharp. Break 3 is a statement about whether its hypotheses hold for Yang–Mills, not about the implication.

What does not survive is any inference from the above to a continuum Yang–Mills mass gap. The program states that plainly, and §17 is the list of what would have to change.

16 The refutation register

A refuted claim is evidence that it was tried. This section reproduces the register of every claim this program made, or inherited from its own sources, and then lost — together with the exact thing that killed it.

The register carries 22 entries in three states. 4 are *falsified*: a counterexample or an exact computation contradicted them. 3 are *superseded*: replaced by a sharper statement. 15 are *resolved*: the apparent conflict dissolved, in most cases because the two sides turned out to be differently anchored quantities rather than competing values for the same quantity.

Three features of how this register is maintained matter to a reader deciding how much to trust the rest of the document.

First, a dispute closes by derivation or it does not close. Both values stay recorded side by side, and no code path is permitted to pick one, average them, or prefer whichever is an exact rational because exactness reads as authority. The fourth-order off-axis coefficient closed that way: every cluster of the rotation record reproduces the historical pipeline’s own number, and the single cluster that does not is explained ordering by ordering (§7). Its two recorded values remain in the registry with their verdicts attached.

Second, a resolved entry is not a win. The largest class here is resolutions in which a contradiction turned out to be a naming collision — two differently anchored coordinates both called m_4 , for instance. Those cost real time and produced no mathematics, and the register’s response was to forbid the ambiguous name rather than to record a victory.

Third, this program has retracted a mechanism of its own. The *tier collapse* — the observation that two of the four fourth-order shape coefficients vanish at every solved rank — was proposed here as a kinematic degree bound with a falsifiable prediction, and the mechanism was withdrawn a fortnight later when one projection turned out to be uncounted (§7). The withdrawal is a numbered decision record rather than a silent edit, because the next attempt at that mechanism needs to know which step failed. The reformulation survives; the mechanism and its prediction do not.

That is a different object from the degree bound of §9, which was never retracted — it was made unnecessary. The two are easy to confuse and this document keeps them apart deliberately.

Falsified (4)

Falsified — killed by a counterexample or an exact computation. These are the four claims this program asserted, or inherited, and then lost.

- **C6 F-2 mobility rule.** The promotion $r_physical = w_min - 2$ is falsified by the pentagonal isotropic cap hop at $r=4$ with $w_min-2=5$. The scoped bound $r \geq w_min - 2$ survives, with survival gates.
- **C7 Stranded-flux zero backend.** Claimed Haar zero for all 20 pentagonal histories refuted by exact Weingarten/network contractions (8/8).
- **C18 Center-only circuits are dynamically dark.** Refuted by the exact fifth-order determinant correction $\delta c_5, \det \neq 0$
- **C19 Global fixed-window firewall.** Disproved by the Bernoulli no-go; replaced by the rooted conditional calculus

Superseded (3)

Superseded — replaced by a sharper statement. The original is not wrong so much as no longer the best available.

- **C5 String-tension sign at $O(u^3)$.** $-61/408$ (the earlier $+61/408$ was an odd-order convention error)
- **C11 Compact-gap constants.** Corrections: $c_0 = -3(N^2-3)/(16N)$; radial moment difference $3N^2+1$; dilation $\epsilon^4 = N/(2\beta)$;
- **C12 Curvature misreads.** Two-stage. The kappa values are RADIAL DIRECTIONAL curvatures, not Hessian entries at Gamma; a genuine Hessian needs $\beta = 2\alpha$, violated by the

historical kernel.

Resolved (15)

Resolved — the collision was dissolved. In most of these the two sides turned out to be differently anchored quantities rather than competing values.

- **C1 Fourth-order rest scalar — dissolved, an anchoring distinction.** NOT a contradiction. $q_{\text{band}}^{(4)}$ and $m_{\text{Gamma}}^{(4)}$ are not two competing estimates of the same scalar coordinate: the first is a band-kernel anchor, the second a vacuum-subtracted physical Gamma-point coefficient. They are differently anchored coordinates related by a translation-local scalar shift, and such a shift cannot change the centered operator, its eigenvectors, the SOS factorization, the mobility coefficients, or the bandwidth.
- **C2 Fourth-order off-axis coefficient C_{shp} — resolved by derivation 2026-09-04.** $C_{\text{shp}} = C_{\text{SHP_HISTORICAL}} + 25/1024 = -13035490122347/550663802582400 = -0.0236723$, the cluster-assembled coefficient read in the kernel's own basis (ADR 0024). Both recorded kernels are wrong on the rotation amplitude: the historical by the sixteen orderings of the adjacent-face cube completion its own Stage-3I ledger lacks (8 of 24, each with the right weight; the shortfall is exactly $25/512$ on ρ), the cold dump on u , π and ρ alike.
- **C3 Status of the fourth-order "theorem".** Newest status controls. The §5.2 values are sealed; everything else is $C1/C2$.
- **C4 Coupling conventions.** The archived $Y = 2*\beta/3 = 4u$ line is a definition/label error, not a rescaling rule. Printed coefficients were already in $u = \beta_{\text{lat}}/6$. Never apply 4^{**n} factors to them.
- **C8 Pentagonal degenerate space.** $E_{\text{cap}} = 10/3 \neq E_{\text{side}} = 8/3$; choose the physical degenerate eigenspace first
- **C9 SU(4) exceptional correction.** Only the flat-branch eigenvalue identity holds, not scalar-ity of the matrix
- **C10 Determinant sectors shift only q_{N} .** True for $N \geq 4$, false at SU(3): $\Delta_{\beta_3} = -25/64$ and $\Delta_{q_3} = -16863189551/76406976000$.
- **C13 C-even second-order hopping.** $-481/612$ superseded by $-11/306$ (the vacuum-mediated route was omitted)
- **C14 Betti "+2".** Reinterpreted as $b_2 - b_3$; the open box has no extra states
- **C15 Tetrahedral local coefficient.** Re-derived 2026-08-21 (tetrahedral Haar-resolvent suite): the asserted $-8/(N(N^2-1))$ is exact at symbolic N . The derivation reconstructs the primitive law's three stated ingredients — $1/N$ per merge from the fundamental-pair Haar moment (independent of shared-path length, derived not assumed), the certified electric convention $E(L) = L*C_F/2$, and exhaustive temporal orderings — and reproduces every certified or printed instance first (sealed-core cube $-5/48$ with the $(-8,-8,-4)$ temporal classes, prism 64 with its cap-sector 24 dissolved as a sector ...
- **C16 SU(5)/SU(6) fourth-order completeness.** The payloads were shipped after all, and are now machine-checked (restored-payloads suite, 2026-08-21): the SU(6) determinant certificate re-derives $\Delta_{q_6} = 6/343$ exactly, momentum-independent with A , B , and the bandwidth

unchanged, and the SU(5) stage-1 summary pins the 895,524-pair scan with zero determinant sectors. Two upstream word manifests remain absent (their SHAs are recorded in the certificates), so the evidence level is output-certified, not cold-reproduced.

- **C17 Fixed-rank holdout overclaim.** Corrected: q_N matches stored values for $N=7..18$, but full kernels only at $N=7,8$; the A/B sample ledger remains open.
- **C20 RUN15 internal display artifact.** Cosmetic; the gate value is the intended exact one
- **C21 Monte Carlo gate integrity.** `next14.json`'s "23/23 gates" is downgraded: one gate is a literal truth value, the JSON is not source-hash bound (non-RFC NaN tokens), and no raw ensemble exists. Numbers stand as structured numerical evidence; the certificate claim does not.
- **C22 Gate-85 equality.** Not a numerical contradiction but a status correction: the final diagonal shift was chosen to produce exactly that equality, so gate 85 certifies internal bookkeeping, not independent agreement.

16.1 Claims withdrawn by this program

Distinct from the register above, the following are claims this repository itself advanced and then withdrew. They are listed separately because their failure mode is different: nobody contradicted them, the argument was found not to close.

- **The tier collapse as a kinematic degree bound.** Proposed with a falsifiable prediction — that the next shape tier would first appear at sixth order — and retracted eleven days later. The count bounded the vertices of H_4 ; the carrier projection $d^\dagger(\cdot)d$ contributes one further power of the Bloch variable that the argument did not carry. Recorded as `docs/decisions/0005-retracting-the-degree-bound.md`. The reformulation stands, the mechanism does not, and the collapse is now established by the range-one argument of §7 instead.
- **An unrestricted exponential-source radius.** Inferred from independent bounded sources; the inference does not hold, and the reformulation is Problem 7.
- **Closure of the source-energy transport budget.** Claimed complete, rejected on review: a vacuum-only skew generator does not control a source that the generator transports rather than annihilates. The corrected partial results are retained and the obligation is Problem 2.

The falsified entries above — the mobility rule, the stranded-flux backend, the dark centre-only circuits and the global fixed-window firewall — are a different category and are not repeated here: each was killed by a counterexample or an exact computation rather than by an argument failing to close.

Each of these was, at some point, written down in this program's own documents as established. That is the reason the machine-verification tier of §2 is computed rather than assigned, and the reason the present document quotes its counts from the ledgers by macro rather than by hand.

17 The obligation register

This is the operative section of the document. It states, as numbered problems, what would have to be proved; and it then reproduces the repository's full route register, including every route that is dead and the reason it died.

The numbered problems come first because they are curated: they are the routes this program judges to be live and decisive. The register that follows is not curated — it is the whole record, 111 routes across 20 open and 5 discharged gaps, of which 64 are complete, 20 live, 12 untried and 15 dead.

17.1 The problems

Ordering expresses research judgement. It is *not* a claim that an earlier problem is a prerequisite for a later one; the ledger records explicit inputs separately, and each problem below states its own.

Problem 1 (Conditional-score domination, uniformly in the coupling). For the synchronized transport of the four-face $SU(2)$ conditional model, establish

$$K_g(w) \leq C_0 + C_1 W_g(w) \quad \nu_g\text{-almost everywhere,} \quad (17.1)$$

with C_0, C_1 independent of g as $g \rightarrow 0$, retaining the chart-complement, rare-fiber and antipodal contributions.

Available. The actual source moment; the interval-obstruction repair; conditional-score continuity; synchronized conditional dilation; the antipodal Hessian spectrum, gauge-normal coercivity, angular reduction and gauge-score invariance — all proved. Exponentially small complement suppression outside a neighbourhood of the complete minimizing set, and a uniform Agmon action gap on the complement, are proved.

Missing. Polynomial relative-amplitude control; the tube moments; the actual normal/angular comparison.

Consequence. With the established source-potential moment, (17.1) makes the uniform weighted score bound applicable. It does *not* supply source-energy transport or interacting-grid uniformity; those are Problems 2 and 3.

Next computation. For the synchronized field, obtain the actual conditional tube moments and the differentiated amplitude bound; then use the established antipodal magnetic normal and angular geometry to prove the true-ground score estimate, controlling the outside deviation against the same reference. Compare the resulting complete variance with $1 + g^{-2}\mathbb{E}(V \mid w)$, or prove a sufficient source-weighted pairing directly.

Dead here. Inferring uniform domination from endpoint requirements on arbitrary cutoffs; differentiating a global fixed-rank physical Coulomb source projection across flat holonomies; bounding the global true-ground Fisher matrix by $C_0 + C_1 \sqrt{u} v(U)$.

Problem 2 (Source–vacuum transport energy jets). Bound the complete compact source and vacuum transport on a common energy domain against a physical clock. A vacuum-only skew generator is insufficient: the source generator transports the ground state, it does not annihilate it.

Available. The ground jets; the ground-frame generator; the score–Poisson identity; the conditional derivative; H_4 derived from H_0 ; the projection-jet recurrence — all proved.

Next computation. Decompose the complete generator into vacuum and conditional-source parts; retain commutator and energy-domain terms through two derivatives; test the proposed powers of g on the full energy graph rather than only on ground vectors.

Recorded failure. An earlier closure of this problem was claimed and rejected on exactly the ground that the vacuum-only generator does not see the source. The rejection is preserved; see §16.

Problem 3 (Interacting-grid comparison with uniform constants). On the actual coupled Wilson grid, with the true vacuum, the full fast space, source metrics, interfaces and a common physical clock, prove a comparison whose constants are uniform in the number of interacting blocks.

Available. The literal coarse form; covariant block sources; the form Schur scale comparison — all proved.

Next computation. Two genuinely coupled adjacent blocks. Compute the full conditional and source form including the interface and the high retained directions, and isolate a local estimate whose constants sum without a block-count factor.

Why uniformity is the whole problem. Fixed-block estimates are available in quantity. A constant that degrades with block count telescopes to nothing, which is the same arithmetic failure as Break 15.1 in a different variable.

Problem 4 (The signed conditional defect in the actual vacuum). On compatible finite periodic SU(2) blocks with the actual positive Wilson vacuum, bound the full signed conditional reference defect, retaining all localization and domain terms.

Available. The first and third residual estimates, proved.

Next computation. On the smallest coupled block, derive outside-pressure and projector derivatives alongside the magnetic and metric terms; check the Gaussian cancellation exactly; then bound the remaining full expression with its cutoff commutators.

Problem 5 (Summable normalized RG and source increments). For the actual normalized RG map and the renormalized observables after marginal matching, with all measures and source spaces identified, prove a cross-scale increment bound with scale-uniform remainder constants that is *summable*.

Available. Contractive drift on a common space, proved.

This is Break 15.1. The required improvement is from $O(g_k^2)$ to $O(g_k^4)$, and (15.4) shows the normalization supplies no small factor, so the improvement must come from an exact cancellation of the leading partition-function covariance.

Next computation. Work out one genuine normalized blocking step with its observable transport; isolate the marginal contribution; and calculate the surviving common-norm increment before claiming a power-improved bound.

Problem 6 (A controlled trajectory out of the proved interval). Connect the fixed-spacing construction of §13, proved on $|u| \leq u_*/(10\,022\,400\,000\,N)$, to the weak-coupling trajectory along which the continuum limit is taken.

This is Break 15.5, and it is load-bearing. The two regimes are disjoint. Borel–Padé consistency of the strong-coupling series is not such a trajectory: agreement of a series with a resummation is not continuation of an estimate.

What a solution looks like. An estimate whose constants survive continuation in the coupling, or a second construction valid at large u together with a proof that the two constructions describe the same theory on an overlapping window. The second route has a named prerequisite already discharged in part — the dimension-five operator irrelevance lemma — and the coupling window between the two constructions is the object to close.

Not a solution. Any improvement of the finite constant 10 022 400 000.

Problem 7 (A valid source-family formulation). Identify the exact source operator, normalization and norm consumed by the selected spectral comparison, and derive a local, connected or normalized replacement *together with* a proof that the replacement preserves that comparison and a nonzero physical source weight.

Refuted approach. Inferring an unrestricted exponential-source radius from independent bounded sources. Linear susceptibility bounds do not bound the centered exponential source.

Available. The raw α -dependent partition bound for bounded sources, established. Scalar Kotecky–Preiss inequalities and raw moment estimates remain useful ingredients but construct neither the tilted activities nor a gap.

Next computation. Compute the candidate source norm and its spectral normalization in the explicit independent product law; compare with $(M(2\alpha)^n - M(\alpha)^{2n})/n$; then state the precise interacting inequality that remains.

Problem 8 (One complete sixth-order cluster). At fixed $SU(3)$, on one smallest admissible connected plaquette support, evaluate the eighteen electric folds with exact Haar and Gram quotients, physical projections, electric denominators and rooted subtraction.

Available. The Hodge–Feshbach splitting; the fourth-order support theorem; the Hermitian folded support; local sixth-order non-cancellation; the one-face sixth order — all proved (§10).

Consequence. The extra rational band-shape component is already forced under the local-shell and third-order hypotheses. Computing the direct term determines its independent polynomial shapes and coefficients, and decides whether the local structure survives. It changes neither the fourth-order support theorem nor the global sixth-order census.

Next computation. Evaluate the eighteen folds on a connected multi-plane support with all Haar weights and rooted subtractions. Reduce the direct term separately; then combine it with the known mixing numerator at common denominator, retaining both the direct polynomial shape coefficients and the invariant remainder.

Unlike the others, this one is finite and priced. It is the cheapest decisive computation on this list.

17.2 The full route register

What follows is generated from `ledger/gaps.yaml`. Dead routes are included deliberately and at length. A route marked *cannot decide* is structurally unable to resolve the named claim, which is a stronger statement than having failed to.

G1 — Ship or restore machine-readable payloads that are only record-backed

Tier 0 (days); partial.

$SU(3)$ 189-record kernel with reference SHA; Q_32 and P_402 ledgers; $SU(5)/SU(6)$ exceptional certificates;

Ledger status. largely discharged 2026-08-21 — three of five families were shipped all along and are now machine-checked (restored-payloads suite): the 189-record kernel with both reference SHAs, the Q_32/P_402 Newton ledgers, and the $SU(5)/SU(6)$ certificates. Still absent: the pentagonal dual-cold bundle (manifest row A60 records the SHA any restoration must hash to) and the tetrahedral certificate, for which no reference SHA exists anywhere — that one was G5’s to close, not this gap’s, and G5 closed it by re-derivation on 2026-08-21 (the artifact itself remains absent). A...

G25 — The charge-even sector is registered and unchecked

Tier 1 (weeks); partial.

What remains. The suite reaches orders 2 and 3, which is where the corpus's C-even ledger stops. Fourth order is untouched on this side: the pre-registered $O(u^4)$ criterion in the corpus is a statement about C-odd flatness, and no C-even fourth-order coefficient is registered here at all, so the sector's fourth order is not T3 – it is absent.

Ledger status. opened and largely discharged 2026-08-28. The C-even sector was registered and inert: LEAK_2, LEAK_2_EVEN and LEAK_3_EVEN were read by no check body, band_
assembly() was dead code, and the only C-even geometry the repository checked was two hand-written matrices at Gamma and R. The 'charge-even band, exactly' suite (15 checks, all T1) closes the second- and third-order half. Both entries of constants.DECLARED_COINCIDENCES are now checked rather than only declared: -11/306 is ell_N...

Depends on. G3

Live routes.

- **fourth-order domino, the U4 falsifier.** PROGRESS 2026-09-12, part (2) DONE: conn_vac is derived and no longer T3. The linked two-face vacuum weight is now computed exactly by two routes that agree (swap-odd suite, new T1 check). At SU(3) the third engine's loop calculus gives the two-face vacuum sector 0, 0, -3/2, 9/16, -54321/837760 against twice the one face, so the linked cumulant is 0, 0, 0, 0, -327/83776: the corpus's omega4, identical for the coplanar and the perpendicular pair, with the linked $O(u^2)$ and $O(u^3)$ weights exactly zero as the v10a.7 gates reported in floats.

G4 — Pentagonal $O(u^4)$ closure in-corpus

Tier 1 (weeks); partial.

Fierz closure of the 20-history cut spaces, Gram quotient, physical resolvents, summed compression.

Ledger status. partial — the raw history Gram/Haar frontier is closed (8/8 gates, cold rerun in settlement/); the Fierz closure, physical quotient, and summed resolvent compression remain (see progress.remaining).

G6 — B_N fixed-rank holdout ledger for N = 7..18

Tier 1 (weeks); open.

G7 — Native torelon reruns of sigma_5 and sigma_6

Tier 1 (weeks); open.

Replace the KPS historical targets.

G8 — Cold one-shot regeneration of the all-rank fourth-order bundle

Tier 1 (weeks); open.

Promotes output-certified to cold-certified.

G10 — Full fifth-order pentagonal coefficient

Tier 2 (months); open.

Support-resolved folds for the 572 return histories before rooted subtraction.

G11 — Interval rigor and near-Gamma touching gates

Tier 2 (months); partial.

Outward-rounded pi. Without this, a fixed-momentum coefficient theorem must NOT be promoted to a uniformly isolated near-Gamma band theorem: for $|k| \lesssim u$ the $O(u^2)$ splitting $\sim u^2|k|^2$ meets the $O(u^4)$ scale.

Ledger status. partially discharged

G12 — Hyperhoneycomb $O(u)$ candidate

Tier 2 (months); open.

Tests the framework beyond cubic cells.

G9 — m_6: global sixth-order geometry census

Tier 2 (months); open.

Contraction with external-memory sharding; folded weights and carrier census pre-cleared.

Depends on. G3

Live routes.

- **Assemble one complete sixth-order cluster and test its extra carrier-shape component.** The September 11 continuation derives all 18 universal Hermitian electric folds, evaluates the complete one-face $SU(3)$ rooted coefficient $-2055143/35123200$, and reduces all 25 H4 mixing pairs. Under the recorded third-order factorization and local-shell hypotheses, an additional rational shape cannot cancel against any local direct H6: $q^3 \lambda_6 \bmod q^2$ is $4 \kappa a e^2 - 3 \kappa q e^3$ for symmetric polynomial direct F. The remaining task is the direct connected multi-plane H6 evaluation, including its Haar/Gram quotient, electric...

G13 — Classification theorem for shortest physical temporal histories

Tier 3 (unbounded); open.

Determine r_{escape} from the energy-decorated incidence/Feshbach graph without exponential enumeration. Would subsume tetra/prism/cube/pentagonal/ hyperhoneycomb as one theorem, and includes proving partition-independence of the linked/folded assembly.

G14 — Fourth-order support mechanism and its higher-order extension

Tier 3 (unbounded); partial.

Derive the actual supported operator words from complete physical histories, with the Haar quotient, electric resolvents and folded assembly retained. Use the established fourth-order support theorem as the baseline for a bounded sixth-order calculation. The Hodge splitting alone does not assert an all-length word classification or a dynamical sixth-order coefficient.

Ledger status. Fourth-order tier collapse is established by `RESULT:TIER_COLLAPSE_ACTUAL_H4_SUPPORT`: the recorded kernel has support $\{I, U, S, S^2, R\}$, with cleared carrier

span $\{q, q^2, e_2\}$. The broader R-degree inference is falsified by $\sigma(\text{UR}) = -2q e_2$. The open successors concern history-to-word support, higher multiplicities and the separate proper-return identification in other geometries; the September channel computations are established inputs.

Live routes.

- **prove channel-wise universality from the Haar contraction – that a channel’s value depends on which links are doubled and in which irreps, not on where on the cluster they sit – for u, the fan and the single-contact dressing.** Opened 2026-09-04 (ADR 0029). With every channel an identity in $Q(N)$, the mechanism of the tier collapse is a statement about the channels alone: B is unpopulated by the carrier projection, $D = 0$ is $u^2 = 2u$, and $u^2 = 2u$ is the L chain equalling the straight chain channel by channel. What is left is a proof of that equality from the contraction rather than a computation of it.

Untried routes.

- **derive the actual fourth-order Hodge support from the histories, including the absence of UR, RU and repeated R insertions.** Reviewed 2026-09-10. The narrower actual support $\{I, U, S, S^2, R\}$ already proves the recorded kernel’s $B=D=0$. Explain this support from the complete electric/Haar and folded histories, including the absence of UR, RU and repeated R insertions;
- **test the Hodge-summand mechanism in a second geometry by computing the tetrahedral Q-projected proper-return contributions, the first clause of U7’s falsifier.**
- **audit the premise that $D - (1/2)\{H_2, V_2\}$ is the Hermitian fourth-order effective Hamiltonian on a cluster with $PVP = 0$ (the pair element at $N \geq 4$), against a direct Bloch or Schrieffer-Wolff expansion on the two-face cluster at $N = 5$.** Taken as given by every lens of the 2026-09-04 adversarial verification of ADR 0027; unrefuted, unaudited.
- **audit the rank-generic resolvent at $N \geq 5$ – link_spectrum roots, the dropping of the E0 component, vacuum intermediates – against the SU(3) Casimir table route continued to $N = 4, 5$ by hand.** Taken as given by every lens of the 2026-09-04 adversarial verification of ADR 0027; unrefuted, unaudited. Partly covered later the same day (ADR 0029): over $Q(N)$ every eigencomponent of every resolvent is verified to be an exact eigenvector of every single-link H_0 with the SU(N) Casimir energies (21,556 components), so the spectra are no longer read off characteristic-polynomial roots at all;
- **the completeness of the eleven-cumulant decomposition of β_N at general N rests on the charge-counting argument alone (a connected four-insertion history from P to Q is $P \bar{Q} X \bar{X}$, the pair, or the cube); enumerate the fourth-order support at $N = 5$ from the engine and compare with the 189 Hodge-form keys.**
- **the $N = 4$ row of the beta record rests on the determinant trick (pseudoinverse Weingarten at $n > N$ plus the epsilon insertion) in the three-face cumulants; audit that path at $N = 4$ against an independent epsilon-network reduction, and make the (4,0)-family pair element feasible the same way.** Sharpened 2026-09-04 (ADR 0029): the closed forms derived over $Q(N)$ are regular at $N = 4$ and agree with the $N = 4$ row of every per-rank record, so whatever the determinant trick contributes there is what the generic rational function gives. The symbolic run also names the $N = 4$ degeneracy: the state P^3 with every link of the plaquette in $\Lambda^3 F$, which is $\bar{F}$ at $N = 4$, has energy E0 exactly there (the one resolvent denominator with an integer root ≥ 4). The audit of the path itself is still open.

Dead routes, and why they died.

- **derive why the fourth-order dynamics is R-linear – why no four-insertion history inserts the cross-plane operator twice – which is what the tier collapse now reduces to.** The proposed reduction is false: UR contains one R but its symbol $-2q e_2$ populates the B tier. The actual fourth-order collapse instead follows from support $\{I, U, S, S^2, R\}$. The step name is retained to preserve its existing route id;

Completed routes (2). read the mechanism off $u(N)$ – decompose the two-hop cumulant by intermediate irrep and match its denominator factors to resolvents; the same channel decomposition for the corner and fan dressings of the rotation orbit.

G15 — Symbolic-N Gram transcript

Tier 3 (unbounded); open.

Promotes c_1, c_2 to unrestricted fixed-N theorems; one more order of the equivariant harmonic-well lemma.

G16 — Rank-uniform control in $\tau = \beta/N^3$

Tier 3 (unbounded); open.

Neither fixed-regime series supplies an overlap theorem. PROGRESS 2026-09-11 (ADR 0046, runs/planar_band_2026-09-11, suite "the planar limit of the fourth-order band (G16)", five T1 checks): the strong-coupling side is now exact through fourth order. Every fourth-order cumulant of the β_N assembly is $O(N^{-7})$ in both sectors while its resolvent channels are $O(N^{-3})$, the N^{-3} and N^{-5} channel totals vanishing identically in all sixteen cluster/sector pairs;

G17 — Source-family stability and a valid source-radius formulation

Tier 3 (unbounded); open; load-bearing.

Identify the exact source operator, normalization and norm consumed by the selected spectral comparison. Derive a local, connected or normalized replacement only with a proof that it preserves that comparison and nonzero physical source weight. Linear susceptibility bounds do not bound the centered exponential source.

Ledger status. The raw α -dependent partition bound is established by DERIV:TRACK_A_SUPPORTED:G17_P with $K_\alpha = \exp(|\alpha|/2)$. The unrestricted centered exponential-source radius is not a consequence of clustering: DERIV:TRACK_A_SUPPORTED:G17_I gives divergence even for independent bounded centered sources. A valid source-family formulation and its actual interacting estimate remain open. The fixed-spacing infinite-volume Wilson band and literal-source...

Untried routes.

- **Reformulate the source family while preserving its downstream spectral comparison.** Choose a precisely stated local, connected or normalized source formulation and prove the required relationship to the actual spectral comparison, including nonzero physical weight. Test it in the independent product model before seeking an interacting estimate. This is a candidate route with a source-identification obligation, not a claim that the original unrestricted exponential radius can be recovered from covariance summability.

Dead routes, and why they died.

- **Infer an unrestricted exponential-source radius from independent bounded sources.** DERIV:TRACK_A_SUPPORTED:G17_I constructs an independent bounded centered source whose exponential variance divided by footprint cardinality tends to infinity. This refutes the stated inference, not every localized or normalized source formulation.

Completed routes (1). Establish the raw alpha-dependent partition bound for bounded sources.

G19 — Continuum limit

Tier 3 (unbounded); open; load-bearing.

The actual infinite-volume Wilson construction and complete physical source band are now established at fixed spatial spacing on a common small- u interval. A controlled scale trajectory connecting it to large u , renormalized continuum correlations and a positive physical mass gap remain open. Borel-Pade consistency is not that trajectory.

Ledger status. Reconciled 2026-09-09 after reviewing 43 distinct G19 document versions. Non-linear finite-cell shells and scores, complete Gaussian fast/source bounds, fixed-spacing matching, and conditional continuum/OS transport are retained. The newly cited Balaban continuum proof stops at (7.6): its $O(1/k)$ expectation increments do not imply the Cauchy conclusion in Theorem 7.1, which invokes a nonexistent $O(1/k^2)$ estimate. Its (6.10) weight $\kappa_k = (3/4)^k \kappa_0$ also fails the positive entropy margin required in (8.14).

Live routes.

- **Prove the actual conditional-score domination M10.** The source-potential moment M8-M9 is established. The open target is the actual small- g conditional score $K_g \leq C_0 + C_1 g^{-2} E(V|w)$, including rare fibers, with uniform nonnegative C_0, C_1 . This sufficient route implies the weighted score/Hardy estimate M11-M15;
- **Control complete source-vacuum transport energy jets R10.** Raw H_g derivatives and vacuum-only ground jets are established. Prove the complete source/vacuum generator bounds R10, $\|A^\wedge(r)(g)\|_{(q_g \text{ to } q_g)} \leq d_r g^{-(r-1)}$ for $r=0,1,2$, or use the source's direct mixed-form R11 alternative. These are alternative sufficient entrances to the complete residual budget, not cumulative prerequisites.
- **Control the interacting quantum comparison in redundant compact coarse variables.** Use the actual smooth matrix observation with its exact coarse gauge action and boundary source lift. Prove interacting true-vacuum marginal and full complementary energy estimates, including nonflat localization, noncommuting compact slow dynamics, the complete direct/exchange/metric/source correction and high retained tails. The ambient redundant projector is not a physical quantum Schur projection.
- **Bound the nonlinear excess above the exact quadratic memory on complete physical source windows.** Combine the established dynamic cubic energy and exact full fast Green identity with quartic magnetic, electric metric, Haar, moving-source, baseline cross and fast-form variation terms. Prove bounds for the actual spatial coefficients and the complete interacting remainder with retained harmonic localization, actual compact dynamics or explicit cancellations. Full source algebra, high retained directions and a summable physical-clock trajectory remain open.

- **Transport the endpoint and localized true-ground estimates through actual interacting scale blocks.** The additive complete endpoint and selected static-score comparisons are established inputs with different scopes. Prove uniform actual interacting fast/source or memory estimates and compare the resulting endpoint law with a consistent coarse quantum theory in one physical clock. For a static Schur route, retain its high-source coupling rather than inferring a full gap from the chosen chart.
- **Connect the actual Wilson fixed-scale construction to a controlled continuum trajectory.** Actual fixed-spacing infrared transfer and sources remain the controlled endpoint. The ultraviolet side now has a general fixed-cell physical first-cluster theorem and all-size planar and three-dimensional harmonic interface bounds, with planar low-mode/source frames and retained periodic harmonic directions. Gaussian Schur elimination retains memory and induced mass, giving a quantitative sorted-frequency approximation without small interface coupling.
- **Control the true interacting Wilson score and forms on the relevant energy space.** The exact $SU(2)$ central true-ground Fisher matrix grows at least linearly in u , so the proposed global $C0+C1 \sqrt{u}v(U)$ estimate is no longer a live target. Prove an actual full- Q energy weight and a resolvent-weighted or coarse-gradient-energy score estimate on the relevant source window. A good-region small ratio plus controlled bad-region gradient-energy leakage yields a conditional small window Schur loss.
- **prove actual interacting W6 and coarse/source matching through spatial refinement.** The exact selected inverse-pairing and compatible source transport theorems remain available. R4a supplies a flat conditioned Gaussian reference budget; R12a identifies its zero complete pressure defect. *Cannot decide: G19.*
- **supply the dimension-five irrelevance lemma, then close the coupling window between the two constructions.** The two rigorous constructions in this repository approach the continuum from opposite ends of the coupling axis, and the target is now the interval between them. From the strong-coupling side SC17 and BA20-BA25 give a volume-uniform physical gap for the actual vacuum, and the thermodynamic continuation gives an infinite-volume vacuum with a closed generator and spectrum in $\{0\} \cup [4\epsilon/3, \infty)$, for k/ϵ at most $\lambda_{c_}$, the root of $6561 l^4 + 1458 l^3 - 81 l^2 - 72 l + 1$ in $(1/73, 1/72)$. From the...
- **Extend the explicit square current to the complete finite-coupling residual.** Prove the complete transported finite- g current remainder against the actual interacting energy, with reference source synthesis and the required uniform parameter bounds. The no-diagonal criterion then supplies the selected-inverse estimate. Compatible source domains and metric transport remain explicit, and coupled-volume extension is a further step.
- **Transfer a physical continuum gap from one growing-time estimate per cutoff and approximate sources.** The original spectral-measure theorem and sharp rate budget are proved. The compatible-kernel successor now constructs the limiting physical semigroup and total centered defining space from finite Gram limits and fixed-time compatibility; positive-time divisibility supplies zero-time continuity, so a separate vague-limit or frame estimate is then unnecessary.

Untried routes.

- **Establish the actual interacting-grid fast and source comparison with uniform constants.** Construct the true interacting-vacuum fast floor and complete local centered/source comparison uniformly in block count, retaining interfaces, Gauss constraints, nonreducing source

directions and the normalized coarse operator. A fixed-square bound, additive tensorization or harmonic floor alone does not prove this target. The chosen physical scale trajectory and summable errors must then satisfy the closed-form iteration theorem’s actual hypotheses.

- **Bound the complete signed SC17 conditional defect in the actual vacuum.** Instantiate the actual uniform smallness input for the Hessian defect $D_B = \text{Hess}(V_B + (H_out - \Omega)/\Omega) - 2\epsilon A^2$, with pressure, metric, projector, moving-source and cutoff/product-domain terms assigned once. Preserve signed cancellations before bounding the complete expression. The conditional Riccati and block-geometry consequences are available, and the thermodynamic/physical-time constructions already stand at fixed spacing;
- **Prove normalized RG and source increments with a summable common-space remainder.** `RESULT:COMMON_SPACE_CONTRACTIVE_DRIFT` supplies the proved abstract convergence criterion; its actual Wilson map and source hypotheses remain open. Construct the actual RG transformation and matched normalized observable maps on common spaces, subtract the running marginal terms, and bound the remaining expectation/source increments by a summable sequence.

Dead routes, and why they died.

- **Infer actual uniform M10 from the endpoint requirements on arbitrary Q8 cutoffs.** A permitted smooth radial cutoff freezes a fixed small regular Q while moving its actual conditional center. Full conditional concentration and a finite-flow speed contradiction rule out M10 for that choice. This is an existence counterexample within Q8, not a proof against every admissible transport or the synchronized successor.
- **Differentiate a global fixed-rank physical Coulomb source projection across flat holonomies.** An explicit SU2 stabilizer change gives a norm-one projection jump. The directions are gauge off identity and physical harmonic at identity, so this does not refute the uniform fast bound. Keep the smooth redundant actual observation and impose Gauss exactly instead of differentiating the changing Coulomb quotient. *Cannot decide:* G19, G23.
- **Bound the global true-ground Fisher matrix by $C0 + C1 \sqrt{u} v(U)$.** The exact actual SU(2) bouquet gives weighted Fisher at least $(64/3)u - O(\sqrt{u})$ at $U = -I$, where $v(U) = 4$. Smoothness makes the violation persist on open neighborhoods, so uniform almost-everywhere variants fail. The generic intrinsic-score and full-gap theorem remains valid under its explicit hypotheses, and other weighted or energy-restricted score estimates remain available. *Cannot decide:* G19, G23.
- **Infer a scale contraction from the forward averaging derivative and a uniform raw fiber diffusion gap.** The preserved EX-014 Section 1 STEP 6 reverses the pullback derivative: a linear arithmetic-average test requires reverse gradient constant m , not $1/m$. Correct quotient metrics remove the artificial gain. The actual Wilson horizontal score also has an order- g geometric term in a physical two-holonomy example; *Cannot decide:* G19, G23.
- **Griffiths-type monotonicity of the electric-flux sector free energies in beta.** proposed 2026-09-03 (Alex): a correlation inequality that holds at every beta and makes the electric-flux free energies $F_e(\beta)$ monotone in beta, sector by sector – the sectors are Z_N -valued, so the non-abelian obstruction to Griffiths’ inequalities for Wilson loops need not apply – would carry the gap from strong to weak coupling the way Griffiths carries Ising order across phases. Closed the same day by a sign, before the inequality itself was attempted: the Ising argument needs the

monotone quantity to be INCREASING in beta, and the certified series says... *Cannot decide: G19.*

- **compare the whole compact Wilson fast form from above by a Gaussian form after finite-rank retention.** W1-W3 use actual compact SU2 spectral growth and finite-codimension min-max to refute this implementation at every fixed positive coupling. Compact eigenvalues grow quadratically in the irrep label while the Gaussian radial reference grows linearly. This does not refute a compatible compact reference or selected-inverse comparison. *Cannot decide: G19.*
- **infer a regulator-uniform scalar single-site angle margin from the full square-root Gaussian precision.** For $M_m = \sqrt{L+m^2}$ on a periodic scalar lattice, $(L+m^2)^{-1} = m^{-2}$ gives $M_m^{-1} = m^{-1}$ exactly, so the row sums are m ; every off-diagonal entry is negative by the heat representation, hence $\sum_{f \neq e} |M_m(e,f)|$ equals $M_m(e,e) - m$ and the conditional maximal-correlation row is exactly $\kappa_m = 1 - m/M_m(e,e)$. Since $M_m(e,e)$ tends to a finite positive constant, κ_m tends to one as m vanishes: numerically .6154 at $m=1$,... *Cannot decide: G19.*

Completed routes (28). Derive the actual conditional continuity score and locate the constrained Agmon centers; Synchronize compact radial dilation with the actual conditional minimum curve; Resolve the antipodal conditional magnetic normal geometry; Control time-integrated conditional cubic energy with retained-mean localization; Identify the actual nonreducing Gaussian fast inverse with its energy prior; Resolve the unlocalized local harmonic cubic exchange target; Keep the exact quadratic endpoint and its complete low-frequency source normalization; Control the entire conditioned quantum Gaussian fiber at each fixed block scale; Determine the first actual Wilson ground and chosen-chart source correction; Extend the actual path-source fast bound over every flat holonomy; Realize the uniform transverse harmonic source as an actual local covariant path observation; Compare the complete actual endpoint spectrum with all retained source directions included; Establish actual local true-ground score loss on complete additive selected source charts; Identify the true-vacuum literal coarse form and additive physical fast space; Lift the full harmonic boundary inequality to the entire Gaussian quantum fast form; Establish the actual two-face vertical, coarse and full-vacuum complementary forms; Retain planar and periodic three-dimensional harmonic interfaces while isolating box-scale fast modes; Derive an energy comparison that retains arbitrary cross coupling and the induced norm; construct actual discrete-time Wilson blocks and cancel disconnected vacuum factors at a fixed observable; control the complete marked shell and onto literal source frame at fixed spatial spacing; sum temporal Wilson matching and transport the carrier with its specified observable overlap; assemble the full spatial Schur excess, second Wilson and source jets and accumulated physical-clock budget; replace the failed global fast-form comparison by an admissible selected-resolvent Schur identity; reconcile the broader G19 corpus and derive the flat reference and Cauchy repair inputs; audit the GPU and Lean closure claims against their actual operators and hypotheses; Construct the complete first residual current for the fixed twelve-edge square; Derive the compatible-kernel reconstruction and moving-time full-space implication; Optimize finite decay channels and time-amplified errors in the physical clock.

G20 — SAFE-region curvature certificate

Tier 3 (unbounded); open.

The notes program's quantitative core: a certified lower bound on the projected physical Hessian H_{phys} over a SAFE ball around the identity (claimed constants $\kappa^* = 0.25$, $\delta = 0.006$,

headline min eigenvalue 0.248), feeding curvature $\rightarrow$ LSI $\rightarrow$ gap. As reviewed, the headline is UN-REPRODUCED: the sole source table is contradicted by the archive’s own scans and independent recomputation (single-link Haar profile is flat at $1/4$), the operator’s code has never been executed, and no lattice size is stated anywhere. What discharging requires: run the imported...

G21 — Davies/Combes-Thomas $O(m)$ decay for the massive Maxwell 1-form kernel

Tier 3 (unbounded); partial.

The strongest result in the notes archive, and the first to survive a full-proof audit: $|G(b, b')| \leq (2/m^2) \exp(-\eta \text{ dist})$ for $G = (m^2 + \alpha d_1^* d_1)^{-1}$ on lattice 1-forms of a finite periodic lattice, with $\eta = \text{arcosh}(1 + m^2/(2\alpha C)) = 2 \text{ arsinh}(m/(2 \sqrt{\alpha C}))$ – an $O(m)$ exponent where naive resolvent bounds give $O(m^2)$. RE-ANCHORED after the audit: the canonical constant C is the WEIGHT-PROFILE row-sum $C_{\text{bdy}}(M_1)$ of Def A.9.4 (Appendix H, Prop H.4.3); the proved...

Ledger status. Appendix H audited in full 2026-08-22, proposition by proposition: the proof is sound end to end, and the no-factor-2 perturbation bound (Lemma H.3.3) is correctly derived and numerically confirmed with near-tight margin (≤ 0.97). Remaining open: the standing T2 sweep at several (m, α) , the H.3.3 mechanism check, the arsinh-form float guard, and the deferred upstream Prop A.9.5.

G22 — Coercivity via Cartan alignment

Tier 3 (unbounded); open.

The notes program’s crux conjecture: on the rough set (average plaquette energy $\geq \epsilon$), configurations far from Cartan alignment have force norm bounded below, volume-uniformly. The original 6-constraints-vs-3-DOF counting argument is refuted (stationarity at a link is 3 equations, and three orthogonal antipodal pairs sum to zero with no common Cartan direction – a registered FINDING check), so whatever truth the conjecture has must come from the gauge-coupled structure across links. Falsifier, required by this register’s own discipline: exhibit a configuration with $B_{\text{avg}} \geq \dots$

Ledger status. The stated finite matrix model has Simon’s transverse confinement bound $H_m \geq T/2 + g \sqrt{m-1} \sum |A_i|$, finite-dimensional compact resolvent and a raw ground-energy floor. These results are retained with their exact scalar split. WILSON_G19_REVIEW identifies the remaining derivation: realize this comparison for the actual gauge-coupled physical operator and prove volume-uniform constants after vacuum subtraction.

Live routes.

- **identify the transverse quantum form and uniform constants for the actual gauge-coupled physical operator.** Identify the actual physical transverse form and retained kinetic allocation. The cited finite-model proof uses finite-dimensional Rellich compactness and bounds E_0 before subtraction; it supplies neither this operator identification nor uniform vacuum-complement coercivity for the coupled lattice.

Completed routes (1). derive transverse confinement along classical flat directions in the stated finite matrix model.

G23 — The OS bridge with uniform constants

Tier 3 (unbounded); open.

Historical notes-program spine as a conditional chain: local curvature $\rightarrow$ LSI $\rightarrow$ diffusion gap $\rightarrow$ exponential clustering $\rightarrow$ OS reconstruction $\rightarrow$ Hamiltonian mass gap, with two named holes that no archive document closes. (1) Uniformity: every implication holds at fixed cutoff and volume once a curvature floor ρ_0 is granted; nothing supplies ρ_0 uniform in a and volume.

Ledger status. The actual fixed-scale Wilson transfer and complete physical odd source band remain established. The exact history theorem gives an isometry and transfer intertwiner for a true reflection/time-compatible pushforward. Gaussian observability now proves that its range is a history-frequency subspace, not an equal-time coordinate count; an arbitrarily small observed residue can make the whole physical two-mode $SU(2)$ history space visible.

Live routes.

- **Transport the nonlinear excess through a compatible actual generated-history hierarchy.** Use the actual full fast inverse and bounded-mean energy input in the complete nonlinear generated coefficient, retaining the harmonic witness and spatial cancellations. Establish uniform interacting remainder and complete source/history matching with a common physical clock. The chosen local source chart need not generate the whole matrix observation sigma algebra;
- **Compare the actual generated history dynamics with a controlled physical low-energy model.** The actual history map, jointly invariant algebra, generated measure, memory and physical clock remain explicit. Literal true-vacuum source ranges now have complete low-window frames and fast floors in additive Wilson models, including common Gauss, and in the regulated full Gaussian boundary model. These equal-time ranges need not be the reducing OS-history range.
- **control the actual physical trial residual after local vacuum cancellation.** WR14-WR16 construct a gauge-invariant trial with a nonnegative shared-link residual, but its global sum is not a centered gap budget. Connected residual and complete boundary-source control are still needed. Independent copies refute a naive extensive residual loss.
- **prove physical conditional assembly and improved coercivity through spatial refinement.** Fixed-spacing substeps are complete: BA20–BA26 Cauchy-Schwarz true-vacuum bound eliminates the multiplicity penalty on single-link partitions ($\kappa \leq 10/229$, gap > 2.8679 epsilon). SC17 bootstrap closes the connected spatial estimate for $\lambda \leq 1/73$, establishing weighted Hessian row $\leq 2/7$ at weight $1025/1024$, projection angle row $\kappa \leq 10658/24923$, $C_{AT} \leq 24923/14265 < 7/4$, and physical gap > 4 epsilon / 3. The rational budgets are machine-certified by the supplied Lean lemmas; *Cannot decide: G19, G23.*

Untried routes.

- **establish tunable physical-time localization on a complete observable class.** Prove the localization and bad-event bounds for every radius with controlled prefactors and complete physical observables. A diffusion gap alone does not identify Euclidean-time spectral decay. A source with an unobserved slow mode falsifies an incomplete-source argument.

Dead routes, and why they died.

- **transfer an isolated physical rotor gap to a fixed finite unreduced boundary frame.**

WR20 embeds the entire single-link electric Dirichlet form using the true patch ground and its Haar marginal. The unreduced boundary gap is at most 3ϵ and the number of modes below a fixed target grows as $\epsilon^{-(3/2)}$. This refutes the fixed finite frame, not a physical gap or assembly with full compact boundary retention. *Cannot decide: G19, G23.*

- **replace physical spatial control by globally positive classical Wilson orbit curvature at weak coupling.** The exact regular SU2 plaquette curvature is $8/3\beta/2$ at $\theta=2\pi/3$. Horizontal geodesics extend the failure to the stated finite cubic lattices. The classical weight is not the physical ground and Wilson flow has an expansion witness $8/7$. *Cannot decide: G19, G23.*
- **obtain a positive physical bound by discarding WR25 local remainders before vacuum subtraction.** VA1-VA8 give an actual connected periodic SU2 Wilson test at $L\geq 3$, $k/\epsilon\geq 32$. The discarded vacuum remainder is at least $15\delta p_0 |E|$ with $p_0>0$ independent of volume. A physical mean-zero local phase has expectation of A-F at most $\delta(4-15p_0 |E|)$, negative at large volume. *Cannot decide: G19, G23.*

Completed routes (15). Identify the actual nonreducing Gaussian fast inverse with its energy prior; Identify exact Gaussian endpoint memory and its first chosen-source nonlinear coefficient; Retain exact endpoint memory while comparing the entire additive physical low source window; Determine what the exact harmonic history block observes before assigning an eliminated space; repair negative Wilson curvature and derive the complete compact rotor gap; retain the full compact boundary and prove the actual Wilson-link fast barrier; identify the finite ground transform and retain the full trial residual gap budget; derive the physical spectral-weight bound and optimized localization exponent; align conditional projections with the true Wilson vacuum and establish their local floor; optimize the actual conditional block gap and derive its connected angle and pressure identities; prove actual true-vacuum block assembly in the explicit electric strong-coupling window; derive an all-coupling background magnetic comparison and charged physical-time covariance bound; prove SC17 spatial decay and actual conditional assembly through its explicit strong-coupling threshold; construct the SC17 thermodynamic ground law and its closed physical form with nonzero plaquette overlap; identify actual SC17 thermodynamic physical-time correlations with the closed ground-law semigroup.

Discharged gaps

Finished work stays recorded rather than re-queued. Each entry names what closed it.

- **G18 The spectral bridge at fixed lattice spacing.** Fixed-spacing closure only; continuum source transport remains G19.
- **G2 Regenerate all displayed tables in canonical u.** discharged 2026-08-21 — the conversion statement is now arithmetic (coupling-erratum suite): the printed towers reproduce $4\Delta(3u/2)$ verbatim in $u = \beta/6$ and would be off by exactly $4r$ under the archived Y reading.
- **G24 Derive the shared-link isotropy premise, the one unproved input of the second-order chain.** discharged 2026-08-28, the day it was opened. The premise is not an assumption: it follows from the order-2 Weingarten values.
- **G3 Fourth-order adjudication — decide C_{shp} where it actually lives.** DISCHARGED 2026-09-04 (ADR 0024). C_{shp} lives in the rotation amplitude ρ , and ρ is now decided:

the historical pipeline’s own Stage-3I ledger, the 2026-09-02 cluster assembly and a third engine agree on every cluster of the rotation record but the adjacent-face cube completion, where the historical ledger holds 8 of 24 orderings;

- **G5 Tetrahedral local Haar-resolvent coefficient.** discharged 2026-08-21 — the $r = 2$ instance of the primitive completion law is re-derived exactly at symbolic N (tetrahedral Haar-resolvent suite, `workhouse.cellular`): $c_2 = -8/(N(N^2-1))$, from two temporal histories through the single 4-link intermediate loop, under the same convention that reproduces the sealed-core cube row, the prism 64/24 sector pair, and the pentagonal 120-history count. The falsification test the mobility theorem proposed did NOT kill the coefficient: the $r = w_{\min} - 2 = 2$ allowance carries nonzero primitive Haar-resolvent...

18 Working the problems

§17 states what must be proved. This section is about how to start, and it is written for an agent with a fresh context window and a week.

18.1 Choosing

Three properties make a target worth taking, and they are independent.

Decisive — solving it resolves a contradiction, discharges a gap, or unblocks another problem. *Ready* — no unresolved input is recorded. *Bounded* — someone has measured, or can cheaply estimate, what it costs.

Most of the problems in §17 are decisive and ready. Almost none are bounded, and that is the honest difficulty: they are open analytic estimates, not computations waiting for a machine. The two exceptions are worth naming, because an agent looking for tractable decisive work should start at one of them.

- **Problem 8**, one complete sixth-order cluster. Finite, priced, and it determines an object that is already proved to exist (§10): the extra rational band shape survives every local direct term, so computing the direct term settles its coefficients rather than gambling on whether there is anything there.
- **Acquiring the unobtained near neighbours** (§12). Days, not weeks. It is the only item that can *subtract* from the program, which is exactly why it should not keep being deferred in favour of items that can only add.

18.2 What the published methods actually supply

Several of the continuum problems have a named external method attached. The attachment is real but narrower than it looks in a citation list, and the narrowing is the useful part.

Quantum-perturbation Riesz projection, and ground-state methods.

Verified against the fixed-spacing bridge and against the source-family problem. This is the machinery behind the discharged fixed-spacing construction; it is method, not result, and it is already spent — it got the program to §13 and does not by itself go further.

Cluster expansion methods.

Verified against the source-family problem and the continuum gap. Break 15.2 is the cautionary

result here: the expansion’s activity and a scale-independent physical mass cannot be read off one schedule. Scalar Kotecky–Preiss inequalities and raw moment estimates remain useful ingredients, but they construct neither the tilted activities nor a gap.

WKB and tunnelling for operators on a vector bundle.

Supplies the local weighted comparison with actual Dirichlet eigenfunctions used in the conditional-transport route. It does *not* supply the all-fiber domination of Problem 1, nor the differentiated coupling-score asymptotics. The scalar one-well application is what has been used; the vector case is what is needed.

Renormalization-group multiscale methods.

Supply the shape of Problem 5 and nothing about its missing power. Break 15.1 localizes the difficulty to an exact cancellation of the leading partition-function covariance.

Osterwalder–Seiler positivity and transfer matrices.

Registered in the evidence map, and the classical background against which the fixed-spacing construction of §13 should be read. Novelty claims in that section must clear it.

Stochastic-analysis approaches at strong coupling.

Present in the reading library, with mass-gap and exponential-clustering results at explicit small-coupling bounds. These live in the *same* regime as §13 — neighbourhoods of zero coupling — and therefore also do not cross Break 15.5. They are comparison, not a route.

The pattern is worth stating plainly: every external method that has been verified against this program supplies control *at fixed scale or small coupling*. None of them supplies continuation in the coupling, which is the one thing Problem 6 needs. An agent that finds a method claiming otherwise should treat that as the most interesting thing it found that week.

18.3 A procedure for one problem

1. **Resolve the target to graph identifiers.** Run `workhouse why` on the gap and on each named input. Retain the briefing. You now have the routes, their states, and the `cannot_decide` annotations.
2. **Read the dead routes first (§17).** If your idea matches one, read why it died before deciding it died for a reason that no longer applies. Sometimes that is true; the register records the reason so the judgement can be made rather than guessed.
3. **Locate the available lemmas as DERIV statements,** not as prose. Appendix D gives each one a document, a digest and a line range. Read the hypotheses; the corpus records them separately from the statement because they are what gets dropped.
4. **Check the literature both ways.** The evidence map for what has been verified against this claim; the reading library (Appendix G) and the extraction index (Appendix H) for whether the estimate you are about to attempt already exists somewhere.
5. **Do the smallest instance first.** Every decisive-next-test in §17 is phrased this way — two coupled blocks, one cluster, one blocking step — because the program’s own experience is that the general case hides which term is the obstruction and the smallest instance exposes it.
6. **Record the attempt, including its failure.** A dead route is a result. The procedure is in the working agreement.

18.4 Failure modes with a track record

Each of these has cost this program real time. They are listed in descending order of how much.

Running an instrument that cannot decide the question.

Four sessions went into a sealed sweep that a static argument had already shown emits only a scalar at one momentum, a quantity structurally incapable of constraining the coefficient it was aimed at. The register now carries `cannot_decide` for exactly this reason. Check it before running anything expensive.

Interpolating without a degree bound.

Sixty-seven exact agreements between exact rationals do not determine a rational function. The fix that worked was not to prove the bound but to remove the need for it by computing in the function field (§9).

Counting vertices and forgetting the projection.

The retracted tier-collapse mechanism (§7) bounded the degree of the operator and omitted the power contributed by the carrier projection itself. Any degree argument in this setting must count the projection.

A correct calculation in the wrong basis.

An assembled amplitude was right and was read in the conjugate basis, producing a third value in a two-value dispute. The signature was visible in the data: every C -odd component negated, every C -even one identical. When a new value appears in an existing dispute, test for conjugation before believing it.

Quoting a tier from a record instead of a run.

A registered axiom set is evidence; it is not a build. If a tier is load-bearing for what you are about to claim, re-establish it — the commands are in Appendix A and take seconds.

Treating a summary as a source.

Measured in this program at thirteen out of thirteen (Appendix C). Cite the primary, and check that the anchor resolves rather than only that the digest matches.

18.5 Where new mathematics is cheap

The mass-gap problems are hard by construction. The structural questions below are not, and several are decisive for the theory even though none of them is on the continuum path. An agent looking to produce new results rather than to close the Clay problem should read this list first.

The general grading law.

§8 establishes $S_m(N) = c_m N^{-(2m-1)}(1+O(N^{-2}))$ for even $m \leq 10$, by a four-channel cancellation with leading coefficients in the ratio $(-4, +2, +1, +1)$ summing to zero. The law for general m is a conjecture. The cancellation is a statement about four specific intermediate representations, which is the kind of statement a character-theoretic argument settles at all orders at once. This is the most likely place in the program for a clean general theorem.

The multi-face cumulant.

The one-face law is proved; the cluster cumulant version is not discharged, and the shared-link pair route carries it. Distinct from the item above and often conflated with it.

The fifth-order pentagonal coefficient.

Registered open, finite, and it interacts with the centre-parity theorem of §8: the odd orders are

determinant families, and the fifth-order determinant correction is what refuted the centre-only-circuits-are-dark claim. Computing the full coefficient is bounded work with a known consumer.

The rank holdout ledger.

Fixed-rank values for a mid-range band of N . Pure computation, and it is the natural holdout against every closed form the symbolic engine has produced — exactly the kind of check that would have caught an interpolation error before the function-field method removed the risk.

Native torelon reruns.

Two string-tension coefficients, by a method the program already has.

The hyperhoneycomb candidate.

An alternative tessellation with a first-order term. The reading library has a substantial Hamiltonian-and-quantum-simulation category, including improved honeycomb and hyperhoneycomb Hamiltonians, so the external groundwork exists.

Classification of shortest physical temporal histories.

Open and unbounded, but it would consolidate the mobility taxonomy — which is where the falsified $F-2$ rule lived, and where a replacement is still missing.

The symbolic Gram transcript.

Would make the function-field method of §9 auditable end to end rather than trusted at its output.

Three of these — the general grading law, the fifth-order coefficient, and the multi-face cumulant — are in the same neighbourhood as results this program has already proved, with the same tools. That is the definition of cheap here, and it is where the next novel theorem most plausibly comes from.

Appendices

A Reproducibility

Every claim in this document that carries a [T0 Lean], [T1 exact] or [T2 numeric] badge can be re-established from a clean checkout in seconds to minutes. This appendix says how, and — more usefully — says what each kind of re-establishment does and does not prove.

A.1 From a clean checkout

```
git clone https://github.com/ats314/WORKHOUSE
cd WORKHOUSE
uv sync --all-extras --frozen
uv run --no-sync workhouse --help
```

Then:

```
make verify      # every registered check: T1 and T2, a few seconds
make lean        # T0: proof-check the Lean core (requires elan)
make status      # the contradiction and gap registers
```



```

workhouse verify --tier 1           # only the exact re-derivations
workhouse verify -v                 # every tolerance, printed
workhouse verify --only 'h_4^side' # one claim, with its numbers

```

A single claim is the useful unit. `workhouse verify --only` takes a claim name and prints the numbers it establishes, which is what makes a disagreement arguable rather than merely reported. A certification a reader cannot cheaply reproduce is a claim of authority, not evidence, and the design of this repository takes that seriously enough to attach the reproducing command to every row of `CERTIFIED.md`.

A.2 Querying a claim

```

workhouse why C2                     # everything recorded about one id
workhouse why t_N                     # a constant: value, formalizations, sources
workhouse branches C2                 # both sides of a dispute, side by side
workhouse derive C2 G3 --out f.md     # evidence chain, registered edges only
workhouse search '5/612'              # search by exact VALUE, not by name
workhouse lit --for G19                # published work bearing on a claim
workhouse brief G19 --json             # target snapshot with freshness

```

`search` matches by value rather than spelling, which matters because the join keys of this corpus are exact rationals and no natural-language query retrieves 109151/249696. It also carries two warnings a text search cannot: names the register forbids because they regenerate a dissolved contradiction, and names coined in this program that its own source corpus never uses.

A.3 What each tier actually establishes

[T0 Lean] — Lean 4, no sorry.

The statement is checked by the Lean kernel from explicit definitions, with axioms restricted to `propext`, `Classical.choice` and `Quot.sound`. The axiom set is exported and validated rather than asserted: `scripts/export_lean_dependencies.py` builds strictly and records elaborated constants and transitive axioms into `ledger/lean_dependencies.json`, and the loader checks these against declaration records and source fingerprints. Identifiers mentioned in comments are not proof dependencies.

What a T0 badge does *not* establish is importance. Several registered theorems formalize arithmetic evaluations of named rationals. The registry records for each theorem whether it proves an entire named check (`promotes`) or one ingredient of a larger statement (`lean_support`); supporting a statement never silently promotes the whole statement's tier.

[T1 exact] — exact re-derivation.

The statement is recomputed symbolically in exact arithmetic from stated definitions. Corpus rationals are `sympy.Rational`; values the corpus records only as floats carry a `_NUM` suffix and are Python floats. A float that reads as exact is the most dangerous defect available in this repository, and the boundary is guarded by a dedicated test.

[T2 numeric] — numerical agreement.

Float agreement within a tolerance the check itself prints. A T2 pass is that and nothing more. Tolerances are never widened to make a disagreement disappear; a disagreement is registered as an explicit finding that asserts the discrepancy.

[T3 analytic] — no whole-statement machine certification.

The default. An analytic result may be fully proved and still be T3: the badge reports machine coverage, not mathematical status. Status, evidence and tier are three independent fields in `ledger/results.yaml` for exactly this reason.

A.4 The one inference this apparatus does not license

A finite check certifies the statement it checks. It does not certify a general theorem of which that statement is an instance, and the converse also holds: a complete analytic proof is not weakened by absent Lean coverage.

This cuts both ways in practice, and both directions have caused errors here. A suite of exact checks passing at ranks $N = 3, 4, 5, 6$ is not a statement about all N — which is why the symbolic- N engine of §9 exists, and why its degree bound was retracted when one projection turned out to be uncounted. Conversely, a separate Lean library in this program’s archive comprises seventy-one modules with no `sorry` and was found on audit to prove mostly tautologies; a theorem count is not a measure of coverage, and no count in this document should be read as one.

A.5 Regenerating this document

```
python paper/master/generate_apparatus.py
tectonic paper/master/workhouse_master.tex      # the monograph
tectonic paper/master/workhouse_band_paper.tex  # Part I, standalone
```

The generator reads `literature/index.yaml` and `ledger/{gaps,results,theorems,contradictions}.yaml` and writes the bibliography, the external-evidence table, the route register, the results table and the count macros. Every number the prose quotes is one of those macros. If a count here disagrees with the repository, the generator has not been re-run; the repository is right.

B Verification index

Table 2 lists every registered analytic result with its recorded status and evidence level. The two columns are independent fields and are meant to be read against each other: **proven** with **record-backed** evidence means the argument exists and the machine artefact does not.

Of 143 registered results, 140 are recorded **proven**, 1 **conditional**, 1 **disputed** and 1 **falsified**. The Lean tree carries 439 registered theorems with 0 occurrences of `sorry`.

A caution on both counts. 140 out of 143 is a high proportion because a claim is not registered here until its argument is written down; the register measures what has been recorded, not what fraction of the problem is solved. Likewise 439 theorems is a count of declarations, not of mathematical content — see §A, which explains why a theorem count is not a coverage measure and cites a case in this program’s own archive where seventy-one `sorry`-free modules turned out to prove tautologies.

Table 2: Analytic results by recorded status and evidence level. *Status* is what the argument establishes; *evidence* is what artefact backs it. The two are deliberately independent: a result can be **proven** in status and **record-backed** in evidence, meaning the proof exists and the machine artefact does not.

Result	Status	Evidence
ANISOTROPY_THREE_COORDINATE_SHARP_BOUND	proven	analytic
ARCHIVE_CENTER_DEFECT_OBSTRUCTIONS	proven	analytic
ARCHIVE_CONCENTRATION_LOCALIZATION	proven	analytic
ARCHIVE_FINITE_RANGE_INVERSE	proven	analytic
ARCHIVE_GLOBAL_PAIRING_OBSTRUCTION	proven	analytic
ARCHIVE_HORIZONTAL_RESTRICTION_BOUNDARY	proven	analytic
ARCHIVE_LYAPUNOV_PATCHING	proven	analytic
ARCHIVE_REFLECTION_PERMANENCE	proven	analytic
ARCHIVE_SIGNED_SU2_DIFFUSION	proven	analytic
ARCHIVE_SMOOTH_PROFILES	proven	analytic
ARCHIVE_SOURCE_TILT_AND_ROOTED_BOUNDS	proven	analytic
ASSEMBLED_Q4_COERCIVITY	proven	analytic
BALABAN_MULTISCALE_CAUCHY_SUMMABILITY	proven	analytic
COMMON_SPACE_CONTRACTIVE_DRIFT	proven	analytic
COMPACT_ROTOR_FAST_GAP	proven	analytic
CONDITIONAL_GRADIENT_REPAIR	proven	analytic
CONDITIONAL_QUANTUM_PATH_COVARIANCE	proven	analytic
CREATOR_PARENT_GAP	proven	analytic
CREATOR_VELOCITY_INVERSION	proven	analytic
DIMENSION_FIVE_OPERATOR_IRRELEVANCE	proven	analytic
DYNAMIC_CONDITIONAL_CUBIC_ENERGY	proven	analytic
EARLIER_B6_RETAINED_KRYLOV_CAMPAIGN	conditional	numerical
EARLIER_CONDITIONAL_SPECTRAL_FLOOR	proven	analytic
EARLIER_CORNER_BLOCKING_REFLECTION_DEFECT	proven	analytic
EARLIER_CUBIC_QUOTIENT_REGULARITY	proven	analytic
EARLIER_D4_DISJOINT_STAPLE_COORDINATES	proven	analytic
EARLIER_DECORATED_HISTORY_MERGING	proven	analytic
EARLIER_DETERMINANT_PARITY_REPAIR	proven	analytic
EARLIER_EQUAL_RAY_IDENTIFICATION	proven	analytic
EARLIER_FINITE_ORDER_SUPPORT	proven	analytic
EARLIER_GRAM_NULL_OPERATOR_QUOTIENT	proven	analytic
EARLIER_ISOLATED_POSITIVE_MATRIX_ATOM	proven	analytic
EARLIER_JOINT_RANK_VOLUME_SCALING	proven	analytic
EARLIER_LOCALIZED_FRAME_OBSTRUCTION	proven	analytic
EARLIER_MOVING_ADJOINT_FORCE_DERIVATIVE	proven	analytic
EARLIER_NESTED_QUOTIENT_REDUCTION	proven	analytic
EARLIER_POSITIVE_SECTOR_PHASE_ISOLATION	proven	analytic
EARLIER_REFLECTION_ADAPTED_BLOCKING	proven	analytic
EARLIER_SHELL_SYMMETRY_RITZ_CONTROL	proven	analytic

Result	Status	Evidence
EARLIER_SPHERE_EQUAL_COUPLING_COVARIANCE	proven	analytic
EARLIER_SU3_POSITIVITY_BOUNDARIES	proven	analytic
EARLIER_VSU_BOUNDED_DOMAIN_WELLPOSEDNESS	proven	analytic
EXPONENTIAL_SOURCE_TOTALITY	proven	analytic
FESHBACH_RESOLVENT_COMPARISON	proven	analytic
FORM_SCHUR_SCALE_COMPARISON	proven	analytic
FULL_GRAPH_SOURCE_CONGRUENCE	proven	analytic
G17_CORRECTED_KP_POLYMER_STABILITY	proven	analytic
G9_COMBINED_MIXING_SUPPORT	proven	analytic
G9_HERMITIAN_FOLDED_SUPPORT	proven	analytic
G9_LOCAL_SIXTH_NONCANCELLATION	proven	analytic
G9_ONE_FACE_SIXTH	proven	analytic
G9_SIXTH_ORDER_RUR_ACTIVATION	disputed	record-backed
GAUSSIAN_FULL_FAST_GREEN	proven	analytic
GAUSSIAN_OS_OBSERVABILITY	proven	analytic
GAUSSIAN_PATH_ENDPOINT_BASELINE	proven	analytic
GAUSSIAN_QUANTUM_FAST_SOURCES	proven	analytic
GAUSSIAN_SCHUR_MEMORY	proven	analytic
GROUND_MARGINAL_SCHUR_SCORE	proven	analytic
HODGE_FESHBACH_SPLITTING	proven	analytic
HODGE_INDUCED_EIGHTH_ORDER_MOMENT	proven	analytic
HODGE_LEAKAGE_ZERO_EXTENSION	proven	analytic
HODGE_RANK_ONE_SECULAR_CUBIC	proven	analytic
INTRINSIC_SCORE_CURRENT	proven	analytic
LITERAL_ENDPOINT_SPECTRAL_COMPARISON	proven	analytic
LOCAL_CLASS_WICK_COEFFICIENTS	proven	analytic
LOCAL_CLASS_WICK_RECURRENCE	proven	analytic
LOCAL_CLASS_WICK_SIGNS	proven	analytic
LOCAL_GRADIENT_EXCITATION_SUPPORT	proven	analytic
MESOSCOPIC_NORMALIZED_SOURCE	proven	analytic
MOVING_TIME_MULTICHANNEL_RATE	proven	analytic
MOVING_TIME_MULTICHANNEL_SHARPNESS	proven	analytic
MOVING_TIME_SPECTRAL_GAP	proven	analytic
ODD_ORDER_CENTRE_PARITY	proven	analytic
ORDERED_CONTOUR_ACTIVITIES	proven	analytic
OS_HISTORY_BLOCK_INTERTWINER	proven	analytic
OS_KERNEL_MOVING_TIME_GAP	proven	analytic
PREDICTABLE_SOURCE_DRIFT	proven	analytic
PROJECTIVE_SOURCE_MARTINGALE	proven	analytic
REVERSE_MARTINGALE_BLOCKING	proven	analytic
RICCATI_RESIDUAL_CERTIFICATE	proven	analytic
ROUGH_GAUGE_COERCIVITY_LOWER_BOUND	proven	analytic
RSM_R_CARRIER_SYMBOL_MASTER_THEOREM	proven	analytic
SELECTED_INVERSE_WITHOUT_DIAGONAL	proven	analytic
SHARP_CUTOFF_SOURCE_GAP_BUDGET	proven	analytic
SOURCE_COMPLEMENT_COMPLETE_WINDOW	proven	analytic

Result	Status	Evidence
SOURCE_CONGRUENCE_METRIC_TRANSPORT	proven	analytic
SOURCE_CURRENT_BOUNDED_W6	proven	analytic
SOURCE_CURRENT_VARIATIONAL_ENERGY	proven	analytic
SQUARE_FIRST_CURRENT_BUDGET	proven	analytic
SQUARE_FIRST_RESIDUAL_CURRENT	proven	analytic
SQUARE_SCALAR_SOURCE_ODD_SCHUR_VANISHING	proven	analytic
TETRAHEDRAL_FLUX_PROJECTION_BOUNDARY	proven	analytic
TETRAHEDRAL_S4_COMMUTANT_RESOLUTION	proven	analytic
TIER_COLLAPSE_ACTUAL_H4_SUPPORT	proven	analytic
TIER_COLLAPSE_IS_R_DEGREE	falsified	analytic
TRUE_GROUND_CENTER_SCORE_OBSTRUCTION	proven	analytic
UNIFORM_MARKED_CLUSTERING	proven	analytic
UNIVERSAL_CELLULAR_HODGE_SPECTRUM	proven	analytic
W6_ACTUAL_CENTERED_COMPLEMENT_SUPPRESSION	proven	analytic
W6_ANGULAR_REFERENCE_SCORE_BUDGET	proven	analytic
W6_R10_CONDITIONAL_DERIVATIVE	proven	analytic
W6_R10_H4_FROM_H0	proven	analytic
W6_R10_PROJECTION_JET_RECURRENCE	proven	analytic
W6_UNIFORM_AGMON_COMPLEMENT_GAP	proven	analytic
WILSON_ACTIVITY_EXTRACTION	proven	analytic
WILSON_ACTUAL_BLOCK_FAST_COMPLEMENT	proven	analytic
WILSON_BLOCK_SCORE_OBSTRUCTION	proven	analytic
WILSON_CARDINALITY_CHART	proven	analytic
WILSON_COEFFICIENT_LOCALITY	proven	analytic
WILSON_COMMON_GAUSS_LITERAL_FAST_COMPLEMENT	proven	analytic
WILSON_COVARIANT_BLOCK_SOURCES	proven	analytic
WILSON_CREATOR_LIMIT	proven	analytic
WILSON_CUBIC_GROUND_TRANSFER	proven	analytic
WILSON_ENDPOINT_COMPLETE_CLUSTER	proven	analytic
WILSON_ENDPOINT_EQUATION	proven	analytic
WILSON_FINITE_CELL_PHYSICAL_GAP	proven	analytic
WILSON_FIXED_ORDER_CHART	proven	analytic
WILSON_FLAT_BACKGROUND_FAST_SOURCES	proven	analytic
WILSON_GLOBAL_VERTICAL_BARRIER	proven	analytic
WILSON_GROUND_BUNDLE_RELATIVE_FORM	proven	analytic
WILSON_HARMONIC_BOUNDARY	proven	analytic
WILSON_HARMONIC_CUBIC_OBSTRUCTION	proven	analytic
WILSON_INFINITE_PHYSICAL_BAND	proven	analytic
WILSON_KINETIC_WINDOW	proven	analytic
WILSON_LITERAL_COARSE_FORM	proven	analytic
WILSON_LITERAL_FAST_COMPLEMENT	proven	analytic
WILSON_LOCALIZED_TRUE_GROUND_SCORE	proven	analytic
WILSON_PARENT_GNS	proven	analytic
WILSON_PHYSICAL_SOURCE_RANK_REPAIR	proven	analytic
WILSON_ROOTED_CONTRACTION	proven	analytic
WILSON_SAME_WEIGHT_OBSTRUCTION	proven	analytic

Result	Status	Evidence
WILSON_SECOND_ORDER_CHART	proven	analytic
WILSON_STRIP_BO_FIRST_TERM	proven	analytic
WILSON_SYMMETRIC_CREATORS	proven	analytic
WILSON_THREE_DIMENSIONAL_HARMONIC_BOUNDARY	proven	analytic
WILSON_TRANSFER_RESOLVENT	proven	analytic
WILSON_TWO_SQUARE_PHYSICAL_SHELLS	proven	analytic
WILSON_TWO_STRIP_PHYSICAL_SPLITTING	proven	analytic
WILSON_UNIFORM_FINITE_SHELL	proven	analytic
WILSON_VACUUM_COMPRESSION	proven	analytic
WILSON_VACUUM_SPECTRAL_FLOW	proven	analytic
WILSON_VERTICAL_FAST_ENERGY	proven	analytic
WILSON_WEIGHTED_ACTIVITIES	proven	analytic

C Provenance

The verification repository is one layer over a larger research archive. Both are needed to read a result’s history, and they answer different questions: the repository says what is currently checked, the archive says what was actually done and when.

C.1 The two layers

The repository (<https://github.com/ats314/WORKHOUSE>) holds the ledgers, the invariant suites, the Lean tree, the generated views (`FRONTIER.md`, `CERTIFIED.md`, `index/`) and the manuscripts. Its generated views are maintained by the verifier commands and are never hand-edited: a test fails if either has drifted, because a generated file that has drifted still reads as current.

The archive holds the primary sources: manuscripts, proof collections, appendices, simulation campaigns, a separate Lean library, reference papers and the raw computational runs. Its organizing rule is that nothing is erased — failed attempts, duplicates, superseded drafts and uncommitted work are all preserved, because deleting a refuted claim destroys the evidence that it was tried.

C.2 Reading a source document

Documents under `theory/`, `corpus-import/` and `settlement/` state what someone believed at the time. They are immutable evidence and are never edited to make a check pass; when a check disagrees with a document, the disagreement is registered as a finding. `theory/SHA256SUMS` pins the contents. `theory/superseded/` holds documents kept for the audit trail, and none of them should be read as current.

This matters more than it usually would, because parts of the corpus were written with AI assistance and some of it is wrong. The repository has already caught a reversed tensor-product identity, a one-ulp transcription error, a tolerance quoted tighter than its own data, and a mechanism proposed here and then retracted. Confidence of phrasing carries no evidential weight in either direction.

C.3 Synthesis documents are not evidence

A specific hazard, recorded because it has cost this program time. Several large synthesis documents in the archive were produced by automated agents summarizing primary sources. Audit found the primary sources honest about their own scope and the synthesis layer systematically stripping their caveats — upgrading conditional statements to proven, attaching Lean support that does not cover the statement it is attached to, and citing a formalization library whose theorems are tautologies.

The rule this produced: cite the primary source, never the synthesis of it, and verify that a claimed anchor resolves rather than only that its hash matches. A correct SHA-256 over a fabricated locator is a failure mode that has actually occurred here.

C.4 A measurement of that hazard

The present edition was assembled with a deliberate test of the effect above, and the result is worth recording because it is quantitative.

Thirteen result clusters of this corpus were summarized independently, each by an agent instructed to extract exact statements with their recorded status, hypotheses and verification tier, and explicitly told not to upgrade anything the ledger records as conditional. Each summary was then audited by a second agent instructed to refute it against the primary files.

All thirteen summaries were found to overstate in at least one respect. The failure modes recurred:

- a status upgraded past what the ledger records;
- a hypothesis present in the ledger and absent from the summary — most often a periodicity condition, a rank restriction, or a regime bound;
- a verification tier asserted with no check behind it, or with a check that tests a tautology;
- novelty asserted against an index the same summary elsewhere notes is incomplete;
- a cited anchor that does not resolve to the named object, with the surrounding hash correct.

Five of those findings changed statements in this document, and each is named at the point of correction rather than silently applied: §8 on the planar reduction, §7 on the rank scope of the operator splitting, §9 on what the formal proof of the coefficient identity actually checks, §13 on chains versus operators, and §13 again on the overlap exponent.

The lesson for the next agent is not that summarization is useless — the extractions found real material that a direct reading missed, including the strongest external corroboration in §12. It is that a summary of this corpus should be treated as a *lead*, never as a citation, and that the audit step is not optional overhead. It changed one statement in every three that mattered.

D Derivation concordance

Every registered derivation statement, grouped by the document it lives in, with the digest of the bytes it was read from and the heading and line range that locate it inside that document.

332 statements across 40 documents. The *Formal* column is **full** when a Lean theorem covers the whole statement, **partial** when a theorem is registered as supporting one ingredient of it, and **--** when neither. Partial support never promotes the whole statement's tier, which is why the two are separate columns of the underlying registry rather than one.

A caution on using this table. The status recorded here describes the *source argument* as the corpus assessed it, not an independent re-derivation. Empty formal coverage means no complete registered formalization exists in this inventory — it is not a judgement that the analytic proof is invalid, and reading it as one would invert the meaning of the default tier (§2).

The concordance below covers all 332 registered derivation statements across 40 source documents. Each document carries the SHA-256 of the bytes its statements were read from; if the file has changed, the digest has changed and the statements must be re-located rather than assumed.

CITE:ARCHIVE_FUNCTIONAL

docs/derivations/archive-functional-inequalities.md

SHA-256 048fbf01cdc55ee4...

Statement	Locator	Status	Formal
F1	## F1. Trace defect, faithfulness and the vacuum set (lines 13-22)	proven	—
F2	## F2. Global gradient domination without a quotient singularity (lines 23-37)	proven	—
F3	## F3. Plaquette lifts preserve link derivatives (lines 38-53)	proven	—
F4	## F4. Aggregate defect comparisons (lines 54-64)	proven	—
F5	## F5. Exact drift identity and the squared-profile budget (lines 65-82)	proven	—
F6	## F6. General profiles with vanishing derivative at zero (lines 83-96)	proven	—
F7	## F7. Linear pairing implies a genuinely uniform small-set drift (lines 97-114)	proven	—
F8	## F8. Lyapunov energy conversion as a completed square (lines 115-129)	proven	—
F9	## F9. Local-to-global Poincare patching (lines 130-144)	proven	—
F10	## F10. Averaging gives a volume-scale gradient bound (lines 145-156)	proven	—
F11	## F11. Herbst concentration and the mean threshold (lines 157-172)	proven	—
F12	## F12. Covariance localization with an explicit error budget (lines 173-190)	proven	—
F13	## F13. Absorbing a volume tail into a distance exponent (lines 191-203)	proven	—
F14	## F14. Averaged smallness does not imply plaquettewise smallness (lines 204-213)	proven	—

Statement	Locator	Status	Formal
F15	## F15. Center configurations refute the proposed Laplacian identity (lines 214-229)	proven	—
F16	## F16. Extensive center defects obstruct uniform pairing coercivity (lines 230-270)	proven	—
F17	## F17. The exact $SU(2)$ diffusion budget retains favorable signs (lines 271-311)	proven	—

CITE:ARCHIVE_SOURCE_TRANSPORT

docs/derivations/archive-source-transport.md

SHA-256 4a9d953d888e1b0e...

Statement	Locator	Status	Formal
S1	## S1. Inhomogeneous source tilting is a partition ratio (lines 10-19)	proven	—
S2	## S2. Source response is covariance and log partition is convex (lines 20-31)	proven	—
S3	## S3. Pointwise envelopes do not order normalized tilts (lines 32-44)	proven	—
S4	## S4. Bounded log likelihood ratios transfer tilted estimates (lines 45-57)	proven	—
S5	## S5. Free energy implies conditional rooted-capacity summability (lines 58-78)	proven	—
S6	## S6. A conservative rooted-animal count (lines 79-94)	proven	—
S7	## S7. Reflection-positive deterministic pushforward (lines 95-107)	proven	—
S8	## S8. Projective and weak limits preserve the stated OS forms (lines 108-122)	proven	—
S9	## S9. Reflection equivariance alone has a counterexample (lines 123-134)	proven	—
S10	## S10. Block Schur bound with row and column control (lines 135-147)	proven	—
S11	## S11. Finite-range inverse transport with an explicit exponent (lines 148-162)	proven	—
S12	## S12. Massive Maxwell row sums from incidence counting (lines 163-178)	proven	—

Statement	Locator	Status	Formal
S13	## S13. Cochain horizontals reduce the massive Maxwell operator (lines 179-186)	proven	—
S14	## S14. Restriction need not preserve an ambient local kernel (lines 187-201)	proven	—

CITE:EARLIER_PROOF_RECOVERY

docs/derivations/earlier-proof-recovery.md

SHA-256 4b5b067ee7697cf5...

Statement	Locator	Status	Formal
CUBIC_QUOTIENT_REGULARITY	## E1. Cubic quotient regularity trichotomy	proven	—
EQUAL_RAY_IDENTIFICATION	## E2. Equal-ray recovery and an independent holdout	proven	—
LOCALIZED_FRAME_OBSTRUCTION	## E3. No uniformly localized finite translation frame	proven	—
JOINT_RANK_VOLUME_SCALING	## E4. Sharp joint rank-volume scaling	proven	—
FINITE_ORDER_SUPPORT	## E5a. Bounded support at a fixed order	proven	—
GRAM_NULL_OPERATOR_QUOTIENT	## E5b. Gram-null decoupling	proven	—
DECORATED_HISTORY_MERGING	## E5c. Exact merging of decorated histories	proven	—
NESTED_QUOTIENT_REDUCTION	## E5d. Finite-order nested-quotient spectral reduction	proven	—
SHELL_SYMMETRY_RITZ_CONTROL	## E6a. Symmetry and residual control for a reduced shell	proven	—
B6_RETAINED_KRYLOV_CAMPAIGN	## E6b. Retained B6 numerical campaign	conditional	—
ISOLATED_POSITIVE_MATRIX_ATOM	## E7. Isolated positive matrix atom persists	proven	—
CONDITIONAL_SPECTRAL_FLOOR	## E8. Conditional spectral floor and defect contraction	proven	—
D4_DISJOINT_STAPLE_COORDINATES	## E9. Six disjoint staple coordinates in four dimensions	proven	—
POSITIVE_SECTOR_PHASE_ISOLATION	## E10. Positive-sector phase isolation	proven	—
VSU_BOUNDED_DOMAIN_WELLPOSEDNESS	## E11. Convex VSU action on a bounded domain	proven	—
DETERMINANT_PARITY_REPAIR	## E12. Determinant reduction with the missing parity restored	proven	—

Statement	Locator	Status	Formal
MOVING_ADJOINT_FORCE_DERIVATIVE	## E13. A moving adjoint force has two derivative terms	proven	—
SPHERE_EQUAL_COUPLING_COVARIANCE	## E14. Equal-coupling SU(3) faces on a sphere have nonnegative covariance	proven	—
SU3_POSITIVITY_BOUNDARIES	## E15. Exact SU(3) boundaries of the covariance argument	proven	—
REFLECTION_ADAPTED_BLOCKING	## E16. Reflection-adapted deterministic blocking preserves positivity	proven	—
CORNER_BLOCKING_REFLECTION_DEFECT	## E17. Literal corner paths fail the required reflection identity	proven	—
DIPOLAR_REMAINDER	## E18. Explicit remaining application obligations	open	—
PHYSICAL_CARRIER_REALIZATION	## E18. Explicit remaining application obligations	open	—
STAPLE_TRANSVERSALITY_REPAIR	## E18. Explicit remaining application obligations	open	—
VSU_WHOLE_SPACE	## E18. Explicit remaining application obligations	open	—
OMITTED_DETERMINANT_PARITY	## E12. Determinant reduction with the missing parity restored	falsified	—
OMITTED_OWN_FORCE_DERIVATIVE	## E13. A moving adjoint force has two derivative terms	falsified	—

CITE:HODGE_POLYMER_SUPPORTED

docs/derivations/hodge-polymer-supported-statements.md

SHA-256 e0221c892ff185e7...

Statement	Locator	Status	Formal
H1	## H1: complementary kernels from Hodge relations (from line 17)	proven	full
H2	## H2: carrier action and absence of Hodge coupling (from line 33)	proven	full
H3	## H3: rank-one normalization, excitation and factorization (from line 45)	proven	full
H4	## H4: finite word patterns and supplied polynomial identities (from line 71)	proven	full
P1	## P1: geometric-series majorant (from line 93)	proven	full
P2	## P2: parameter bounds and the separate tree budget (from line 107)	proven	full

CITE:LOCAL_CLASS_WICK_SPECTRUM

docs/derivations/local-class-wick-spectrum.md

SHA-256 5cde9ee4fe88bd47...

Statement	Locator	Status	Formal
RECURRENCE	## 3. All-order constructive theorem	proven	—
COEFFICIENTS	## 4. Recovered coefficients and the next odd term	proven	—
SIGNS	## 4.2 Sign corollary	proven	—

CITE:MOVING_TIME_SPECTRAL_GAP

docs/derivations/moving-time-spectral-gap.md

SHA-256 d0a20c25e023bff7...

Statement	Locator	Status	Formal
PROJECTION	## 1. Premises and the finite-cutoff projection estimate (MT1) (lines 17-43)	proven	—
VAGUE_LIMIT	## 2. Moving-time exclusion and totality (MT2) (lines 44-91)	proven	—
SHRINKING_TILTS	## 3. Shrinking exponential sources without a Taylor rate penalty (MT3) (lines 92-126)	proven	—
OPTIMAL_BUDGET	## 4. Optimal power-law budget and finite observation horizon (MT4) (lines 127-179)	proven	—
SHARPNESS	## 5. Sharpness and falsifiers (MT5) (lines 180-234)	proven	—

CITE:OS_KERNEL_GAP

docs/derivations/os-kernel-moving-time-gap.md

SHA-256 d7088d36f67a1f8c...

Statement	Locator	Status	Formal
KERNEL	## K1. Construct the physical semi-group from compatible limiting kernels (lines 10-61)	proven	—
LATE_TIME	## K2. A single late-time bound controls every limiting fixed time (lines 62-110)	proven	—

Statement	Locator	Status	Formal
WEIGHT	## K3. Nontrivial finite-energy weight from one separated-time lower bound (lines 111-132)	proven	—
LIMITATIONS	## K4. Falsifiers and actual Wilson application check (lines 133-175)	proven	—
POSITIVE_HISTORY	## K5. Positive-time history separation supplies zero-time continuity (lines 176-221)	proven	—

CITE:PBH_WILSON_TEST

docs/derivations/yangmills-pbh-wilson-test.md

SHA-256 3122ef912820b5a0...

Statement	Locator	Status	Formal
PBH1_PBH3	## PBH-1. The actual measure and horizontal form (lines 8-57)	proven	—
PBH4	## PBH-2. Explicit regular-point counterexample (lines 58-113)	proven	—
PBH5	## PBH-3. The obstruction is not confined to a one-plaquette lattice (lines 114-163)	proven	—
PBH6	## PBH-4. Classical Wilson weight is not the quantum ground-state weight (lines 164-198)	proven	—
FLOW_COUNTEREXAMPLE	## PBH-5. The stated gradient-flow shortcut also fails an exact test (lines 199-230)	proven	—

CITE:REVIEW_SPECTRAL_BUDGETS

docs/derivations/review-derived-spectral-budgets.md

SHA-256 96037da9bfc02943...

Statement	Locator	Status	Formal
Q1	## Q1. The assembled fourth-order polynomial is coercive (lines 11-42)	proven	—
M1	## M1. Several decay channels and time-amplified remainders (lines 43-89)	proven	—
M2	## M2. Closed optimal rate and a witnessing observation time (lines 90-125)	proven	—

Statement	Locator	Status	Formal
M3	## M3. Sharpness, failure cases and physical scope (lines 126-171)	proven	—

CITE:SEPTEMBER10_UPSTREAM_SCALARS

docs/derivations/september10-upstream-scalar-inventory.md

SHA-256 75a6b78ebaed289f...

Statement	Locator	Status	Formal
KP_SCALARS	## U1 Tilted KP scalar ingredients	proven	partial
COERCIVITY_SCALARS	## U2 Coercivity scalar ingredients	proven	partial

CITE:SEPT_DOC_YANGMILLS_FORMAL_BRIDGES

docs/derivations/yangmills-formal-bridges.md

SHA-256 bb3245e82e5dc857...

Statement	Locator	Status	Formal
F1	## Exact finite ground-state identification (lines 14-78)	proven	full
F2	## Exact finite ground-state identification (lines 14-78)	proven	full
F3	## Exact finite ground-state identification (lines 14-78)	proven	full
F4	## A trial residual needs a floor and a mean (lines 79-101)	proven	full
F5	## The pure quartic Gaussian certificate (lines 102-137)	proven	full
F6	## The pure quartic Gaussian certificate (lines 102-137)	proven	full
F7	## Algebra for Simon's transverse mechanism (lines 138-169)	proven	full
SLICE_RADIAL_ALGEBRA	## Algebra for Simon's transverse mechanism (lines 138-169)	proven	—

CITE:SEPT_DOC_YANGMILLS_RESOLVENT_LOCALIZATION_AND_DIRICHLET_GAP

docs/derivations/yangmills-resolvent-localization-and-dirichlet-gap.md

SHA-256 876b76c1aee3eb8e...

Statement	Locator	Status	Formal
SUBMITTED_W6_CLOSURE	## 1. Summary of Results (lines 20-39)	disputed	—
RESOLVENT_IDENTITY	### 2.1 The Resolvent Identity (lines 42-54)	conditional	partial
LOCAL_KERNEL_IMPLICATION	### 2.3 Volume-Uniformity of the Pairing (lines 62-80)	conditional	partial
GROUND_TRANSFORM	### 3.2 True Ground Conjugation and Exact Cancellation (lines 104-123)	proven	—
SUBMITTED_CLOSURES	## 4. Status and Graph Integration (lines 131-135)	superseded	—

CITE:SOURCE_CURRENT_SPECTRAL_BRIDGES

docs/derivations/source-currents-and-spectral-totality.md

SHA-256 860c81a3fb665d6b...

Statement	Locator	Status	Formal
SCB0A	## SCB0A. Degenerate energy variational certificate (lines 21-34)	proven	partial
SCB0B	## SCB0B. Bounded-inverse W6 composition (lines 35-59)	proven	partial
SCB1	## SCB1. The complete first current in the actual square convention (lines 60-114)	proven	—
SCB2	## SCB2. Exact current moments and an all-source constant (lines 115-172)	proven	partial
SCB3	## SCB3. A selected-inverse implication without a separate diagonal bound (lines 173-197)	proven	partial
SCB4	## SCB4. Intrinsic score currents and exact centering corrections (lines 198-257)	proven	—
SCB5	## SCB5. Tiny exponential sources imply totality by differentiation (lines 258-298)	proven	partial
SCB6	## SCB6. Uniform normalized mesoscopic source amplitudes (lines 299-316)	proven	—

CITE:THEORY_GEOMETRY_RICCATI_BRIDGES

docs/derivations/theory-geometry-and-riccati-bridges.md

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Statement	Locator	Status	Formal
TG1	## TG1: The sharp three-coordinate anisotropy constant (lines 14-60)	proven	partial
TG2	## TG2: Zero extension and integrated energy bounds (lines 61-93)	proven	—
TG3	## TG3: One secular cubic for the recorded Hodge support (lines 94-136)	proven	partial
TG4	## TG4: The induced eighth-order moment in the same truncation (lines 137-166)	proven	partial
TG5	## TG5: Complete source motion is a Schur congruence (lines 167-222)	proven	partial
TG6	## TG6: The metric and finite source resolvent must travel with the form (lines 223-274)	proven	partial
TG7	## TG7: Scalar square-source parity eliminates formal odd Schur jets (lines 275-313)	proven	—
TG8	## TG8: A residual certificate for the abstract Riccati fixed point (lines 314-350)	proven	partial

CITE:TRACK_A_SUPPORTED

docs/derivations/track-a-supported-statements.md

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Statement	Locator	Status	Formal
W6_V	## W6-V: actual variational residual identity and dual energy (from line 18)	proven	full
W6_F	## W6-F: bounded-operator factorization controls the residual functional (from line 43)	proven	full
W6_E	## W6-E: sufficient selected-inverse error certificate (from line 61)	proven	full
W6_C	## W6-C: exact Gaussian coefficient substitutions (from line 76)	proven	full
SC17_R	## SC17-R: Riccati roots and the continuous barrier (from line 92)	proven	full
SC17_B	## SC17-B: constructed noncommutative Banach-algebra fixed point (from line 112)	proven	full
SC17_P	## SC17-P: exact interval arithmetic under the proposed inputs (from line 132)	proven	full

Statement	Locator	Status	Formal
G17_P	## G17-P: corrected raw partition estimate for actual probability measures (from line 151)	proven	full
G17_V	## G17-V: centered L2 source, variance, and bounded footprints (from line 168)	proven	full
G17_O	## G17-O: strict obstruction to a constant equal to one (from line 185)	proven	full
G17_I	## G17-I: independent product sources have unbounded radius (from line 206)	proven	full

CITE:UNIVERSAL_CELLULAR_HODGE_TETRAHEDRAL

docs/derivations/universal-cellular-hodge-tetrahedral.md

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Statement	Locator	Status	Formal
SPECTRUM_EIGENVALUE	### 2.2 Universal spectral identities (lines 45-58)	proven	partial
UP_HARMONIC_EXCURSION	### 2.3 Universal Up-Harmonicity of Excursions (lines 59-67)	proven	full
TOTAL_LAPLACIAN_DIAGONAL	### 2.4 Total Laplacian Diagonal & Regularity Criterion (lines 68-83)	proven	—
TETRAHEDRAL_S4_COMMUTANT	### 3.1 Tetrahedral Hodge duality and 3.2 The S4 commutant (lines 86-119)	proven	partial
PROPER_RETURN_Q_ANNIHILATION	### 3.4 Open physical-history identification (lines 133-156)	open	partial
RETAINED_VECTOR_RANK_ONE	### 3.3 Feshbach Intermediate Annihilation of Retained Vectors (lines 120-132)	proven	full
RSMR_MASTER_THEOREM	### 4.3 The carrier symbol (lines 196-221)	proven	partial
SHIFTED_OPERATOR_POWERS	### 4.2 Polynomial recurrence and operator powers (lines 172-195)	proven	full

CITE:W6_ANTIPODAL_MAGNETIC_GEOMETRY

docs/derivations/w6-antipodal-magnetic-geometry.md

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Statement	Locator	Status	Formal
HESSIAN_SPECTRUM	A1-A6; exact fixed-Q chart, quadratic form and full spectrum	proven	—
GAUGE_NORMAL_COERCIVITY	A7-A11; complete minimum sphere, tangent kernel and original electric metric	proven	—
ANGULAR_REDUCTION	A12-A13; exact old-orbit restriction and compact normal-bundle implicit function	proven	—
GAUGE_SCORE_INVARIANCE	A14-A15; actual ground and Haar half-density symmetry	proven	—

CITE:W6_CONDITIONAL_SCORE_TAIL_CONTROL

docs/derivations/w6-conditional-score-tail-control.md

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Statement	Locator	Status	Formal
INTERVAL_OBSTRUCTION_REPAIR	## 1. What the actual compact end-points do prove (lines 67-284)	proven	—
SCORE_DOMINATION_M10	### M.4. The precise remaining conditional-score inequality (lines 473-495)	open	—
ACTUAL_SOURCE_MOMENT	## 6. Uniform actual source potential moment and sharper score criterion (lines 325-586)	proven	—
SCORE_WEIGHT_FROM_M10	### M.4. The precise remaining conditional-score inequality and M.5 (lines 497-563)	conditional	—

CITE:W6_CONDITIONAL_TRANSPORT_OBSTRUCTION

docs/derivations/w6-conditional-transport-obstruction.md

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Statement	Locator	Status	Formal
SCORE_CONTINUITY	## S1 Exact conditional score (lines 15-72)	proven	—
AGMON_GEOMETRY	## S2 Exact constrained geometry (lines 73-140)	proven	—
CONDITIONAL_CONCENTRATION	## S3 Actual conditional concentration (lines 141-213)	proven	—
ARBITRARY_Q8_OBSTRUCTION	## S4 A transport obstruction (lines 214-284)	proven	—

Statement	Locator	Status	Formal
SYNCHRONIZED_TANGENCY	## S5 Synchronized tangency (lines 285-316)	proven	—
LOCAL_GLOBAL_SCORE_CRITERION	## S6 Remaining all-fiber estimate (lines 317-375)	conditional	—
M10_SYNCHRONIZED_SCORE	## S6 Remaining all-fiber estimate (lines 317-375), explicitly open synchronized M10 target	open	—

CITE:W6_GROUND_JETS_TRANSPORT_BUDGET

docs/derivations/w6-ground-jets-and-transport-budget.md

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Statement	Locator	Status	Formal
GROUND_JETS	## 2. Raw electric and magnetic derivatives already have the required scaling (lines 51-198)	proven	—
SOURCE_ENERGY_JETS_R10	## 4. An explicit bridge from the actual transport generator to M1,M2,M3 (lines 199-218); section 5 (lines 271-309)	open	—
COMPLETE_BUDGET	## 4. An explicit bridge from the actual transport generator to M1,M2,M3 (lines 199-367)	conditional	—
INTERACTING_GRID_COMPARISON	## 6. Source energies, subdivision, interacting grids, and the physical clock (lines 351-366)	open	—

CITE:W6_SOURCE_ENERGY_JETS_R10

docs/derivations/w6-source-energy-jets-r10.md

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Statement	Locator	Status	Formal
H0_ENERGY_CONTRACTION	## 6. Scaling defects and unresolved estimates (C6)	open	—
H1_FIBERWISE_SCORE_VARIANCE	## 6. Scaling defects and unresolved estimates (C6)	open	—
VACUUM_CROSS_DERIVATIVES	## 6. Scaling defects and unresolved estimates (C6)	open	—
KATO_DERIVATIVES	## 6. Scaling defects and unresolved estimates (C6)	open	—

Statement	Locator	Status	Formal
CONDITIONAL_DERIVATIVE	## 2. Correct conditional differentiation (C1)	proven	—
HORIZONTAL_SCORE_SUFFICIENT	## 3. A sufficient horizontal-score estimate for H0 (C3)	proven	—
H4_FROM_H0	## 4. H4 follows from H0 and the established ground jet (C4)	proven	—
PROJECTION_JET_RECURRENCE	## 5. Complete projection-jet recurrence (C5)	proven	—
SCALING_OBSTRUCTIONS	## 6. Scaling defects and unresolved estimates (C6)	proven	—

CITE:W6_SOURCE_GENERATOR_SCORE_FRAME

docs/derivations/w6-source-generator-score-frame.md

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Statement	Locator	Status	Formal
GROUND_FRAME_GENERATOR	## 2. The ground-state frame (SF1-SF2) and ## 3. The complete R9 generator in the ground-state frame (SF3-SF5)	proven	—
SCORE_POISSON	## 4. The score Poisson equation (SF6-SF7)	proven	—
FIBER_ENERGY_IDENTITY	## 4. The score Poisson equation (SF6-SF7), conditional divergence identity and fiber identity	proven	—
R10_ORDER_ZERO_REDUCTION	## 5. Exact reduction of the order-zero R10 bound (SF8-SF9)	conditional	—
CONDITIONAL_ENERGY_H0_H4	## 5. Exact reduction of the order-zero R10 bound (SF8-SF9), conditional-energy hypotheses	open	—

CITE:W6_SYNCHRONIZED_M10_DOMINATION

docs/derivations/w6-synchronized-m10-domination.md

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Statement	Locator	Status	Formal
TANGENCY_DRIFT_CANCELLATION	## 2. Synchronized tangency (lines 34-51)	proven	—
TUBE_VARIANCE_DOMINATION	## 3. Small-angle tube implication (lines 52-84)	proven	—

Statement	Locator	Status	Formal
ANTIPODAL_SCORE_REGULARITY	## 4. Antipodal geometry and a uniform angular reference calculation (lines 85-143)	proven	—
AMPLITUDE_GRADIENT_SCALING	## 5. Correct parameter-derivative criterion (lines 144-171)	open	—
OUTSIDE_COMPLEMENT_SUPPRESSION	## 6. Correct complement argument and compact action gap (lines 172-261)	proven	—
FULL_SCORE_DOMINATION_M10	## 7. Assembly criterion and actual status (lines 262-286)	open	—
LOG_PARAMETER_CRITERION	## 5. Correct parameter-derivative criterion (lines 144-171)	proven	—
COMPACT_ACTION_GAP	## 6. Correct complement argument and compact action gap (lines 172-261)	proven	—
NORMALIZED_COMPLEMENT_CRITERION	## 6. Correct complement argument and compact action gap (lines 172-261)	proven	—
ANGULAR_REFERENCE_BUDGET	## 4. Antipodal geometry and a uniform angular reference calculation (lines 85-143)	proven	—
ASSEMBLY_CRITERION	## 7. Assembly criterion and actual status (lines 262-286)	proven	—
ACTUAL_WEIGHTED_JET_AND_LOWER_BOUND	## 6. Correct complement argument and compact action gap (lines 172-261)	proven	—

CITE:WILSON_BACKGROUND

docs/derivations/wilson-background-covariance-continuation.md

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Statement	Locator	Status	Formal
BF1_BF3	## BF1. The energy-dependent relative-form statement (lines 48-102)	conditional	partial
BF4_BF5	## BF4. An actual Wilson small-field estimate, uniform in volume (lines 103-151)	proven	—
BF6	## BF6. Centering at an actual constrained minimizer (lines 152-188)	proven	—
SC1_SC4	## 1. Setup and exact equations (lines 282-343)	proven	—
SC5_SC9	## 2. A gauge-orbit gap that does not assume the physical gap (lines 344-415)	proven	—

Statement	Locator	Status	Formal
SC10_SC15	## 3. Uniform pointwise first and mixed derivatives at every coupling (lines 416-468)	proven	—
SC16	## 4. Direct finite-time form, without an implicit long-time interchange (lines 469-493)	proven	—
SC17	## 5. The neutral channel and the exact remaining spatial bound (lines 494-534)	conditional	—
BC1	## BC1. Why a fast inverse alone cannot finish the transport (lines 535-544)	proven	—
BC2_AUDIT	## BC2. The additional multiscale document is evidence, not a completed input (lines 545-569)	proven	—

CITE:WILSON_MARKED

docs/derivations/wilson-marked-transfer.md

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Statement	Locator	Status	Formal
WT1	## WT-1. Exact physical-time blocks (lines 14-49)	proven	—
WT2	## WT-2. An all-orders activated-block bound (lines 50-88)	proven	—
WT3	## WT-3. Configurations and exact vacuum cancellation (lines 89-121)	proven	—
WT4	## WT-4. A uniform convergent vacuum polymer expansion (lines 122-171)	proven	—
WT5	## WT-5. Uniform control of free propagation and marked contour denominators (lines 172-208)	proven	—
WT6	## WT-6. The remaining complete-shell estimate and conditional transport (lines 209-248)	open	—

CITE:WILSON_SC17

docs/derivations/wilson-sc17-spatial-closure.md

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Statement	Locator	Status	Formal
REANCHORING	## Endpoint reanchoring removes the neutral obstruction for norm bounds (lines 54-65)	proven	—
CHARGED_EVOLUTION	## Why the charged contraction survives the time-dependent drift (lines 66-105)	proven	—
H1_H2	## Actual weighted Volterra inequality (lines 106-131)	proven	—
SMALL_WINDOW	## Explicit new regime (lines 132-166)	proven	—
BARE_MAJORANT_WALL	## Exact wall for this majorant (lines 167-189)	proven	—
CURVATURE	## Actual-vacuum curvature corollary (lines 190-212)	proven	—
ANGLES	## Sharper projection angles via covariance and the derived conditional gaps (lines 213-256)	proven	—
OPTIMIZED_INTERVAL	## Maximized bare-Hessian bootstrap: a larger interval (lines 257-298)	proven	—
RATIONAL_WINDOW	### Explicit rational larger-window corollary (lines 299-320)	proven	—
GAUSSIAN_PRECISION	## 1. The actual flat quadratic ground precision (lines 323-358)	proven	—
FRACTIONAL_KERNEL	## 2. Exact lattice fractional kernel and a sufficient weighted norm (lines 359-395)	proven	—
GAUSSIAN_ANGLE	## 3. An exact scalar Gaussian angle test (lines 396-422)	proven	—
FAST_PRECISION_CUSP	## 4. Exact actual averaged-path fast conditional precision, ell=2 (lines 423-466)	proven	—
R4A	## 6. The existing dynamic covariance supplies the flat reference budget (lines 489-534)	proven	—
R1_R3	## 1. Exact Euclidean equation and its projected cross terms (lines 537-573)	proven	—
R4_R9	## 2. A rigorous conditional small-defect criterion (lines 574-648)	conditional	partial
R10_R11	## 3. An explicit true-angle consequence if the full Hessian defect is known (lines 649-697)	conditional	—
R12_R14	## 4. The actual quantum pressure and cutoff defects are part of the input (lines 698-756)	conditional	—

Statement	Locator	Status	Formal
ACTUAL_REFERENCE_DEFECT_BOUND	## 4. The actual quantum pressure and cutoff defects are part of the input (lines 698-756), with the R5-R7 budget in section 2	open	—
R15	## 5. The averaged-path cusp is an analytic compression, not an eigenvalue crossing (lines 757-780)	proven	—
PROPOSED_QUANTUM_PARAMETER_WINDOW	## 6. Formal repair of the spatial closure for $\lambda \geq 1/73$ via the quantum reference defect (lines 781-821)	conditional	partial

CITE:WILSON_SC17_LIMIT

docs/derivations/wilson-sc17-thermodynamic-limit.md

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Statement	Locator	Status	Formal
T1	## T1. Uniform setting including an actual cut interpolation (lines 28-95)	proven	—
T2_T3	## T2. Weighted gradient contraction from the actual charged equations (lines 96-131)	proven	—
T4_T10	## T3. The normalized actual vacuum response has no boundary-area loss (lines 132-195)	proven	—
T11	## T4. Actual exponential covariance localization (lines 196-210)	proven	—
T12	## T5. Fixed-lambda thermodynamic Cauchy convergence (lines 211-264)	proven	partial
T13	## T7. Endpoint continuity also gives a thermodynamic limit at λ_c (lines 280-311)	proven	—
SCORE_LIMIT	## Uniform score limits discharge the form construction's local input (lines 312-337)	proven	partial
IF3	## 2. Exact infinite-volume integration by parts (lines 371-392)	proven	partial
IF4_CLOSABLE	## 3. Closability and the canonical closed form (lines 393-436)	proven	partial
IF4_IF7	## 3. Closability and the canonical closed form (lines 393-436)	proven	partial
PHYSICAL_RESTRICTION	## 4. Gauge invariance and the physical closed subspace (lines 437-460)	proven	—

Statement	Locator	Status	Formal
IF8_LIMIT	## 5. Passing the full and physical Poincare floor (lines 461-479)	proven	partial
IF8_IF9	## 5. Passing the full and physical Poincare floor (lines 461-479)	proven	partial
IF10_IF13	## 6. Actual nonzero local observable overlap with a bounded energy band (lines 480-528)	proven	partial

CITE:WILSON_SC17_TIME

docs/derivations/wilson-sc17-physical-time-limit.md

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Statement	Locator	Status	Formal
P4_P6	## P2. A limiting influence kernel and weighted Lipschitz estimate (lines 75-133)	proven	—
P7_P13	## P3. A global extension with exact sphere invariance (lines 134-220)	proven	—
P14	## P4. Uniform Hilbert convergence of the actual finite drifts (lines 221-258)	proven	—
P15	## P5. Strong finite-time process comparison (lines 259-293)	proven	—
P16	## P6. Stationary path laws and actual multitime vacuum correlations (lines 294-332)	proven	—
P17_P18	## P7. Identification with the already closed gradient form (lines 333-440)	proven	—
PHYSICAL_TIME_GAP	## P8. Physical restriction and the fixed-spacing time limit (lines 441-470)	proven	—
P19_P20	## P9. Endpoint extension by a summable nonexponential weight (lines 471-525)	proven	—

CITE:WILSON_SELECTED

docs/derivations/wilson-selected-inverse-wall.md

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Statement	Locator	Status	Formal
W1_W3_OBSTRUCTION	## 2. First attempted step: global relative Gaussian fast-form control (lines 33-100)	proven	—
W4_W5	## 3. Repair: only compare inverses on the graph force (lines 101-157)	conditional	partial
SIGNED_PAIRING_OBSTRUCTIONS	## 4. Attempt to establish the repaired estimate for Wilson (lines 158-220)	proven	—
W6	## 5. The exact point at which this argument cannot proceed (lines 221-264)	open	partial

CITE:WILSON_SHELL

docs/derivations/wilson-marked-shell-transport.md

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Statement	Locator	Status	Formal
S2_S7	## 2. The expansion with a fixed observable (lines 76-169)	proven	—
S8_S9	## 3. Complex anchored operator bound (lines 170-218)	proven	—
S10_S15	## 4. Actual complete shell and a holomorphic literal-source frame (lines 219-311)	proven	—
S16_S17	## 5. From finite-order locality to a uniform weighted Taylor bound (lines 312-390)	proven	—
S18_S20	## 6. Sum the matching and transport the carrier (lines 391-461)	conditional	—
S21_S22	## 7. Observable overlap, with the quantifiers retained (lines 462-493)	proven	—

CITE:WILSON_SPATIAL

docs/derivations/wilson-spatial-schur-excess.md

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Statement	Locator	Status	Formal
SP1_SP6	## 2. Exact excess about the full reference graph (lines 42-114)	conditional	partial

Statement	Locator	Status	Formal
SP7_SP10	## 3. The complete second-order coefficient and a uniform remainder (lines 115-182)	conditional	partial
SP11_SP16	## 4. What W_1 and W_2 mean for the actual Wilson operator (lines 183-306)	proven	—
SP17_SP19	## 5. Second-order literal-source normalization and harmonic control (lines 307-411)	proven	—
SP20_SP24	## 6. Iterating energies and observable amplitudes in the physical clock (lines 412-477)	conditional	—
SP25	## 7. Why the complete second-order subtraction matters (lines 478-519)	proven	—

CITE:WILSON_TRUE_BLOCK

docs/derivations/wilson-true-vacuum-block-estimates.md

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Statement	Locator	Status	Formal
BA1	## BA1. A specified block partition (lines 10-28)	proven	—
BA2_BA5	## BA2. Optimizing the true conditional gap (lines 29-94)	proven	partial
BA6_BA8	## BA3. Reduce the actual projection angle to a mixed ground-amplitude ratio (lines 95-141)	proven	partial
BA9_BA10	## BA4. What a frozen-block spectral computation would additionally require (lines 142-168)	conditional	—
BA11_BA12	## BA5. Exact finite-time derivation of the connected quantity (lines 169-201)	proven	—
BA13_BA16	## BA6. The first connected coefficients for the actual coupled lattice (lines 202-269)	proven	—
BA20_BA26	## BA7. Actual angle closure in an explicit strong-coupling window (lines 270-405)	proven	—

CITE:WILSON_TRUE_BLOCK_CHECK

docs/derivations/wilson-true-blocks-independent-check.md

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Statement	Locator	Status	Formal
ENVELOPE_AND_CROSS_RATIO	## CB1. What was re-derived independently (lines 17-56)	proven	—
CB1	## CB2. The single-link partition is stronger, and closes VA12 directly (lines 57-75)	proven	—
FREE_BUDGET_LOOSENESS	## CB3. The angle budget is not graded in lambda (lines 76-87)	proven	—
CB2	## CB4. The BA26 wall is more robust than its own derivation (lines 88-109)	proven	—
NONCHAINING	## CB5. The two boxed constants must not be chained (lines 110-121)	proven	—

CITE:WILSON_VACUUM_ASSEMBLY

docs/derivations/wilson-vacuum-aligned-assembly.md

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Statement	Locator	Status	Formal
VA1_VA4	## VA1. A local finite-energy bound for the actual quantum vacuum (lines 17-74)	proven	partial
VA5_VA7	## VA2. The WR25 remainder is extensive on actual connected lattices (lines 75-130)	proven	partial
VA8	## VA3. WR26 is false on the physical vacuum complement at large volume (lines 131-164)	proven	—
VA9_VA11	## VA4. Exact cancellation before comparison (lines 165-210)	proven	partial
VA12_VA13	## VA5. Projection geometry, with the Gauss law retained (lines 211-251)	conditional	partial
VA14_VA19	## VA6. The next estimate and the attempts to establish it (lines 252-415)	conditional	partial

CITE:WILSON_WEIGHTED_REPAIR

docs/derivations/wilson-weighted-repair-and-rotor-gap.md

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Statement	Locator	Status	Formal
WR1_WR3	## WR1. Absorb negative curvature in the derivative energy (lines 15-66)	proven	—
WR4_WR6	## WR2. A smooth non-Gaussian trial for the physical rotor (lines 67-99)	proven	—
WR7_WR12	## WR3. A physical gap on the complete rotor space, at every coupling (lines 100-161)	proven	—
ROTOR_VOLUME	## WR4. Volume passage that is already justified (lines 162-175)	proven	—
WR13_WR16	## WR5. Actual shared-link residual, and a positive interacting repair (lines 176-228)	proven	—
WR17_WR21	## WR6. The precise remaining derivation (lines 229-308)	conditional	—
WR22_WR24	## WR7. Retain the full boundary: an actual-link fast barrier (lines 309-356)	proven	—
WR25_WR28	## WR8. Where the repaired local bounds still require a new proof (lines 357-443)	proven	—

CITE:YM_BALABAN_MULTISCALE

docs/derivations/yangmills-continuum-balaban-multiscale-proof.md

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Statement	Locator	Status	Formal
LEMMA_3_2_ADJOINT	### 3.1 Geometric Covariant Averaging Operator Q_k (lines 84-110)	proven	—
PROPOSITION_3_4	### 3.2 The Background Field Variational Problem (lines 111-139)	disputed	—
THEOREM_4_2	### 4.2 Super-Exponential Suppression of Large Fields (lines 157-180)	conditional	—
GAUGE_HESSIAN	### 5.1 Fluctuation Gauge Fixing (lines 183-197)	disputed	—
GHOST_5_8	### 5.2 Faddeev–Popov Ghost Determinant (lines 198-208)	conditional	—
THEOREM_5_1	### 5.3 Combes–Thomas Resolvent Localization for Continuous $\mathcal{H}_{B_k}$ (lines 209-248)	disputed	—
BANACH_ALGEBRA	### 6.2 The Polymer Banach Algebra $\mathcal{E}_k$ (lines 260-268)	proven	—

Statement	Locator	Status	Formal
THEOREM_6_2	### 6.3 The Mayer Tree-Graph Expansion & Cluster Contraction (lines 269-295)	conditional	—
THEOREM_6_3	### 6.4 Asymptotic Freedom and Infinite Scale Stability (lines 296-314)	conditional	—
THEOREM_7_1	### 7.1 Cauchy Sequence of Effective Actions (lines 317-340)	open	—
THEOREM_7_2	### 7.2 Construction of the Limiting Continuum Measure μ_{YM} (lines 341-363)	conditional	—
OS0	### 8.1 OS0: Analyticity and Temperedness (lines 368-374)	conditional	—
OS1	### 8.2 OS1: Euclidean Covariance (lines 375-382)	open	—
THEOREM_8_1	### 8.3 OS2: Reflection Positivity (lines 383-408)	conditional	—
OS3	### 8.4 OS3: Permutation Symmetry (lines 409-411)	conditional	—
THEOREM_8_2	### 8.5 OS4: Exponential Clustering and Mass Gap (lines 412-436)	open	—
OS_RECONSTRUCTION	### 9.1 The Physical Hilbert Space and Hamiltonian (lines 439-453)	conditional	—
THEOREM_9_1	### 9.2 The Physical Spectral Mass Gap (lines 454-478)	conditional	—
SCATTERING	### 9.3 Relativistic Wightman Fields and Non-Trivial Scattering ($\neq I$) (lines 479-501)	conditional	—

CITE:YM_FLAT_DIRECTIONS

docs/derivations/yangmills-simon-flat-directions.md

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Statement	Locator	Status	Formal
OSCILLATOR	## 1. The scalar oscillator, including its constant (lines 102-138)	proven	—
TRANSVERSE	## 2. SU(2) has two transverse oscillator modes per slice (lines 139-167)	proven	—
SLICE_AGGREGATION	## 3. Allocate the pair potential without losing longitudinal control (lines 168-212)	proven	—
COMPACT_RESOLVENT	## 4. Compact resolvent follows from the retained gradient and radial tail (lines 213-233)	proven	—

Statement	Locator	Status	Formal
GROUND_FLOOR	## 5. An explicit ground-energy lower bound (lines 234-275)	proven	—
JACOBIAN_VALLEY	## 6. What Simon's Jacobian condition measures (lines 276-324)	proven	—
SCALING	## 7. Energy scale, vacuum subtraction and the research gaps (lines 325-368)	proven	—

CITE:YM_GPU_AUDIT

docs/derivations/yangmills-gpu-resolution-audit.md

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Statement	Locator	Status	Formal
GA1_GA4	## GA1. Phase 1 is a massive scalar model with an exact uniform theorem (lines 24-76)	proven	—
GA5_GA7	## GA2. Phase 2's alleged domino is an exact tensor sum (lines 77-129)	proven	—
RADIAL_DISCRETIZATION	## GA3. Phase 3 compares two radial discretizations, not a transfer matrix (lines 130-161)	proven	—
GA8_GA9	## GA4. The additional Phase 4 has a valid conditional scalar limit (lines 162-209)	conditional	—
GA10	## GA5. Phase 5 keeps an admissible cube basis but changes its magnetic operator (lines 210-249)	proven	partial
GA11	## GA6. What the supplied Lean additions actually formalize (lines 250-286)	proven	—
GA12	## GA7. The incoming smearing script measures spatial power, not spectral overlap (lines 287-327)	proven	—
GA13	## GA8. The later G17 script inserts both its cluster coefficients and its mass (lines 328-363)	proven	—
GA14_GA19	## GA9. Phase 8 supplies an assumed block kernel, not conditional projections (lines 364-500)	proven	—
GA20_GA21	## GA10. Phase 9 verifies a Gram identity, an inserted exponential and a Haar moment (lines 501-563)	proven	partial

CITE:YM_GROUND_STATE

docs/derivations/yangmills-weighted-curvature.md

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Statement	Locator	Status	Formal
GST1	## GST-1: exact conjugation with a trial-state residual (lines 32-70)	proven	partial
GST2	## GST-2: weighted energy and the needed domain statement (lines 71-91)	proven	partial
GST3	## GST-3: curvature normalization and a spectral argument without a discrete excited state (lines 92-130)	conditional	partial
GST4	## GST-4: the finite matrix identity and its exact hypotheses (lines 131-146)	proven	full
GST5	## GST-5: oscillator calibration (lines 147-166)	proven	—
GST6	## GST-6: the residual-mean certificate (lines 167-190)	conditional	—
GST7	## GST-7: a complete algebraic certificate for the pure quartic oscillator (lines 191-244)	proven	—
GST8	## GST-8: falsifiers and the Yang–Mills boundary (lines 245-266)	proven	—
GST9	## GST-9: an exact volume-accumulation falsifier (lines 267-300)	proven	—

CITE:YM_RECONSTRUCTION

docs/derivations/yangmills-reconstruction.md

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Statement	Locator	Status	Formal
R1	## 1. Source statements and the operator to be bounded (lines 5-27)	proven	partial
R2	## 2. A finite reflection-positive model one can inspect exactly (lines 28-38)	proven	full
R3	## 3. Observable completeness is a testable interface (lines 39-51)	proven	full
R4_R5	## 4. What a localization plateau actually proves (lines 52-89)	proven	partial
R6_R7	## 5. A precise sufficient repair, with its remaining hypothesis exposed (lines 90-111)	conditional	partial

E Statements that do not close

63 of the 332 registered statements are recorded as open, conditional, or disputed. They are extracted here with their hypotheses and the corpus’s own sentence on what remains, because this is the shortest complete list of places where an agent can do mathematics that the program has already established is missing.

Read alongside §17, which curates the subset the program judges decisive, and §18, which is about how to start on one.

63 of the registered derivation statements are not recorded as proven. Each is listed with the document it lives in, its locator, and the corpus’s own sentence on what remains. These are the statements an agent can act on directly.

DERIV:EARLIER_PROOF_RECOVERY:DIPOLAR_REMAINDER open

Establish the archived $O(|x|^{-4})$ remainder of the dipolar lattice kernel with the required Fourier-cutoff and derivative bounds.

docs/derivations/earlier-proof-recovery.md — ## E18. Explicit remaining application obligations

Hypotheses. The exact source assumptions and required estimates described in E18.

Remaining. Explicit pending application or extension; neither the exact finite controls nor the narrower recovered theorem discharge it.

DERIV:EARLIER_PROOF_RECOVERY:PHYSICAL_CARRIER_REALIZATION open

Construct actual cutoff sources, residue floor, collapsing isolated intervals, true-sector gap, physical clock and limiting identifications needed to apply the positive matrix atom theorem.

docs/derivations/earlier-proof-recovery.md — ## E18. Explicit remaining application obligations

Hypotheses. The exact source assumptions and required estimates described in E18.

Remaining. Explicit pending application or extension; neither the exact finite controls nor the narrower recovered theorem discharge it.

DERIV:EARLIER_PROOF_RECOVERY:STAPLE_TRANSVERSALITY_REPAIR open

Repair the force-rank and tube estimates with the full moving-adjoint derivative.

docs/derivations/earlier-proof-recovery.md — ## E18. Explicit remaining application obligations

Hypotheses. The exact source assumptions and required estimates described in E18.

Remaining. Explicit pending application or extension; neither the exact finite controls nor the narrower recovered theorem discharge it.

DERIV:EARLIER_PROOF_RECOVERY:VSU_WHOLE_SPACE open

Establish compatible normalization and uniform local estimates for the whole-space VSU limit, then justify the claimed far-field law.

docs/derivations/earlier-proof-recovery.md — ## E18. Explicit remaining application obligations

Hypotheses. The exact source assumptions and required estimates described in E18.

Remaining. Explicit pending application or extension; neither the exact finite controls nor the narrower recovered theorem discharge it.

DERIV:UNIVERSAL_CELLULAR_HODGE:PROPER_RETURN_Q_ANNIHILATION open

The 60 paths flagged by the submitted endpoint-or-zero flux predicate identify physical tetrahedral retained-sector returns that vanish individually under intermediate Q projection.

docs/derivations/universal-cellular-hodge-tetrahedral.md — ### 3.4 Open physical-history identification (lines 133-156)

Hypotheses. An actual Haar/Fierz Hilbert space, physical retained projector, insertion channels, and projected resolvent states matching the enumerated paths.

Remaining. Open: construct and identify the physical states/projector and calculate each projected contribution. The 60/36 flux count and $Q_{\psi=0}$ do not establish this identification.

DERIV:W6_CONDITIONAL_SCORE_TAIL_CONTROL:SCORE_DOMINATION_M10

open

For the actual compact square and specified source-preserving dilation, prove $K_g(w) \leq C_0 + C_1 W_g(w)$ almost everywhere in ν_g , where $W_g(w) = g^{-2} E_{\mu_g}(V|w)$, with finite nonnegative C_0, C_1 independent of $0 < g < g_*$. This sufficient conditional-score domination is open.

`docs/derivations/w6-conditional-score-tail-control.md` — ### M.4. The precise remaining conditional-score inequality (lines 473-495)

Hypotheses. Actual fixed twelve-edge, four-face compact Wilson square, true normalized quantum ground Ψ_g , $d\mu_g = \Psi_g^2 dU$, literal source $w = \text{Tr}(U_2 U_3)/2$ and its marginal ν_g ; $\sigma_g = (\partial_g \Psi_g + D \Psi_g/g)/\Psi_g$ uses the specified compact dilation of Q_8 , including its Haar divergence; $K_g = \text{Var}_{\mu_g}(\sigma_g|w)$ retains all compact corrections and rare coarse fibers.; The established actual fixed-square gap and ground-energy bounds hold on the specified small-coupling interval. Averaged conditional-score control is not assumed to imply pointwise domination.

Remaining. Prove M10 for the actual conditional law, or retain its open status if a different sufficient source-energy estimate is proved. The phase-adapted local implication requires whole-fiber tails, conditional-mean gluing and consistent global dilation before it can address this target.

DERIV:W6_CONDITIONAL_TRANSPORT_OBSTRUCTION:M10_SYNCHRONIZED_SCORE

open

The selected synchronized realization of the generic M10 target asks whether the actual score for the specified S13 compact field obeys $K_g(w) \leq C_0 + C_1 g^{-2} E_{\mu_g}[V|w]$ with finite nonnegative constants uniform in sufficiently small positive g and almost every retained trace w .

`docs/derivations/w6-conditional-transport-obstruction.md` — ## S6 Remaining all-fiber estimate (lines 317-375), explicitly open synchronized M10 target

Hypotheses. Actual positive four-face $SU(2)$ square ground and literal retained trace $w = \text{tr}(U_2 U_3)/2$.; The particular synchronized four radial profiles S13 and their full Haar unitary generator, not an arbitrary Q_8 field.; All conditional source regimes, including rare fibers and the antipodal degeneration, are retained in the requested uniform estimate.

Remaining. Prove the actual full conditional score bound, or a sufficient direct source-weighted pairing estimate. The original M11-M15 consequence then becomes applicable;

DERIV:W6_GROUND_JETS_TRANSPORT_BUDGET:INTERACTING_GRID_COMPARISON

open

For the actual interacting growing-grid family, obtain a volume-independent fast floor and complete source/transport energy estimates using linked or locally centered energy bounds, and control the coarse-theory matching discrepancy after the prescribed injection. The fixed-square constants depending on total ground energy E and the Gaussian grid floor do not prove this extension.

`docs/derivations/w6-ground-jets-and-transport-budget.md` — ## 6. Source energies, subdivision, interacting grids, and the physical clock (lines 351-366)

Hypotheses. Actual coupled Wilson grids and specified block/source injections, with true vacuum subtraction, common physical energy normalization and the required retained, fast and spectral-energy domains.; Constants must be independent of growing spatial volume in the intended small-coupling regime; the existing fixed-square E and γ are not assumed to have that uniformity.; Actual source-energy comparison factors and the discrepancy between the generated coarse form and prescribed coarse theory are retained; the Gaussian covariance-weighted source is not identified with the nonlinear quantum source without proof.

Remaining. Prove the interacting fast-floor and connected energy/source comparison in the specified grid family.

Controlling the local coupling remainder does not control coarse matching, the explicit physical-clock change in R14, or convergence of the exact transported jets and observables.

DERIV:W6_GROUND_JETS_TRANSPORT_BUDGET:SOURCE_ENERGY_JETS_R10

open

For the actual complete compact-square source/vacuum generator $A(g)$ in R9, prove $\|A^{\wedge(r)}(g)\|_{(q_g \rightarrow q_g)} \leq d_r g^{\wedge(-r-1)}$ for $r=0,1,2$ with finite constants independent of small positive g . This is the open model input R10, not the already proved conditional implication R10 to R12.

`docs/derivations/w6-ground-jets-and-transport-budget.md` — ## 4. An explicit bridge from the actual transport generator to M1,M2,M3 (lines 199-218); section 5 (lines 271-309)

Hypotheses. Actual fixed compact-square Hamiltonian and common electric H1 form domain; actual simple normalized positive ground, $0 \leq e_g \leq E$ and physical gap $\gamma > 0$ on $0 < g < g_*$; P_g is the complete true-ground literal-source projection from C1, and $A(g)$ is exactly the R9 source/vacuum generator; $q_g[u] = H_g[u] + \gamma \|u\|^2$, including vacuum components.; Parameter derivatives are taken at positive g on the fixed compact form domain. The target covers all source and fast energies, including spatial derivatives and generator derivatives through order two.

Remaining. Establish the conditional-source projection energy bounds beyond the proved vacuum-only generator. A direct proof of mixed-form R11 is an alternative sufficient route and need not prove R10 itself.

DERIV:W6_SOURCE_ENERGY_JETS_R10:H0_ENERGY_CONTRACTION

open

Establish the proposed bound $b_g[P_g \Phi] \leq q_g[\Omega_g \Phi]$ with $\kappa_P = 1$ on the actual model; the weaker uniform finite- κ_P H0 also suffices for R10.

`docs/derivations/w6-source-energy-jets-r10.md` — ## 6. Scaling defects and unresolved estimates (C6)

Hypotheses. Actual fixed compact square at positive g ; normalized positive ground, $q_g = H_g + \gamma I$, common form domain.; Constants must be independent of g on $0 < g < g_*$.

Remaining. The submitted proof is not a discharge. C1-C5 repair identities and conditional reductions;

DERIV:W6_SOURCE_ENERGY_JETS_R10:H1_FIBERWISE_SCORE_VARIANCE

open

Establish $\sup_w \text{Var}_{\mu_g}(\sigma_g|w) \leq \kappa_0 g^{\wedge-2}$ uniformly at small positive coupling.

`docs/derivations/w6-source-energy-jets-r10.md` — ## 6. Scaling defects and unresolved estimates (C6)

Hypotheses. Actual fixed compact square at positive g ; normalized positive ground, $q_g = H_g + \gamma I$, common form domain.; Constants must be independent of g on $0 < g < g_*$.

Remaining. The submitted proof is not a discharge. C1-C5 repair identities and conditional reductions;

DERIV:W6_SOURCE_ENERGY_JETS_R10:KATO_DERIVATIVES

open

Establish $\|\partial^r_g [P_g \text{prime}, P_g]\|_{(q_g \rightarrow q_g)} \leq d_{P,r} g^{\wedge(-r-1)}$, $r=0,1,2$, for the actual source projection.

`docs/derivations/w6-source-energy-jets-r10.md` — ## 6. Scaling defects and unresolved estimates (C6)

Hypotheses. Actual fixed compact square at positive g ; normalized positive ground, $q_g = H_g + \gamma I$, common form domain.; Constants must be independent of g on $0 < g < g_*$.

Remaining. The submitted proof is not a discharge. C1-C5 repair identities and conditional reductions;

DERIV:W6_SOURCE_ENERGY_JETS_R10:VACUUM_CROSS_DERIVATIVES

open

Establish $\|\partial^r_g A_{\nu}\|_{(q_g \rightarrow q_g)} \leq d_{\nu,r} g^{\wedge(-r-1)}$, $r=0,1,2$, for the actual vacuum-cross source vector.

`docs/derivations/w6-source-energy-jets-r10.md` — ## 6. Scaling defects and unresolved estimates (C6)

Hypotheses. Actual fixed compact square at positive g ; normalized positive ground, $q_g = H_g + \gamma I$, common form domain.; Constants must be independent of g on $0 < g < g_*$.

Remaining. The submitted proof is not a discharge. C1-C5 repair identities and conditional reductions;

DERIV:W6_SOURCE_GENERATOR_SCORE_FRAME:CONDITIONAL_ENERGY_H0_H4

open

Prove, uniformly on $0 < g < g_*$, the fiberwise conditional-energy estimates H0-H4 for the undilated ground score on level sets of w . Their averaged forms are established (SF7b and the integrated J_g bound conditional on H4); the fiberwise forms are open.

`docs/derivations/w6-source-generator-score-frame.md` — ## 5. Exact reduction of the order-zero R10 bound (SF8-SF9), conditional-energy hypotheses

Hypotheses. Actual fixed square, undilated score $\partial_g \log \Omega_g$; K^0_g is not the dilated K_g of M10.

Remaining. A fiberwise Poincare-type inequality for $-\Delta_\mu$ restricted to level sets of w , uniform in g , would give H1 and H2 from SF6. H0, H3 and H4 concern the source energy of conditional expectations and need a separate argument.

DERIV:W6_SYNCHRONIZED_M10_DOMINATION:AMPLITUDE_GRADIENT_SCALING

open

The actual uniform amplitude-score gradient bound ca/g remains open; changing parameter to $h=g^2$ and a spatial WKB expansion do not alone prove it.

`docs/derivations/w6-synchronized-m10-domination.md` — ## 5. Correct parameter-derivative criterion (lines 144-171)

Hypotheses. Actual positive ground amplitude in a valid phase tube.

Remaining. Establish uniform spatial control of $h \partial_h \log A$ and $Z \log A$ for the actual ground; the logarithmic-parameter criterion gives the sufficient implication.

DERIV:W6_SYNCHRONIZED_M10_DOMINATION:FULL_SCORE_DOMINATION_M10

open

Actual full-fiber M10 for S13 remains open. The earlier completion claim is retracted; the repaired assembly implication retains all unsupplied model hypotheses.

`docs/derivations/w6-synchronized-m10-domination.md` — ## 7. Assembly criterion and actual status (lines 262-286)

Hypotheses. Actual four-face compact $SU(2)$ quantum ground law and S13 dilation.

Remaining. Actual complement is discharged. Supply the polynomial relative-amplitude and tube moment estimates, intermediate controls, and actual normal/angular comparison and score error.

DERIV:WILSON_MARKED_TRANSFER:WT6

open

The proposed rooted coefficient-operator bound WT6 is an explicit sufficient route to complete-shell transport; this historical argument did not establish it. Its continuation uses a different full-Hilbert estimate and finite-order locality.

`docs/derivations/wilson-marked-transfer.md` — ## WT-6. The remaining complete-shell estimate and conditional transport (lines 209-248)

Hypotheses. Finite cubic link Hilbert product; positive free generators kill the product vacuum and have gap $\delta > 0$ on its complement.; Bounded plaquette interactions $\|V_p\| \leq v$; physical character-clock step $\tau > 0$ and blocks of duration $\ell = m \tau$.; Actual vacuum-dressed transfer on the specified exact-support coefficient space must first be constructed.

Remaining. Formalize or discharge the particular WT6 rooted norm if used; do not equate the later sufficient route with a proof of this specific estimate.

DERIV:WILSON_SC17_SPATIAL_CLOSURE:ACTUAL_REFERENCE_DEFECT_BOUND

open

On the specified actual Wilson conditional source chart, establish a volume-uniform weighted bound on the complete signed Hessian defect $D_B = \text{Hess}(V_B + (H_{\text{out}} \Omega_g) / \Omega_g) - 2\epsilon A^2$, including the actual kinetic metric, moving source projector and localization terms, with $\text{norm}(D_B)_\infty \leq \delta$ and $2\beta_s(A)^2 \delta / \epsilon < 1$.

`docs/derivations/wilson-sc17-spatial-closure.md` — ## 4. The actual quantum pressure and cutoff

defects are part of the input (lines 698-756), with the R5-R7 budget in section 2

Hypotheses. Actual positive Wilson vacuum, compatible source and block coordinates, and multiplication terms assigned exactly once; the complete coupled Gaussian reference is subtracted.; The reference semigroup budget $\beta_{s(A)}$ is finite in the chosen weighted norm at the fixed blocking scale; $\epsilon > 0$ and all conditional, metric and cutoff domains are retained.; The desired uniformity concerns the stated block family and surrounding volume; a separate source-compatible reference-angle margin is still needed for global assembly.

Remaining. Prove the actual complete weighted smallness estimate. The proved Gaussian pressure cancellation, an isolated magnetic Taylor coefficient, or formalizing the conditional Riccati implication does not discharge this model obligation.

DERIV:WILSON_SELECTED_INVERSE_WALL:W6

open

The needed actual interacting Wilson bound is $\sup_{(\Lambda, b[p]=1)} |d_g, QQ[R_0 t_1 p, R_g t_1 p]| \leq C|g|$ uniformly in volume, coupling, backgrounds and required spectral interval.

`docs/derivations/wilson-selected-inverse-wall.md` — ## 5. The exact point at which this argument cannot proceed (lines 221-264)

Hypotheses. The operator chart, vacuum subtraction and fast/retained spaces are those of the spatial Schur derivation; domains are explicitly checked rather than identified by notation.; $0 < g \leq g_0$; complete relevant retained energy space and high-retained contribution controlled.

Remaining. Prove the actual coupled signed resolvent pairing with the complete dressed force, physical energy normalization and an interacting fast floor; no existing Gaussian control discharges it.

DERIV:YANGMILLS_CONTINUUM_BALABAN_MULTISCALE_PROOF:OS1

open

Full Euclidean covariance requires restoration of $SO(4)$ beyond cubic block symmetry; the repair record ties this to the unproved irrelevance estimate.

`docs/derivations/yangmills-continuum-balaban-multiscale-proof.md` — ### 8.2 OS1: Euclidean Covariance (lines 375-382)

Hypotheses. Submitted $SU(N)$, $N \geq 2$ continuum multiscale construction; read together with its in-document Repair record audited 2026-09-09.; The source title and summary claim completion, but its repair record retains explicit averaging/interpolation, irrelevance, physical-scale and scattering obligations. No machine certification is inferred from prose.

Remaining. Prove rotational restoration for the actual continuum limit, then translation/rotation covariance of its Schwinger distributions.

DERIV:YANGMILLS_CONTINUUM_BALABAN_MULTISCALE_PROOF:THEOREM_7_1

open

The submitted Cauchy theorem needs a summable cross-scale expectation increment. The printed $O(1/k)$ bound alone is not summable; the repair record names the missing irrelevance factor a_k^θ .

`docs/derivations/yangmills-continuum-balaban-multiscale-proof.md` — ### 7.1 Cauchy Sequence of Effective Actions (lines 317-340)

Hypotheses. Submitted $SU(N)$, $N \geq 2$ continuum multiscale construction; read together with its in-document Repair record audited 2026-09-09.; The source title and summary claim completion, but its repair record retains explicit averaging/interpolation, irrelevance, physical-scale and scattering obligations. No machine certification is inferred from prose.; Actual gauge-invariant cylinder expectations and uniform operator normalization; a positive irrelevance exponent or another summable increment bound.

Remaining. Prove the actual summable remainder after marginal matching, then construct the Cauchy limit with its observable norms and scale maps.

DERIV:YANGMILLS_CONTINUUM_BALABAN_MULTISCALE_PROOF:THEOREM_8_2

open

The claimed continuum exponential clustering requires a strictly positive rate in common physical units

at a fixed physical scale; UV cell-scale polymer decay alone does not provide that conclusion.

`docs/derivations/yangmills-continuum-balaban-multiscale-proof.md` — ### 8.5 OS4: Exponential Clustering and Mass Gap (lines 412-436)

Hypotheses. Submitted $SU(N)$, $N \geq 2$ continuum multiscale construction; read together with its in-document Repair record audited 2026-09-09.; The source title and summary claim completion, but its repair record retains explicit averaging/interpolation, irrelevance, physical-scale and scattering obligations. No machine certification is inferred from prose.; The repair record explicitly requires the missing bridge to a scale with $g=O(1)$; uniform cutoff/volume estimates are retained.

Remaining. Prove the actual nonperturbative KP or equivalent correlation estimate at a fixed physical reference scale, then transport it through the constructed continuum limit.

DERIV:YANGMILLS_CONTINUUM_BALABAN_MULTISCALE_PROOF:GAUGE_HESSIAN disputed

The submission gives the background gauge Hessian (5.3)-(5.5); the operator must include the correct covariant 1-form curvature term and the repaired averaging map consistently.

`docs/derivations/yangmills-continuum-balaban-multiscale-proof.md` — ### 5.1 Fluctuation Gauge Fixing (lines 183-197)

Hypotheses. Submitted $SU(N)$, $N \geq 2$ continuum multiscale construction; read together with its in-document Repair record audited 2026-09-09.; The source title and summary claim completion, but its repair record retains explicit averaging/interpolation, irrelevance, physical-scale and scattering obligations. No machine certification is inferred from prose.; Smooth background and gauge-fixed fluctuation domain; repaired deterministic block constraint.

Remaining. Formalize the covariant exterior-derivative/Weitzenböck identity with signs and domains; reconcile Q_k versus Q_B before using the Hessian.

DERIV:YANGMILLS_CONTINUUM_BALABAN_MULTISCALE_PROOF:PROPOSITION_3_4 disputed

Proposition 3.4 asserts a unique gauge-equivariant constrained background minimizer with the Sobolev/coercivity bounds (3.9)-(3.12); the repair record says the corrected affine adjoint constraint and transported interpolation have not been propagated.

`docs/derivations/yangmills-continuum-balaban-multiscale-proof.md` — ### 3.2 The Background Field Variational Problem (lines 111-139)

Hypotheses. Submitted $SU(N)$, $N \geq 2$ continuum multiscale construction; read together with its in-document Repair record audited 2026-09-09.; The source title and summary claim completion, but its repair record retains explicit averaging/interpolation, irrelevance, physical-scale and scattering obligations. No machine certification is inferred from prose.; Small curvature $g_{(k+1)} \leq a_{(k+1)}^{-2}$, $0 < \kappa < 1/6$, sufficiently small coupling; all correct gauge and boundary domains still required.

Remaining. First implement the actual repaired deterministic constraint and equivariant interpolation, then prove coercivity, existence, uniqueness and smooth parameter dependence on its genuine domain.

DERIV:YANGMILLS_CONTINUUM_BALABAN_MULTISCALE_PROOF:THEOREM_5_1 disputed

Theorem 5.1 asserts a uniform positive background fluctuation floor and exponential pointwise resolvent kernel bound. The actual averaging conjugation, covariant Poincaré and kernel regularity estimates remain required.

`docs/derivations/yangmills-continuum-balaban-multiscale-proof.md` — ### 5.3 Combes–Thomas Resolvent Localization for Continuous $\mathcal{H}_{B_k}$ (lines 209-248)

Hypotheses. Submitted $SU(N)$, $N \geq 2$ continuum multiscale construction; read together with its in-document Repair record audited 2026-09-09.; The source title and summary claim completion, but its repair record retains explicit averaging/interpolation, irrelevance, physical-scale and scattering obligations. No machine certification is inferred from prose.; $\delta \leq 1/(4224\ell^2)$; $\alpha_0 \geq 8\delta + 1$; the domain, cutoff and averaging operator must be specified.

Remaining. Formalize the corrected covariant averaged operator and exponential conjugation including its non-local averaging term; replace or justify the all-x,y bounded continuum resolvent kernel claim, which has a diagonal singularity without an explicit smoothing cutoff.

DERIV:YANGMILLS_RESOLVENT_LOCALIZATION_AND_DIRICHLET_GAP:SUBMITTED_W6_CLOSURE disputed

The proposal asserts actual Wilson W6 closure from massive spatial Green decay; its own source audit records that the computed massive scalar operator is not identified with Wilson.

`docs/derivations/yangmills-resolvent-localization-and-dirichlet-gap.md` — ## 1. Summary of Results (lines 20-39)

Hypotheses. This submitted proposal is explicitly preserved under its September 9 source audit; closure claims are not the current graph verdict.; The actual model/operator identification is an obligation, separate from a scalar finite replay.

Remaining. Establish the actual Wilson fast floor, connected signed pairing and physical source identification; the submission does not discharge these.

DERIV:EARLIER_PROOF_RECOVERY:B6_RETAINED_KRYLOV_CAMPAIGN conditional

The retained SU(3) B6 certificate reports radial dimensions 67,155,133 and maximum full-space Ritz residual 8.775675697349887e-14 on 201 g-values in [1,2], with crossing $g=1.3039546641713$.

`docs/derivations/earlier-proof-recovery.md` — ## E6b. Retained B6 numerical campaign

Hypotheses. Historical source $K6(u)=E-uM$, source coupling conversion and branch choices; level-eight 511-column word spaces; relative SVD cutoff $1e-10$.; The certificate is retained and hash-verified; the B6 eigensolver was not rerun in this integration.

Remaining. The original numerical certificate is preserved. A fresh B6 eigensolver replay and any uniform-in-coupling or full-shell conclusion are not supplied.

DERIV:W6_CONDITIONAL_SCORE_TAIL_CONTROL:SCORE_WEIGHT_FROM_M10 conditional

Conditional on the actual M10 domination, every centered finite-energy source satisfies integral $K_g[f]^2$ $dnu_g \leq [C1+(C0+C1 E)/\gamma] b_g[f]$, and the actual median Hardy constant obeys $\mathfrak{B}_g \leq C1+2(C0+C1 E)/\gamma$ uniformly at small coupling. The implication is analytically proved; its M10 model input remains open.

`docs/derivations/w6-conditional-score-tail-control.md` — ### M.4. The precise remaining conditional-score inequality and M.5 (lines 497-563)

Hypotheses. Actual compact-square source form $b_g[f]=g^2 \int (1-w^2)|f|^2 dnu_g$, actual ground-energy bound $e_g \leq E$ and physical gap $\gamma > 0$, with the true-ground source identity M7.; The actual conditional-score domination M10 holds with finite nonnegative $C0, C1$ uniform in g .; Sources belong to the full compact retained form domain; centered sources give M12, and the median-anchored half-source argument retains the uncentered vacuum-energy term for M14-M15.

Remaining. Formalize this already established conditional implication if needed. Actual application requires M10 or another sufficient score estimate;

DERIV:W6_CONDITIONAL_TRANSPORT_OBSTRUCTION:LOCAL_GLOBAL_SCORE_CRITERION conditional

The stated uniform conditional tube moments, differentiated amplitude-score bound and synchronized phase remainder imply the local S14 score estimate. S15 glues it to the full conditional variance using the actual outside-tube squared deviation from the same reference β_g . Uniform potential-weighted complement and remaining-regime bounds would supply synchronized M10.

`docs/derivations/w6-conditional-transport-obstruction.md` — ## S6 Remaining all-fiber estimate (lines 317-375)

Hypotheses. Conditional tube moments $E|\eta|^{(2j)} \leq c_{(2j)} g^{(2j)}$, $j=1,2,3$, with uniform constants.; True rel-

ative amplitude score has fast derivative at most c_a/g ; the phase is smooth with bounded third fast derivatives and the stated oscillator quadratic jet.; Synchronized tangency holds. The same reference β_g is used inside and outside the tube, so conditional mean differences remain in the outside deviation.; Full M10 additionally needs a uniform potential-weighted outside deviation bound and a degeneration-uniform actual score bound in the remaining source regimes.

Remaining. Establish the actual uniform differentiated amplitude, tube-moment, conditional complement and antipodal estimates. Undifferentiated local comparison used for the obstruction does not discharge these hypotheses.

DERIV:W6_GROUND_JETS_TRANSPORT_BUDGET:COMPLETE_BUDGET

conditional

For the complete actual positive-reference source/vacuum transport, energy-operator bounds $\|A^{(r)}(g)\|_{(q_g > q_g)} \leq d_r g^{(-r-1)}$, $r=0,1,2$, imply explicit all-energy $M_j(s) \leq c_j s^{-j}$ for $j=1,2,3$ and hence the complete compact Schur residual budget. The composite budget retains changing source energies and physical clock factors.

docs/derivations/w6-ground-jets-and-transport-budget.md — ## 4. An explicit bridge from the actual transport generator to M1,M2,M3 (lines 199-367)

Hypotheses. Complete actual common-domain source/vacuum transport $R'=A R$ and nonreducing graph from the positive-reference source theorem.; For $q_g=H_g+\gamma$, full energy-space $\|A^{(r)}(g)\| \leq d_r g^{(-r-1)}$, $r=0,1,2$, with finite uniform constants.; Local t in $[s/2,s]$, fixed spectral window below the actual gap. The composite source growth and subdivision inputs are the prior S2-S5 theorem.; Positive physical clock τ_n ; physical matching discrepancies are separately retained, not assumed zero.

Remaining. Prove the actual complete conditional source-generator energy jets or exact mixed-form substitute; supply interacting-grid, coarse-matching and physical-clock convergence inputs.

DERIV:W6_SOURCE_GENERATOR_SCORE_FRAME:R10_ORDER_ZERO_REDUCTION

conditional

Under the conditional-energy hypotheses H0-H4 (uniform q_g -boundedness of P_g , $\sup_w K^0_g \leq \kappa_0 g^{-2}$, $J_g \leq \kappa_1 g^{-2} (1+g^{-2} E[V|w])$, the fast-to-source conditional energy bound, and $b_g[m_g] \leq \kappa_3 g^{-2}$), the complete R9 generator satisfies $\|A(g)\|_{(q_g > q_g)} \leq d_0 g^{-1}$ with the...

docs/derivations/w6-source-generator-score-frame.md — ## 5. Exact reduction of the order-zero R10 bound (SF8-SF9)

Hypotheses. H0-H4 with constants independent of g on $0 < g < g^*$; each is stated explicitly in section 5.; The rank-one q_g operator bound and R4 from the transport-budget note; $0 \leq e_g \leq E$ and gap $\gamma > 0$.

Remaining. Prove H0-H4. The $r=1,2$ cases of R10 are not reduced;

DERIV:WILSON_BACKGROUND_COVARIANCE_CONTINUATION:BF1_BF3

conditional

A common fast form with relative defect norm $\theta_z < 1$ gives the energy-weighted inverse difference $\leq \theta_z/(1-\theta_z)$, the selected signed pairing bound BF1, and the exact energy-dependent Schur/source identity BF3.

docs/derivations/wilson-background-covariance-continuation.md — ## BF1. The energy-dependent relative-form statement (lines 48-102)

Hypotheses. Finite periodic cubic SU(2) lattice with nondegenerate plaquettes and side $L \geq 3$; Product unit-S3 electric metric; $\epsilon > 0$, $k \geq 0$, $\lambda = k/\epsilon$; actual positive normalized Wilson ground Ω_g ; $A_0(z)=F_0-z \geq f_z I > 0$; V is a bounded A_0 -relative form with norm $\leq \theta_z < 1$.

Remaining. The normalized bounded inverse, selected pairing and Schur minimization are formalized. Identify the actual common closed form, inverse-square-root coordinates and compatible cross-functionals for the Wilson operator.

DERIV:WILSON_BACKGROUND_COVARIANCE_CONTINUATION:SC17

conditional

The exact distant mixed derivative is $-R_{xy}(F_e \text{ tensor } V_f + V_e \text{ tensor } F_f) - F_e \text{ tensor } F_f$; the original all-coupling proof does not supply a summable distance envelope.

`docs/derivations/wilson-background-covariance-continuation.md` — ## 5. The neutral channel and the exact remaining spatial bound (lines 494-534)

Hypotheses. Finite periodic cubic SU(2) lattice with nondegenerate plaquettes and side $L \geq 3$; Product unit-S3 electric metric; $\epsilon > 0$, $k \geq 0$, $\lambda = k/\epsilon$; actual positive normalized Wilson ground Ω ; Distinct distant tails; $V_{ef}=0$; R_{xy} is the actual charged inverse.

Remaining. Formalize this identity and the singlet obstruction; the all-coupling spatially summable estimate still needs its complete quantum reference-defect bound.

DERIV:WILSON_MARKED_SHELL_TRANSPORT:S18_S20 conditional

The common weighted Taylor majorant sums finite-order Wilson/Hamiltonian matching; the relative shell gap stays $> (t3/4)u^2 q(k)$ away from Γ and carrier projections converge by S20.

`docs/derivations/wilson-marked-shell-transport.md` — ## 6. Sum the matching and transport the carrier (lines 391-461)

Hypotheses. SU(3), fixed spatial spacing and small magnetic coupling on the S1 common domain; Calibrated kinetic-window and the actual September 5 Hamiltonian G18 creator/activity/complete-band hypotheses and literal source charts are explicit inputs; The Hamiltonian G18 gap and kinetic-window coefficient comparison apply on the same weighted literal-source frame; ϵ is sufficiently small.

Remaining. Formalize power-series tail convergence, the common polar source chart, relative spectral perturbation and exact symmetry cancellation at Γ .

DERIV:WILSON_SC17_SPATIAL_CLOSURE:PROPOSED_QUANTUM_PARAMETER_WINDOW conditional

Under the proposed numerical reference and normalized-defect inputs, the scalar Riccati parameter satisfies $p(\lambda) \leq p(3/100) = 96228\sqrt{3}/625000 < 27/100 < 1$, and the small root satisfies $0 \leq r_{\min} < a_{\min}$ throughout $0 \leq \lambda \leq 3/100$. These are conditional parameter conclusions; the section's complete Wilson defect estimate and exponential spatial decay are not established by this arithmetic.

`docs/derivations/wilson-sc17-spatial-closure.md` — ## 6. Formal repair of the spatial closure for $\lambda \geq 1/73$ via the quantum reference defect (lines 781-821)

Hypotheses. Real coupling $0 \leq \lambda \leq 3/100$; $\epsilon > 0$ when interpreting the normalized defect $c = \delta/(2\epsilon)$; The companion script's proposed defaults are $L=3$, $C_s=6/5$ and reference velocity one; $\beta = (108/25)\sqrt{33}$, $c = \lambda\sqrt{\lambda}/48$ and $a_{\min} = 1/(3\sqrt{33})$; $p(\lambda) = 4\beta^2 c$ and $r_{\min} = (1 - \sqrt{1 - p(\lambda)})/(2\beta)$; These numerical inputs are supplied parameters. Their identification with a uniform semigroup budget, the complete conditional quantum-pressure defect and the actual reference floor is a separate model obligation, not a consequence of the quartic Taylor coefficient.

Remaining. The scalar comparison is proved in the supported successor SC17_P. To apply it to the source section's spatial-closure claim, establish the actual weighted reference budget and complete outside-pressure, metric, projector and cutoff defect bound, derive the ground-state Riccati comparison, and justify an exponential spatial weight.

DERIV:WILSON_SC17_SPATIAL_CLOSURE:R10_R11 conditional

Uniform block Hessian floors $2a_B, 2a_C$ and cross norm $2h_{BC}$ yield $c_{BC} \leq h_{BC}/\sqrt{a_B a_C}$; a reference angle budget plus full Hessian error r yields physical gap $\geq 2\epsilon[a_0(1 - \kappa_A) - 2r]$.

`docs/derivations/wilson-sc17-spatial-closure.md` — ## 3. An explicit true-angle consequence if the full Hessian defect is known (lines 649-697)

Hypotheses. Finite periodic cubic SU(2) lattice with nondegenerate plaquettes and side $L \geq 3$; Product unit-S3 electric metric; $\epsilon > 0$, $k \geq 0$, $\lambda = k/\epsilon$; actual positive normalized Wilson ground Ω ; $a_B, a_C > 0$; $h_{BC}^2 < a_B a_C$; $\kappa_A < 1$; $2r < a_0(1 - \kappa_A)$.

Remaining. Formalize Brascamp-Lieb conditional variance, marginal Schur curvature and the actual projection-angle assembly.

DERIV:WILSON_SC17_SPATIAL_CLOSURE:R12_R14

conditional

The necessary full conditional Hessian defect is $\text{Hess}(V_B + ((H_{\text{out}} \Omega)/\Omega)) - 2\epsilon A^2$; exact Gaussian pressure cancels the reference BB^* term, while cutoffs create the explicit gradient and commutator defects.

`docs/derivations/wilson-sc17-spatial-closure.md` — ## 4. The actual quantum pressure and cutoff defects are part of the input (lines 698-756)

Hypotheses. Finite periodic cubic $SU(2)$ lattice with nondegenerate plaquettes and side $L \geq 3$; Product unit-S3 electric metric; $\epsilon > 0$, $k \geq 0$, $\lambda = k/\epsilon$; actual positive normalized Wilson ground Ω ; Actual vacuum and compatible Euclidean block chart; all multiplication terms assigned once.

Remaining. Formalize the outside-pressure derivative, Gaussian cancellation and every cutoff/product-domain term; the actual uniform smallness estimate remains unproved.

DERIV:WILSON_SC17_SPATIAL_CLOSURE:R4_R9

conditional

If $\beta_s(A) < \infty$ and the full conditional potential Hessian defect is $\leq \delta$ with $2\beta_s(A)^2 \delta/\epsilon < 1$, the true log-ground Hessian error is $\leq r_{\text{minus}}$ and conditional gap $\geq 2\epsilon(a_{\text{r_minus}})$.

`docs/derivations/wilson-sc17-spatial-closure.md` — ## 2. A rigorous conditional small-defect criterion (lines 574-648)

Hypotheses. Finite periodic cubic $SU(2)$ lattice with nondegenerate plaquettes and side $L \geq 3$; Product unit-S3 electric metric; $\epsilon > 0$, $k \geq 0$, $\lambda = k/\epsilon$; actual positive normalized Wilson ground Ω ; $A \geq aI > 0$; the full conditional potential includes outside quantum pressure; smooth ground convergence and domain conditions in the section hold.

Remaining. Formalize the noncommutative weighted norm, Duhamel Riccati comparison, ground limit and curvature-to-gap passage.

DERIV:WILSON_SELECTED_INVERSE_WALL:W4_W5

conditional

On admissible reference graph vectors, $S_g - S_0 = A_{g,r} R_{g,r}$; only the selected inverse bound W5 plus direct/force remainders is sufficient for the displayed $O(g^3)$ comparison.

`docs/derivations/wilson-selected-inverse-wall.md` — ## 3. Repair: only compare inverses on the graph force (lines 101-157)

Hypotheses. The operator chart, vacuum subtraction and fast/retained spaces are those of the spatial Schur derivation; domains are explicitly checked rather than identified by notation; z lies below both full fast spectra; J_z p and the interacting minimizer are admissible; W5 includes $f > 0, a_3, t_2, t_0, c_{\text{sel}}$.

Remaining. The bounded Hilbert-space completion/minimum and weighted pairing are formalized. Construct the actual unbounded fast-form domain, source identification and reduced closed-form Schur operator.

DERIV:WILSON_SPATIAL_SCHUR_EXCESS:SP1_SP6

conditional

The exact full-reference Schur excess is $A_{g,V} g^*(I + C_g)^{-1} V_g$; the minimizing graph, two-sided form bounds and $-dS_g/dz = J_{g,z}^* J_{g,z}$ retain the induced metric.

`docs/derivations/wilson-spatial-schur-excess.md` — ## 2. Exact excess about the full reference graph (lines 42-114)

Hypotheses. Closed self-adjoint reference and interacting forms on the stated compatible P/Q domains, already expressed in a common physical source chart and vacuum-subtracted; Real energy z below the full fast restriction; positive retained energy weight B ; $F_z \geq f_z I > 0$; $\|C_g\| \leq \theta < 1$; all weighted A_g, V_g extend boundedly.

Remaining. The normalized inverse, positive fast form and unique bounded Schur minimum are formalized.

Identify actual Wilson form domains and sources;

DERIV:WILSON_SPATIAL_SCHUR_EXCESS:SP20_SP24

conditional

Complete Schur comparisons give the reciprocal-gap recursion, product/inverse-fast-energy lower bound SP21-SP22 and source-overlap telescoping SP24 in common physical clock units.

`docs/derivations/wilson-spatial-schur-excess.md` — ## 6. Iterating energies and observable amplitudes in the physical clock (lines 412-477)

Hypotheses. Closed self-adjoint reference and interacting forms on the stated compatible P/Q domains, already expressed in a common physical source chart and vacuum-subtracted.; Real energy z below the full fast restriction; positive retained energy weight B .; $0 < \alpha_j \leq 1$; full fast floors f_j ; summable $\epsilon_j = 1 - \alpha_j$ bounded below 1; summable inverse fast floors; actual source maps and errors η_j .

Remaining. Formalize iteration of the closed-form gap theorem, positive infinite products and complete carrier/source projection transport.

DERIV:WILSON_SPATIAL_SCHUR_EXCESS:SP7_SP10

conditional

$K1 = J_z^* W1 J_z$ and $K2 = J_z^* W2 J_z - V1^* V1$ are the complete Schur coefficients; SP8 gives the explicit $O(|g|^3)$ remainder SP9, including the dressed force correction SP10.

`docs/derivations/wilson-spatial-schur-excess.md` — ## 3. The complete second-order coefficient and a uniform remainder (lines 115-182)

Hypotheses. Closed self-adjoint reference and interacting forms on the stated compatible P/Q domains, already expressed in a common physical source chart and vacuum-subtracted.; Real energy z below the full fast restriction; positive retained energy weight B .; SP8 remainder constants $a3, v2, v1, c1, g0$ are finite and $c1|g| \leq \theta < 1$.

Remaining. The genuine Neumann operator expansion and cubic norm/sandwich remainder are formalized. Assemble the g -dependent direct term, moving forces, fast defect and spatial coefficients on the actual compatible Wilson form domain.

DERIV:WILSON_TRUE_VACUUM_BLOCK_ESTIMATES:BA9_BA10

conditional

The true conditional operator includes outside pressure $W_B = (H_{\text{out}} \Omega) / \Omega$ and has gap $\geq \text{gap}(h_B(b)) - \text{osc}_B W_B$; a frozen-block gap alone omits this term.

`docs/derivations/wilson-true-vacuum-block-estimates.md` — ## BA4. What a frozen-block spectral computation would additionally require (lines 142-168)

Hypotheses. Finite periodic cubic SU(2) lattice with nondegenerate plaquettes and side $L \geq 3$.; Product unit-S3 electric metric; $\epsilon > 0$, $k \geq 0$, $\lambda = k/\epsilon$; actual positive normalized Wilson ground Ω .; All conditional blocks use the actual vacuum law $\mu = \Omega^2 dU$, not isolated-block spectral data.; The frozen-block gap and finite pressure oscillation are available on compatible form domains.

Remaining. Formalize the conditional ground transform and bounded-potential min-max comparison, including the domain of $H_{\text{out}} \Omega$.

DERIV:WILSON_VACUUM_ALIGNED_ASSEMBLY:VA12_VA13

conditional

For physical conditional-angle row $\kappa < 1$, $G_{\mu}^2 = (1 - \kappa) G_{\mu}$ and $\text{gap}_{\text{phys}}(H) \geq \gamma_{\mu} (1 - \kappa)$; the single plaquette physical restriction has $E_e = E0$ and zero pair angle.

`docs/derivations/wilson-vacuum-aligned-assembly.md` — ## VA5. Projection geometry, with the Gauss law retained (lines 211-251)

Hypotheses. Finite periodic cubic SU(2) lattice with nondegenerate plaquettes and side $L \geq 3$.; Product unit-S3 electric metric; $\epsilon > 0$, $k \geq 0$, $\lambda = k/\epsilon$; actual positive normalized Wilson ground Ω .; c_{ef}^{phys} is the actual L2 conditional-projection norm, and $\sup_e \sum_{(f=e)} c_{ef}^{\text{phys}} \leq \kappa < 1$.

Remaining. The arbitrary-dimensional bounded projection-sum implication and kernel-complement gap are formalized. Construct actual conditional projections, derive the local anticommutator bound from their Friedrichs angles,

identify the common kernel as constants and establish the required row budget below one.

DERIV:WILSON_VACUUM_ALIGNED_ASSEMBLY:VA14_VA19

conditional

Approximate tensorization or a summable block-angle row gives the stated physical gap with covering multiplicity; the original VA10 local floor tends to zero on the displayed logarithmic continuum trajectory.

`docs/derivations/wilson-vacuum-aligned-assembly.md` — ## VA6. The next estimate and the attempts to establish it (lines 252-415)

Hypotheses. Finite periodic cubic $SU(2)$ lattice with nondegenerate plaquettes and side $L \geq 3$; Product unit- S_3 electric metric; $\epsilon > 0$, $k \geq 0$, $\lambda = k/\epsilon$; actual positive normalized Wilson ground Ω ; Actual conditional blocks, $\gamma_{\min} > 0$ and finite covering multiplicity m_* ; a spatial envelope is an additional hypothesis, not assigned data.

Remaining. Actual-measure covariance minorization and explicit-domain unitary gap transport are formalized. Establish the actual Wilson conditional density floor, conditional disintegration, covariance-to-projection-angle norm and spatial row-budget bounds.

DERIV:WILSON_WEIGHTED_REPAIR_AND_ROTOR_GAP:WR17_WR21

conditional

The actual trial comparison needs $\gamma_{\Lambda}(\text{mean } R - \inf R) > 0$ uniformly; a gauge-invariant patch has full-boundary trial excitations of energy 3ϵ , so an unrestricted boundary fast floor of order $\sqrt{\epsilon k}$ cannot be inferred.

`docs/derivations/wilson-weighted-repair-and-rotor-gap.md` — ## WR6. The precise remaining derivation (lines 229-308)

Hypotheses. Each operator is identified separately: classical quotient diffusion, complete compact S_3 quantum rotor, actual coupled link lattice or independent product model as stated in the section.; Positive normalized overlapping-plaquette trial; γ_{Λ} is an actual weighted generator gap.

Remaining. Formalize the common form domains and mean-residual comparison; construct Haar-link marginal trial vectors proving the full-boundary obstruction.

DERIV:YANGMILLS_CONTINUUM_BALABAN_MULTISCALE_PROOF:GHOST_5_8

conditional

Rescaling $z = g^* \zeta$ exposes $g^{(1-\kappa')}$ in (5.8), but the required volume/scale-uniform covariant inverse-gradient bound is not proved by that algebra.

`docs/derivations/yangmills-continuum-balaban-multiscale-proof.md` — ### 5.2 Faddeev–Popov Ghost Determinant (lines 198-208)

Hypotheses. Submitted $SU(N)$, $N \geq 2$ continuum multiscale construction; read together with its in-document Repair record audited 2026-09-09.; The source title and summary claim completion, but its repair record retains explicit averaging/interpolation, irrelevance, physical-scale and scattering obligations. No machine certification is inferred from prose.; $0 < \kappa' < 1/2$; $a \|\zeta\|_{\infty} \leq g^{-\kappa'}$; the first operator inequality in (5.8) is an additional input.

Remaining. Construct the precise ghost operator, determinant class and uniform inverse-gradient estimate on its constrained domain; scalar rescaling alone does not discharge them.

DERIV:YANGMILLS_CONTINUUM_BALABAN_MULTISCALE_PROOF:OSO

conditional

The asserted Schwinger growth bound (8.2) supplies OS regularity only after its actual uniform moment/Sobolev estimates and continuum fields are constructed.

`docs/derivations/yangmills-continuum-balaban-multiscale-proof.md` — ### 8.1 OS0: Analyticity and Temperedness (lines 368-374)

Hypotheses. Submitted $SU(N)$, $N \geq 2$ continuum multiscale construction; read together with its in-document Repair record audited 2026-09-09.; The source title and summary claim completion, but its repair record retains explicit averaging/interpolation, irrelevance, physical-scale and scattering obligations. No machine certification is

inferred from prose.

Remaining. Formalize the continuum Schwinger distributions and prove the factorial/test-seminorm bound from the actual regulated measures.

DERIV:YANGMILLS_CONTINUUM_BALABAN_MULTISCALE_PROOF:OS3 conditional

Bosonic multiplication observables have permutation-symmetric Schwinger functions, provided the claimed continuum distributions exist.

`docs/derivations/yangmills-continuum-balaban-multiscale-proof.md` — ### 8.4 OS3: Permutation Symmetry (lines 409-411)

Hypotheses. Submitted $SU(N)$, $N \geq 2$ continuum multiscale construction; read together with its in-document Repair record audited 2026-09-09.; The source title and summary claim completion, but its repair record retains explicit averaging/interpolation, irrelevance, physical-scale and scattering obligations. No machine certification is inferred from prose.

Remaining. Formalize products of the constructed fields/observables and the permutation action on their distributional arguments.

DERIV:YANGMILLS_CONTINUUM_BALABAN_MULTISCALE_PROOF:OS_RECONSTRUCTION conditional

With corrected OS hypotheses and totality, the reflection-positive quotient completion has a physical Hilbert space, vacuum and nonnegative self-adjoint time generator.

`docs/derivations/yangmills-continuum-balaban-multiscale-proof.md` — ### 9.1 The Physical Hilbert Space and Hamiltonian (lines 439-453)

Hypotheses. Submitted $SU(N)$, $N \geq 2$ continuum multiscale construction; read together with its in-document Repair record audited 2026-09-09.; The source title and summary claim completion, but its repair record retains explicit averaging/interpolation, irrelevance, physical-scale and scattering obligations. No machine certification is inferred from prose.

Remaining. Formalize the OS quotient, completion, translations and corrected reconstruction regularity; actual OS inputs remain separate obligations.

DERIV:YANGMILLS_CONTINUUM_BALABAN_MULTISCALE_PROOF:SCATTERING conditional

Haag-Ruelle scattering is conditional on an isolated positive single-particle mass shell; the source now explicitly states that dispersion hypothesis. Nonzero off-shell/tree-level four-point data alone do not supply the required on-shell amplitude lower bound.

`docs/derivations/yangmills-continuum-balaban-multiscale-proof.md` — ### 9.3 Relativistic Wightman Fields and Non-Trivial Scattering ($\S \neq \S$) (lines 479-501)

Hypotheses. Submitted $SU(N)$, $N \geq 2$ continuum multiscale construction; read together with its in-document Repair record audited 2026-09-09.; The source title and summary claim completion, but its repair record retains explicit averaging/interpolation, irrelevance, physical-scale and scattering obligations. No machine certification is inferred from prose.; Isolated mass hyperboloid separated below multiparticle threshold; locality, spectrum and actual LSZ/Haag-Ruelle assumptions.

Remaining. Formalize the isolated-shell hypothesis, asymptotic states and a controlled on-shell nonzero amplitude; establish these for the constructed theory before concluding $S \neq I$.

DERIV:YANGMILLS_CONTINUUM_BALABAN_MULTISCALE_PROOF:THEOREM_4_2 conditional

The large-field damping claim (4.4)-(4.5) is conditional on a positive coarse-action subtraction constant and the unproved irrelevance gain identified in the repair record.

`docs/derivations/yangmills-continuum-balaban-multiscale-proof.md` — ### 4.2 Super-Exponential Suppression of Large Fields (lines 157-180)

Hypotheses. Submitted $SU(N)$, $N \geq 2$ continuum multiscale construction; read together with its in-document

Repair record audited 2026-09-09.; The source title and summary claim completion, but its repair record retains explicit averaging/interpolation, irrelevance, physical-scale and scattering obligations. No machine certification is inferred from prose.; $0 < \kappa_{\text{prime}} < \kappa < 1/6$; actual regularized functional measure and corrected background construction; uniform constants.

Remaining. Prove the quantitative irrelevance/subtraction constant and the actual large-field measure estimate, including the averaging constraint and fluctuation direction invisible to a coarse penalty.

DERIV:YANGMILLS_CONTINUUM_BALABAN_MULTISCALE_PROOF:THEOREM_6_2

conditional

The claimed RG contraction (6.5) has the repaired induction envelope $\|K_k\| \leq 4 C_{\text{ind}} g_k^2$, but its single-block contraction gain still requires the irrelevance lemma.

docs/derivations/yangmills-continuum-balaban-multiscale-proof.md — ### 6.3 The Mayer Tree-Graph Expansion & Cluster Contraction (lines 269-295)

Hypotheses. Submitted $SU(N)$, $N \geq 2$ continuum multiscale construction; read together with its in-document Repair record audited 2026-09-09.; The source title and summary claim completion, but its repair record retains explicit averaging/interpolation, irrelevance, physical-scale and scattering obligations. No machine certification is inferred from prose.; $\lambda_{\text{contraction}}$ in $(0, 1/2)$; sufficiently small initial g_0 ; the corrected averaging, fluctuation and large-field inputs hold.

Remaining. Formalize the actual tree expansion and prove the residual local-operator irrelevance gain after marginal subtraction; then prove the model-dependent contraction.

DERIV:YANGMILLS_CONTINUUM_BALABAN_MULTISCALE_PROOF:THEOREM_6_3

conditional

The source records $b_0 = 11N/(48\pi^2)$, the perturbative running formula (6.11)-(6.12) and a uniform polymer conclusion (Corollary 6.4); its repaired induction requires quantitative remainder control and a propagated envelope.

docs/derivations/yangmills-continuum-balaban-multiscale-proof.md — ### 6.4 Asymptotic Freedom and Infinite Scale Stability (lines 296-314)

Hypotheses. Submitted $SU(N)$, $N \geq 2$ continuum multiscale construction; read together with its in-document Repair record audited 2026-09-09.; The source title and summary claim completion, but its repair record retains explicit averaging/interpolation, irrelevance, physical-scale and scattering obligations. No machine certification is inferred from prose.; Perturbative UV regime; controlled beta remainder and actual RG map; this does not place the hadronic scale in the small- g domain.

Remaining. Formalize the actual RG beta coefficient and remainder assumptions before asserting monotonicity for all g_0 ; prove the stated envelope with the repaired constants.

DERIV:YANGMILLS_CONTINUUM_BALABAN_MULTISCALE_PROOF:THEOREM_7_2

conditional

The continuum measure claim requires a continuous positive-definite characteristic functional on the full nuclear test space; convergence only of gauge-invariant observables is not itself this construction.

docs/derivations/yangmills-continuum-balaban-multiscale-proof.md — ### 7.2 Construction of the Limiting Continuum Measure $\mathbb{d}\mu_{\{\text{YM}\}}$ (lines 341-363)

Hypotheses. Submitted $SU(N)$, $N \geq 2$ continuum multiscale construction; read together with its in-document Repair record audited 2026-09-09.; The source title and summary claim completion, but its repair record retains explicit averaging/interpolation, irrelevance, physical-scale and scattering obligations. No machine certification is inferred from prose.; Theorem 7.1 with the repaired summable bound; actual uniform regularity and characteristic-functional estimates.

Remaining. Construct the characteristic functional with continuity/positivity and a consistent gauge-invariant observable realization, then apply Minlos-Bochner.

DERIV:YANGMILLS_CONTINUUM_BALABAN_MULTISCALE_PROOF:THEOREM_8_1

conditional

Centered reflection-adapted deterministic blocking preserves reflection positivity under its exact support/equivariance hypotheses, and a convergent compatible limit retains the inequality (8.5).

`docs/derivations/yangmills-continuum-balaban-multiscale-proof.md` — ### 8.3 OS2: Reflection Positivity (lines 383-408)

Hypotheses. Submitted $SU(N)$, $N \geq 2$ continuum multiscale construction; read together with its in-document Repair record audited 2026-09-09.; The source title and summary claim completion, but its repair record retains explicit averaging/interpolation, irrelevance, physical-scale and scattering obligations. No machine certification is inferred from prose.; Reflection-adapted deterministic block map, positive-time support and actual convergent correlations; the cited centered-blocking construction applies.

Remaining. Formalize the deterministic reflected blocking and limit passage for the actual continuum measure; general reflection-equivariant Markov blocking is not this theorem.

DERIV:YANGMILLS_CONTINUUM_BALABAN_MULTISCALE_PROOF:THEOREM_9_1

conditional

A common strictly positive exponential physical-time correlation rate on the dense centered reconstructed family implies spectrum(H) subset $\{0\}$ union $[\Delta_{\text{phys}}, \infty)$, with the simple vacuum; this conditional spectral argument is confirmed sound by the source repair record.

`docs/derivations/yangmills-continuum-balaban-multiscale-proof.md` — ### 9.2 The Physical Spectral Mass Gap (lines 454-478)

Hypotheses. Submitted $SU(N)$, $N \geq 2$ continuum multiscale construction; read together with its in-document Repair record audited 2026-09-09.; The source title and summary claim completion, but its repair record retains explicit averaging/interpolation, irrelevance, physical-scale and scattering obligations. No machine certification is inferred from prose.; Actual reconstructed physical space with dense observable family; common rate $\Delta_{\text{phys}} > 0$ in physical time; simple vacuum.

Remaining. Formalize the spectral-measure positivity, infinite-time exclusion and density extension; derive the actual physical clustering input (9.7) separately.

DERIV:YANGMILLS_GPU_RESOLUTION_AUDIT:GA8_GA9

conditional

The inserted scalar recursion has explicit positive product/tail bounds GA8 when its assumptions hold; the Phase 4 four-plaquette builder remains a tensor sum GA9.

`docs/derivations/yangmills-gpu-resolution-audit.md` — ## GA4. The additional Phase 4 has a valid conditional scalar limit (lines 162-209)

Hypotheses. The preserved September 9 scripts are inspected as the operators they actually implement; they are not assumed to represent Wilson dynamics.; Analytic matrix identities, replayed floating observations and reported-only runs remain distinct.; $\epsilon_j = C(\ell + jh)^{-3/2} < 1$; positive initial gap and fast floors; these are supplied inputs in the script.

Remaining. Formalize the scalar positive-product and geometric tail bounds; derive actual coarse Wilson parameters before applying them.

DERIV:YANGMILLS_RECONSTRUCTION:R6_R7

conditional

If $C_F(t) \leq A_F \exp(-\eta t + \alpha R) + B_F \exp(-\beta R)$ for every $R, t \geq 0$, choose $R = \eta t / (\alpha + \beta)$ to obtain common rate $\eta \beta / (\alpha + \beta)$.

`docs/derivations/yangmills-reconstruction.md` — ## 5. A precise sufficient repair, with its remaining hypothesis exposed (lines 90-111)

Hypotheses. A nonnegative self-adjoint physical Hamiltonian with a specified vacuum and its positive spectral measures; physical Euclidean time is used.; OS reconstruction and any limit theorem are explicit analytic inputs, not consequences of the finite controls.; $\eta, \beta > 0, \alpha \geq 0$; one common rate and controlled cutoff/volume constants on a dense centered physical family.

Remaining. Real-radius exponent balancing, actual-measure spectral exclusion and dense-family extension are formalized. Establish the actual uniform Yang-Mills two-parameter localization estimate, physical spectral identification and total family;

DERIV:YANGMILLS_RESOLVENT_LOCALIZATION_AND_DIRICHLET_GAP:LOCAL_KERNEL_IMPLICATION conditional

Given uniform massive exponentially decaying kernels and a finite-range perturbation of size $C2|g|$, the localized scalar pairing has a volume-independent geometric-sum bound.

`docs/derivations/yangmills-resolvent-localization-and-dirichlet-gap.md` — ### 2.3 Volume-Uniformity of the Pairing (lines 62-80)

Hypotheses. This submitted proposal is explicitly preserved under its September 9 source audit; closure claims are not the current graph verdict.; The actual model/operator identification is an obligation, separate from a scalar finite replay.; Kernel decay constants and perturbation range/row bounds uniform in volume; finite localized source with the source normalization explicitly retained.

Remaining. Infinite kernel-pairing summability, exact integer-lattice sums and the full source l1 factor are formalized. Identify actual Wilson fast kernels and connected signed sources and prove the required uniform decay and perturbation envelopes.

DERIV:YANGMILLS_RESOLVENT_LOCALIZATION_AND_DIRICHLET_GAP:RESOLVENT_IDENTITY conditional

Compatible invertible fast operators obey $R_g R_0 = R_0 (F_g - F_0) R_g$ and the stated selected-pairing identity.

`docs/derivations/yangmills-resolvent-localization-and-dirichlet-gap.md` — ### 2.1 The Resolvent Identity (lines 42-54)

Hypotheses. This submitted proposal is explicitly preserved under its September 9 source audit; closure claims are not the current graph verdict.; The actual model/operator identification is an obligation, separate from a scalar finite replay.; Real z below both spectra; compatible domains and adjoint pairings.

Remaining. Formalize the resolvent identity on the compatible closed-form domains.

DERIV:YANGMILLS_WEIGHTED_CURVATURE:GST3 conditional

$\text{Ric} + (2/\hbar) \text{Hess } S \geq \kappa$ gives $\|A_S f\|^2 \geq c \kappa \langle f, A_S f \rangle$, hence spectrum in $\{0\}$ union $[c \kappa, \infty)$ and the Poincare bound when the kernel is constants.

`docs/derivations/yangmills-weighted-curvature.md` — ## GST-3: curvature normalization and a spectral argument without a discrete excited state (lines 92-130)

Hypotheses. Finite-dimensional smooth Riemannian configuration space or the explicitly specified finite matrix/oscillator model.; $H = -c \Delta + V$, $c = \hbar^2/(2m) > 0$; $\psi = N \exp(-S/\hbar) > 0$ normalized; common operator/form domains as stated.; $\kappa > 0$; closed Bochner identity/inequality and simple constant kernel.

Remaining. The bounded positive-operator square-to-gap implication is formalized in arbitrary complex Hilbert dimension. Construct the actual weighted operator, verify its quadratic inequality and manage the unbounded-domain/spectral realization required by the source.

DERIV:YANGMILLS_WEIGHTED_CURVATURE:GST6 conditional

The closed-form min-max comparison yields $E_1(H) - E_0(H) \geq \gamma - (\text{mean } R - r_{\min})$.

`docs/derivations/yangmills-weighted-curvature.md` — ## GST-6: the residual-mean certificate (lines 167-190)

Hypotheses. Finite-dimensional smooth Riemannian configuration space or the explicitly specified finite matrix/oscillator model.; $H = -c \Delta + V$, $c = \hbar^2/(2m) > 0$; $\psi = N \exp(-S/\hbar) > 0$ normalized; common operator/form domains as stated.; A_S is nonnegative self-adjoint with simple normalized constant ground and first variational level $\geq \gamma > 0$; $R \geq r_{\min}$, finite mean; 1 belongs to the sum-form domain; min-max applies.

Remaining. Formalize the lower first-excited variational comparison and the trial-vector Rayleigh upper bound on common form domains.

F The Lean tree

Every registered theorem, by module, with its line and what the registry links it to.

439 theorems, 0 occurrences of **sorry**, axioms restricted to **propext**, **Classical.choice** and **Quot.sound**. The axiom set is exported and validated rather than asserted: the strict build records elaborated constants and transitive axioms, and the loader checks them against the declaration records and source fingerprints. Identifiers mentioned in comments are not proof dependencies.

Read the third column carefully. **promotes** appears on 49 of the 439 theorems and means the theorem proves an *entire* named check, so that check’s tier is the Lean tier. Everything else is an ingredient. A theorem count is not a coverage measure, and the proportion of ingredients to whole-statement proofs is the honest shape of the tree rather than a defect in it — see §A for the case in this program’s own archive where seventy-one **sorry**-free modules proved tautologies.

439 registered theorems across 23 modules. *Promotes* means the theorem proves an entire named check and so carries its verification tier; *formalizes* means it is linked to a catalogue record. Only 49 of 439 carry a **promotes** link, and reading the difference as a defect would be a mistake: most theorems are ingredients, and the registry keeps ingredient and whole-statement coverage apart on purpose.

(unlocated)

Theorem	Line	Formalizes / promotes
<code>dim_Z2</code>		promotes <code>dim Z_2 = (L^3 - 1) + 3 = L^3 + 2</code>
<code>d3_ledger</code>		<code>CONST:D_3; CONST:B_3; CONST:LEAK_3; promotes d_3 = 7/32 + 12*leak_3 - 4*b_3</code>

AnisotropyVariance.lean

Theorem	Line	Formalizes / promotes
<code>anisotropy_axis_node</code>	105	<code>RESULT:RECENT_ANISOTROPY_NODAL_SET</code>
<code>anisotropy_body_node</code>	115	<code>RESULT:RECENT_ANISOTROPY_NODAL_SET</code>
<code>anisotropy_centered_moment_identity</code>	46	<code>RESULT:RECENT_ANISOTROPY_VARIANCE_CORE</code>
<code>anisotropy_face_node</code>	110	<code>RESULT:RECENT_ANISOTROPY_NODAL_SET</code>
<code>anisotropy_generic_holdout</code>	120	<code>RESULT:RECENT_ANISOTROPY_VARIANCE_CORE; promotes anisotropy variance is exactly the simplex pair sum and cubic shape invariant</code>
<code>anisotropy_induced_coefficient</code>	126	<code>RESULT:RECENT_ANISOTROPY_INDUCED_SIXTH</code>

Theorem	Line	Formalizes / promotes
<code>anisotropy_shape_identity</code>	37	RESULT:RECENT_ANISOTROPY_VARIANCE_CORE; RESULT:RECENT_ANISOTROPY_INDUCED_SIXTH; promotes anisotropy variance is exactly the simplex pair sum and cubic shape invariant
<code>anisotropy_variance_identity</code>	21	RESULT:RECENT_ANISOTROPY_VARIANCE_CORE; promotes anisotropy variance is exactly the simplex pair sum and cubic shape invariant
<code>anisotropy_variance_nonnegative</code>	30	RESULT:RECENT_ANISOTROPY_VARIANCE_CORE; promotes anisotropy variance is exactly the simplex pair sum and cubic shape invariant
<code>anisotropy_variance_upper_bound</code>	57	RESULT:RECENT_ANISOTROPY_VARIANCE_CORE
<code>anisotropy_variance_zero_iff</code>	73	RESULT:RECENT_ANISOTROPY_NODAL_SET

BakryEmeryCoercivity.lean

Theorem	Line	Formalizes / promotes
<code>bochner_gamma2_lower_bound</code>	51	—
<code>coercivityConstant_lt_one</code>	69	—
<code>coercivityConstant_mono</code>	77	—
<code>coercivityConstant_pos</code>	62	—
<code>form_coercivity_bound</code>	87	—
<code>quantum_curvature_restoration</code>	36	—
<code>quantum_restoration_ratio_gt_one</code>	112	—
<code>ricInfy_pos_of_quantum_restoration</code>	44	—
<code>volume_uniform_coercivity_hierarchy</code>	128	—
<code>volume_uniform_rho_pos</code>	121	—

Basic.lean

Theorem	Line	Formalizes / promotes
<code>C_from_beta</code>	244	CONST:C_shp (historical); CONST:BETA_PEN_3; promotes $C_{old} = (\beta_{pen_3} - 2\alpha_3)/16$
<code>adjoint_shelf_numerator</code>	188	G17; promotes the second-Casimir shelf: every irrep beyond F and F-bar clears $5C_F/4$
<code>alphaPen_eq_neg_four_cube</code>	381	CONST:alpha_pen_N; CONST:CUBE_COMPLETION_4
<code>alphaPen_five</code>	216	CONST:alpha_pen_N; promotes axial law reproduces $\alpha_4, \alpha_5, \alpha_6$
<code>alphaPen_four</code>	215	CONST:alpha_pen_N; promotes axial law reproduces $\alpha_4, \alpha_5, \alpha_6$

Theorem	Line	Formalizes / promotes
alphaPen_six	217	CONST:alpha_pen_N; promotes axial law reproduces alpha_4, alpha_5, alpha_6
alphaPen_three	214	CONST:alpha_pen_N; promotes alpha_3 = 4*A_shp = 5/12
alphaPen_three_eq_four_A	220	CONST:A_shp_3; CONST:alpha_pen_N; R7; promotes alpha_3 = 4*A_shp = 5/12
assembly_three_cumulants	718	G14; C10
bandVar_pi	490	SYM:bloch_scalars
bandVar_pi_div_two	493	SYM:bloch_scalars
bandVar_zero	487	SYM:bloch_scalars
betaN_assembled_numerator	683	G14; C10; R2; promotes the final polynomial identity: -16 u + 32 d - 16 corner - 8 K_adj = P17(N^2)/(N R20(N^2)) as rational functions, for the closed forms of the two-hop weight, the single-contact dressing, the corner dressing and the adjacent-face cube completion
betaN_four	743	R2
betaN_from_three_cumulants	727	G14; C10; R2; CONST:TWO_HOP_WEIGHT_ODD_N; CONST:CORNER_DRESSING_ODD_N; promotes the final polynomial identity: -16 u + 32 d - 16 corner - 8 K_adj = P17(N^2)/(N R20(N^2)) as rational functions, for the closed forms of the two-hop weight, the single-contact dressing, the corner dressing and the adjacent-face cube completion
betaN_three	739	C10; CONST:BETA_PEN_3_ASSEMBLED
beta_from_A_and_C	252	G14; CONST:A_shp_3; CONST:C_shp (historical)
beta_shift_from_cShp	428	C10; C2
blind_holdout	230	G3
cIso_assembled	580	G11
cPrimTwo_forms	353	SYM:c_2_tet
cShpAssembled_neg	576	CONST:C_SHP_ASSEMBLED
cShp_assembled_value	424	C2; CONST:C_SHP_ASSEMBLED
cShp_from_rho_shift	418	C2
channel_resolvent_assembly	148	CONST:t_N; R2; promotes the per-channel resolvent equation closes the projector-to-hopping step
combes_thomas_geometric_bound	1248	G19; CITE:YM_GPU_AUDIT
combes_thomas_pairing_volume_independent	1260	G19; CITE:YM_GPU_AUDIT
cornerDen_cof	674	G14; CONST:CORNER_DRESSING_ODD_N
cornerDressing_over_R20	702	CONST:CORNER_DRESSING_ODD_N
cubeCompletionAdjacent_over_R20	708	CONST:CUBE_COMPLETION_ADJACENT_N
cubeCompletionAdjacent_ratio	405	G14
cubeCompletionAdjacent_split	396	G14; CONST:CUBE_COMPLETION_ADJACENT_N
cubeCompletionAdjacent_three	401	CONST:CUBE_COMPLETION_ADJACENT_4; G3
cubeCompletion_three	372	CONST:CUBE_COMPLETION_4; CONST:A_shp_3

Theorem	Line	Formalizes / promotes
<code>cubeDen_cof</code>	677	G14; CONST:CUBE_COMPLETION_ADJACENT_N
<code>cube_shortfall</code>	414	C2; CONST:CUBE_COMPLETION_ADJACENT_HISTORICAL_4
<code>delta_M</code>	526	SYM:bloch_scalars; promotes checkpoint values at X, M, P, R; promotes the four extraction formulas invert the ansatz
<code>delta_P</code>	532	SYM:bloch_scalars; promotes checkpoint values at X, M, P, R; promotes the four extraction formulas invert the ansatz
<code>delta_R</code>	540	SYM:bloch_scalars; promotes checkpoint values at X, M, P, R; promotes the four extraction formulas invert the ansatz
<code>delta_X</code>	521	SYM:bloch_scalars; promotes checkpoint values at X, M, P, R; promotes the four extraction formulas invert the ansatz; promotes X is blind to B, C, D — it fixes A alone
<code>evenHopping_three</code>	467	CONST:T_PLUS_2; R2
<code>even_cubic_at_sixteen</code>	284	G25; CONST:BAND_EVEN_BOTTOM; CONST:M3_EVEN_K0
<code>even_cubic_at_zero</code>	278	G25; CONST:BAND_EVEN_TOP; CONST:M3_EVEN_BANDTOP
<code>even_cubic_derivative_factors</code>	291	G25
<code>even_gram_minors</code>	271	G25
<code>extraction_A</code>	310	SYM:bloch_scalars; promotes the four extraction formulas invert the ansatz
<code>extraction_B</code>	312	SYM:bloch_scalars; promotes the four extraction formulas invert the ansatz
<code>extraction_C</code>	316	SYM:bloch_scalars; promotes the four extraction formulas invert the ansatz
<code>extraction_D</code>	320	SYM:bloch_scalars; promotes the four extraction formulas invert the ansatz
<code>finite_ground_state_dirichlet_nonneg</code>	845	G23; STUDY:YM:operator-identification; promotes finite symmetric matrix ground-state residual identity
<code>finite_ground_state_eigenform_identity</code>	832	G23; STUDY:YM:operator-identification
<code>finite_ground_state_gap_transfer</code>	865	G23; STUDY:YM:operator-identification; STUDY:YM:weighted-curvature
<code>finite_ground_state_ratio_identity</code>	822	G23; STUDY:YM:operator-identification; promotes finite symmetric matrix ground-state residual identity
<code>finite_ground_state_residual_identity</code>	774	G23; STUDY:YM:operator-identification
<code>finite_transverse_valley_identity</code>	944	G22; STUDY:YM:matrix-commutator-model
<code>geom_sum_mul_sub</code>	1239	G19; CITE:YM_GPU_AUDIT
<code>groundStateDirichlet_const</code>	897	G23; CITE:YM_GPU_AUDIT

Theorem	Line	Formalizes / promotes
ground_state_frustration_elimination	906	G23; CITE:YM_GPU_AUDIT
hardy_linear_lower_bound	990	G22; STUDY:YM:matrix-commutator-model
hardy_linear_remainder	984	G22; STUDY:YM:matrix-commutator-model
hopping_deficit_numerator	48	CONST:t_N; promotes deficit identity $1/4 - N^3 t_N$
hopping_three	41	CONST:t_N; R2; promotes $t_2 = 0$ and $t_3 = 5/612$
hopping_two	44	CONST:t_N; R2; promotes $t_2 = 0$ and $t_3 = 5/612$
isolation_condition_assembled	571	CONST:C_SHP_ASSEMBLED; G11
isolation_switch_numerator	196	G17; promotes the exact isolation constant: $\gamma = C_F$, except $5/4$ at $SU(4)$, attained
kotecky_preiss_cluster_sum_bound	1285	G17; CITE:YM_GPU_AUDIT
kotecky_preiss_polymer_activity_bound	1303	G17; CITE:YM_GPU_AUDIT
lambda2_shelf_numerator	180	G17; promotes the second-Casimir shelf: every irrep beyond F and $F\text{-bar}$ clears $5C_F/4$
like_family_sum	129	CONST:t_N; R2; promotes A_N and B_N are the channel sums, not transcriptions
mixed_family_sum	118	CONST:t_N; R2; promotes A_N and B_N are the channel sums, not transcriptions
newton_three	300	G14; ADR:0004; ADR:0005
pentCompletion_three	376	CONST:PENT_COMPLETION_5_SU3
piTilde_neg	566	CONST:PI_ORBIT; CONST:X_QUANTUM; G11
polymer_free_energy_cauchy_bound	1310	G17; CITE:YM_GPU_AUDIT
prismCompletion_three	368	CONST:PRISM_COMPLETION_3_SU3
q_at_checkpoints	512	SYM:bloch_scalars; promotes q at the four high-symmetry points is 0, 4, 8, 12
quartic_gaussian_frequency_optimal	1056	G23; STUDY:YM:weighted-curvature; promotes quartic Gaussian certificate is optimized at frequency cubed $6 \lambda \hbar$ over mass squared
quartic_gaussian_frequency_optimal_iff	1065	G23; STUDY:YM:weighted-curvature; promotes quartic Gaussian certificate is optimized at frequency cubed $6 \lambda \hbar$ over mass squared
quartic_gaussian_frequency_remainder	1049	G23; STUDY:YM:weighted-curvature; promotes quartic Gaussian certificate is optimized at frequency cubed $6 \lambda \hbar$ over mass squared
quartic_gaussian_gap_budget	1043	G23; STUDY:YM:weighted-curvature
quartic_trial_residual_completion	1019	G23; STUDY:YM:weighted-curvature
quartic_trial_residual_floor	1032	G23; STUDY:YM:weighted-curvature
rank_low_numerator	36	CONST:t_N; R2; promotes $t_N = B_N - A_N$
relative_gap_pos	588	G11
residual_mean_gap_budget	1012	G23; STUDY:YM:weighted-curvature; promotes trial residual gap budget uses its mean above its floor
residual_mean_slack_identity	1006	G23; STUDY:YM:weighted-curvature; promotes trial residual gap budget uses its mean above its floor
residual_oscillation_gap_budget	922	G23

Theorem	Line	Formalizes / promotes
<code>residual_oscillation_positive_gap</code>	928	G23; promotes trial residual gap budget uses its mean above its floor
<code>resolventWeight_adjoint</code>	80	CONST:t_N; R2; promotes the four channel weights follow from dimension and Casimir
<code>resolventWeight_antisym</code>	89	CONST:t_N; R2; promotes the four channel weights follow from dimension and Casimir
<code>resolventWeight_singlet</code>	72	CONST:t_N; R2; promotes the four channel weights follow from dimension and Casimir
<code>resolventWeight_sym</code>	104	CONST:t_N; R2; promotes the four channel weights follow from dimension and Casimir
<code>shell_margin_five</code>	174	G17; promotes five charged links clear the shell by exactly $C_F/2$, at every volume
<code>singleContactDen_cof</code>	670	G14; CONST:SINGLE_CONTACT_DRESSING_ODD_N
<code>singleContactEven_eq</code>	455	G14; CONST:SINGLE_CONTACT_DRESSING_EVEN_N
<code>singleContactEven_three</code>	463	CONST:SINGLE_CONTACT_DRESSING_EVEN_N
<code>singleContactOdd_over_R20</code>	696	CONST:SINGLE_CONTACT_DRESSING_ODD_N
<code>stencil_zero_mode</code>	234	—
<code>sym2_shelf_numerator</code>	184	G17; promotes the second-Casimir shelf: every irrep beyond F and F-bar clears $5C_F/4$
<code>tetraCompletion_three</code>	364	C15; CONST:TETRA_COMPLETION_2_SU3
<code>tetra_from_count</code>	358	C15; G5; CONST:c_2^tet (primitive)
<code>transverse_slice_form_aggregation</code>	970	G22; STUDY:YM:matrix-commutator-model; promotes YM flat directions: balanced slice aggregation retains half the kinetic energy
<code>transverse_slice_form_identity</code>	978	G22; STUDY:YM:matrix-commutator-model; promotes YM flat directions: balanced slice aggregation retains half the kinetic energy
<code>twoHopDen_cof</code>	667	G14; CONST:TWO_HOP_WEIGHT_ODD_N
<code>twoHopWeight_over_R20</code>	690	CONST:TWO_HOP_WEIGHT_ODD_N
<code>uStarSq_isolation</code>	584	CONST:T_MINUS_2; G11
<code>w4_old_is_alpha_plus_beta</code>	436	CONST:BETA_PEN_3; CONST:ALPHA_PEN_3
<code>w4_shift_from_beta</code>	433	C2; G11; CONST:W4_ASSEMBLED
<code>width_eq_alpha_add_beta</code>	248	CONST:W4_HISTORICAL_NUM; CONST:BETA_PEN_3
<code>wilson_complex_gram_budget</code>	1199	CITE:WILSON_SHELL
<code>wilson_damped_time_denominator</code>	1121	G19; CITE:WILSON_MARKED
<code>wilson_disconnected_vacuum_cancel</code>	1106	G17; G19; CITE:WILSON_MARKED
<code>wilson_geometric_telescope</code>	1112	G19; CITE:WILSON_MARKED
<code>wilson_polymer_kp_budget</code>	1139	G17; G19; CITE:WILSON_MARKED
<code>wilson_relative_gap_budget</code>	1153	G18; G19; CITE:WILSON_MARKED
<code>wilson_relative_gap_positive</code>	1158	G18; G19; CITE:WILSON_MARKED
<code>wilson_sc17_barrier_polynomial</code>	1405	CITE:WILSON_SC17
<code>wilson_sc17_broad_barrier</code>	1377	CITE:WILSON_SC17

Theorem	Line	Formalizes / promotes
wilson_sc17_broad_curvature_and_cat	1391	CITE:WILSON_SC17
wilson_sc17_discriminant_negative_ at_seventy_two	1411	CITE:WILSON_SC17
wilson_sc17_sharp_barrier	1349	CITE:WILSON_SC17
wilson_sc17_sharp_curvature_and_gap	1364	CITE:WILSON_SC17
wilson_slab_block_row_and_cat	1339	CITE:WILSON_TRUE_BLOCK; CITE:WILSON_TRUE_BLOCK_CHECK
wilson_slab_pinned_activity	1331	CITE:WILSON_TRUE_BLOCK; CITE:WILSON_TRUE_BLOCK_CHECK
wilson_source_weight_defect	1165	G18; CITE:WILSON_MARKED
wilson_source_weight_positive	1170	G18; CITE:WILSON_MARKED
wilson_spatial_budget_telescope	1222	G19; CITE:WILSON_SPATIAL
wilson_spatial_weighted_step	1212	G19; CITE:WILSON_SPATIAL
wilson_symmetric_block_identity	1092	G19; CITE:WILSON_MARKED
wilson_taylor_polynomial_envelope	1182	CITE:WILSON_SHELL
wilson_time_kernel_split	1132	G19; CITE:WILSON_MARKED
wilson_uniform_carrier_coefficient	1204	CITE:WILSON_SHELL; G18; G19

CreatorParent.lean

Theorem	Line	Formalizes / promotes
commuting_pair_square	19	G18; RUN:wilson_creator_parent_2026-09-05
idempotent_square_defect	10	G18; RUN:wilson_creator_parent_2026-09-05
wilson_gap_constant	27	G18; RUN:wilson_creator_parent_2026-09-05

GlobalWilsonVertical.lean

Theorem	Line	Formalizes / promotes
far_affine	73	RUN:nonlinear_wilson_block_2026-09-05
half_potential_threshold	129	RUN:nonlinear_wilson_block_2026-09-05
near_affine	54	RUN:nonlinear_wilson_block_2026-09-05
near_cap	39	RUN:nonlinear_wilson_block_2026-09-05
spectral_from_split	86	RUN:nonlinear_wilson_block_2026-09-05
strip_factor	13	RUN:nonlinear_wilson_block_2026-09-05
two_sector_form	112	RUN:nonlinear_wilson_block_2026-09-05

GroundStateAssembly.lean

Theorem	Line	Formalizes / promotes
assembly_energy	504	—

Theorem	Line	Formalizes / promotes
assembly_kernel_iff	516	—
assembly_square_lower	544	—
assembly_symmetric	509	—
bounded_gap_on_range_of_square	708	—
bounded_projection_assembly_gap	773	—
bounded_spectral_gap_of_square	737	—
finite_assembly_gap	631	—
finite_spectral_gap_of_square	590	—
gram_form_nonnegative	841	—
gram_quadratic_eq_squares	825	—
integral_cauchy_schwarz_sq	318	—
integral_mul_eq_L2_inner	308	—
integral_remainder_measure	329	—
joint_measure_minorization_covariance	383	—
joint_withDensity_covariance_variance	441	—
joint_withDensity_minorization_covariance	421	—
local_floor_tensorization	648	—
measure_minorization_covariance	348	—
measure_poincare_comparison	243	—
measure_variance_density_lower	214	—
measure_variance_density_upper	193	—
measure_variance_le_deviation	184	—
poincare_density_comparison	91	—
product_minorization_covariance	128	—
projection_energy	488	—
projection_energy_nonneg	494	—
symmetric_coupling_sum	528	—
unitary_ground_state_gap	669	—
variance_density_upper	77	—
variance_eq_second_moment	45	—
variance_le_deviation	59	—
weighted_cauchy_schwarz_sq	114	—
withDensity_poincare_comparison	267	—
withDensity_poincare_constant	288	—
withDensity_variance_floor	232	—

HodgeFeshbach.lean

Theorem	Line	Formalizes / promotes
carrier_homological_protection	123	—
carrier_s_eigenvalue	134	—

Theorem	Line	Formalizes / promotes
cellular_excursion_up_harmonic	384	DERIV:UNIVERSAL_CELLULAR_HODGE:UP_HARMONIC_EXCURSION
feshbach_hodge_no_coupling	141	—
feshbach_intermediate_return_annihilation	391	DERIV:UNIVERSAL_CELLULAR_HODGE:RETAINED_VECTOR_RANK_ONE
feshbach_q_annihilates_l_up	377	—
hodge_algebra_relations	33	—
hodge_decomposition	101	—
hodge_feshbach_words_count	268	—
hodge_ker_disjoint	55	—
hodge_pi_cleared	447	DERIV:UNIVERSAL_CELLULAR_HODGE:SHIFTED_OPERATOR_POWERS
hodge_sandwiched_power	472	DERIV:UNIVERSAL_CELLULAR_HODGE:SHIFTED_OPERATOR_POWERS
hodge_shifted_power	458	DERIV:UNIVERSAL_CELLULAR_HODGE:SHIFTED_OPERATOR_POWERS
hodge_vector_decomposition	73	—
hodge_word_is_scalar	355	—
normalized_carrier_projection_fixes_carrier	173	—
r_excitation_carrier_projection_zero	195	—
r_excitation_in_ker_up	154	—
r_excitation_orthogonal	180	—
r_excitation_up_harmonic	188	—
rank_one_laplacian_carrier_eigenvalue	167	—
rank_one_rur_carrier_factorization	215	—
rank_one_rur_cleared_defect_zero	224	—
rank_one_rur_vector_factorization	207	—
rr_carrier_polynomial_remainder	318	—
rur_carrier_polynomial_cleared_identity	327	—
rur_two_r_no_u_false	304	—
tetrahedral_hodge_duality	405	—
traceless_compression_vanishes	431	—
two_r_no_u_words_count	288	—
two_r_no_u_words_enum	291	—
ur_carrier_polynomial_value	333	—

PlateauObstruction.lean

Theorem	Line	Formalizes / promotes
admissible_logarithmic_minimum	71	—

Theorem	Line	Formalizes / promotes
arbitrarily_small_plateau_ obstruction	266	—
bound_time_rescaling	142	—
endpoint_minimum	87	—
interior_minimum_value	99	—
logarithmic_time_is_stationary	47	—
physical_time_minimum_value	150	—
positive_slow_mode_prevents_faster_ bound	239	—
r4_r5_interior_optimum	162	—
r4_r5_optimized_upper_bound	194	—
rescaled_correlation_identity	228	—
slow_mode_fits_positive_plateau	213	—
stationary_global_minimum	20	—
theta_minimum_value	122	—

PolymerCluster.lean

Theorem	Line	Formalizes / promotes
betaStabilityWindow_pos	221	—
beta_le_window_implies_tiltedKp_le	231	—
beta_le_window_implies_tree_bound	246	—
clusterMassGap_ge_log_kp	299	—
clusterMassGap_pos	289	—
clusterMassGap_pos_of_strict_kp	321	—
cluster_correlation_exponential_ decay	329	—
cluster_free_energy_density_bound_ le	258	—
cluster_free_energy_density_bound_ le_window	266	—
cluster_free_energy_density_bound_ nonneg	273	—
geometric_majorant_le_iff	59	—
geometric_majorant_mono	50	—
geometric_majorant_nonneg	38	—
geometric_majorant_pos	44	—
geometric_series_from_one_hasSum	24	—
geometric_series_from_one_tsum	33	—
log_one_add_div_pos	312	—
strict_kotecky_preiss_criterion	170	—
strict_kotecky_preiss_iff	186	—
strict_kp_criterion_lt_one	162	—
tiltedKpParameter_at_alpha_zero	156	—

Theorem	Line	Formalizes / promotes
tiltedKpParameter_nonneg	149	—
tilted_kotecky_preiss_convergent	206	—
tilted_kotecky_preiss_tree_bound	199	—
uniform_source_radius_bound_le	340	—
uniform_source_radius_bound_le_	373	—
window		
uniform_source_radius_bound_nonneg	367	—
unweightedActivityPower_le_weighted	85	—
weightedActivityBase_beta_zero	115	—
weightedActivityBase_geometric_	99	—
hasSum		
weightedActivityBase_nonneg	73	—
weightedActivityBase_small_beta_	122	—
regime		
weighted_geometric_majorant_nonneg	108	—

ResolventLocalization.lean

Theorem	Line	Formalizes / promotes
cubicWeight_nonneg	329	—
cubic_geometric_hasSum	337	—
cubic_localized_pairing	351	—
finite_source_propagation	244	—
integer_geometric_hasSum	301	—
localized_finite_volume_bound	289	—
localized_kernel_pairing	262	—
mul_neumann	29	—
neumann_mul	34	—
neumann_quadratic_expansion	389	—
neumann_quadratic_remainder_	378	—
identity		
neumann_quadratic_remainder_	431	—
pairing_bound		
neumann_summable	24	—
norm_neumann_le	39	—
norm_neumann_quadratic_remainder_le	396	—
norm_neumann_sub_one_le	44	—
norm_neumann_sub_one_le_budget	59	—
norm_sandwiched_neumann_quadratic_	409	—
remainder_le		
normalized_fast_coercive	183	—
normalized_schur_unique_minimizer	203	—
normalized_selected_pairing_bound	92	—
schur_minimizer_lower_bound	173	—

Theorem	Line	Formalizes / promotes
<code>schur_minimizer_value</code>	164	—
<code>schur_square_completion</code>	150	—
<code>signed_resolvent_identity</code>	79	—
<code>signed_resolvent_pairing</code>	133	—
<code>weighted_inverse_difference_bound</code>	121	—
<code>weighted_selected_pairing_bound</code>	111	—

RootedScalarBounds.lean

Theorem	Line	Formalizes / promotes
<code>normalized_taylor_growth</code>	13	G18; RUN:wilson_rooted_contraction_2026-09-05

SC17Riccati.lean

Theorem	Line	Formalizes / promotes
<code>continuous_riccati_barrier</code>	79	—
<code>defaultBeta_pos</code>	244	—
<code>defaultBeta_sq</code>	246	—
<code>default_parameter_endpoint</code>	268	—
<code>default_parameter_identity</code>	252	—
<code>default_parameter_interval</code>	285	—
<code>default_parameter_monotone</code>	260	—
<code>default_reference_budget</code>	307	—
<code>default_reference_product</code>	276	—
<code>default_smallRoot_and_gap</code>	315	—
<code>default_small_defect_criterion</code>	301	—
<code>exists_riccati_fixedPoint</code>	158	—
<code>exists_riccati_solution</code>	199	—
<code>exists_scalar_reference_riccati</code>	212	—
<code>quadratic_forbidden_interval</code>	64	—
<code>riccatiMap_lipschitz_on_ball</code>	131	—
<code>riccatiMap_norm_le</code>	114	—
<code>smallRoot_contraction_margin</code>	51	—
<code>smallRoot_equation</code>	36	—
<code>smallRoot_lt_half_inverse</code>	45	—
<code>smallRoot_lt_reference_floor</code>	59	—
<code>smallRoot_nonneg</code>	28	—

SourceRadiusGrowth.lean

Theorem	Line	Formalizes / promotes
boundedTwoPointSource_abs_le	147	—
independent_exp_moment_difference	38	—
independent_exp_radius_squared_tendsto	77	—
independent_exp_sum_integral	25	—
independent_exp_variance	104	—
independent_nonzero_source_radius_unbounded	124	—
independent_sum_abs_le	94	—
independent_twoPoint_radius_unbounded	160	—
power_difference_lower	51	—
power_difference_per_site_tendsto	64	—

SourceTilt.lean

Theorem	Line	Formalizes / promotes
bounded_footprint_radius	169	—
bounded_footprint_radius_sharp	197	—
centeredExpL2_norm_le	153	—
centeredExpL2_norm_sq	109	—
centeredExpL2_orthogonal_constants	128	—
centered_exp_integral_sq	78	—
exp_source_bounds	28	—
exp_source_integral_le	47	—
exp_source_memLp	40	—
exp_source_second_moment_gap	313	—
exp_source_variance_le	93	—
exp_source_variance_pos	285	—
finite_source_abs_le	56	—
finite_source_partition_bound	67	—
one_lt_integral_exp_of_centered_nonzero	235	—
raw_tilt_gt_one_of_su2_moments	274	—
symmetricTwoPoint_integral	336	—
symmetricTwoPoint_raw_tilt_gt_one	359	—
symmetricTwoPoint_source_bound	347	—

SpectralReconstruction.lean

Theorem	Line	Formalizes / promotes
balanced_rate_pos	127	—

Theorem	Line	Formalizes / promotes
balancing_radius_nonneg	122	—
deleted_frame_invisible_sector	342	—
laplace_decay_excludes_low_energy	70	—
laplace_decay_support_gap	99	—
laplace_integrable	28	—
localization_balanced_decay	133	—
localization_excludes_low_energy	152	—
low_energy_laplace_lower_bound	41	—
low_energy_mass_bound	55	—
observable_frame_correlation	331	—
observable_frame_energy	314	—
observable_frame_gram	306	—
observable_frame_lower_bound	322	—
projection_zero_from_total_family	171	—
reflection_kernel_gram	237	—
reflection_kernel_nonnegative	259	—
total_family_localization_	202	—
projection_gap		
total_family_spectral_projection_	182	—
gap		
transfer_energy_nonnegative	274	—
transfer_energy_reconstructs_ratio	279	—
transfer_power_eq_physical_decay	287	—

TheoryCurrentBridges.lean

Theorem	Line	Formalizes / promotes
factored_inverse_energy_without_	76	—
uniform_gap		
factored_variational_energy_	114	—
without_inverse		
full_graph_source_ward	213	—
gapless_factored_w6_bound	140	—
joint_resolvent_budget	240	—
residual_controls_inverse_without_	250	—
diagonal		
riccati_default_contraction_bound	47	—
riccati_residual_certificate	33	—
rotated_two_level_schur_numerator	195	—
schur_connection_cancellation	179	—
schur_metric_pairing_on_shell	187	—

ThermodynamicLimit.lean

Theorem	Line	Formalizes / promotes
<code>closable_of_integration_by_parts</code>	55	—
<code>closed_extension_of_integration_by_parts</code>	92	—
<code>exists_limit_of_cut_response</code>	22	—
<code>gap_bound_on_graph_limit</code>	75	—
<code>integral_under_weak_and_uniform_convergence</code>	111	—
<code>integration_by_parts_under_weak_limit</code>	135	—
<code>limit_preserves_gap_bound</code>	30	—
<code>poincare_under_weak_convergence</code>	157	—
<code>sc17_plaquette_interval</code>	214	—
<code>spectral_interval_mass</code>	182	—

VacuumChart.lean

Theorem	Line	Formalizes / promotes
<code>corrected_hermitian</code>	72	G18; RUN:wilson_vacuum_chart_2026-09-05
<code>corrected_vacuum_column</code>	63	G18; RUN:wilson_vacuum_chart_2026-09-05
<code>corrected_vacuum_legs</code>	80	G18; RUN:wilson_vacuum_chart_2026-09-05
<code>endpoint_disjoint_cancellation</code>	99	G18; G19; RUN:wilson_vacuum_chart_2026-09-05
<code>fixed_vacuum_row</code>	50	G18; RUN:wilson_vacuum_chart_2026-09-05
<code>generator_antihermitian</code>	39	G18; RUN:wilson_vacuum_chart_2026-09-05
<code>generator_vacuum</code>	44	G18; RUN:wilson_vacuum_chart_2026-09-05
<code>rankOne_adjoint</code>	35	G18; RUN:wilson_vacuum_chart_2026-09-05
<code>scalar_vacuum_cancellation</code>	56	G18; RUN:wilson_vacuum_chart_2026-09-05

VacuumCompression.lean

Theorem	Line	Formalizes / promotes
<code>corrected_eq_compression</code>	55	G18; RUN:wilson_vacuum_compression_2026-09-05
<code>corrected_eq_sub_vacuum_terms</code>	40	G18; RUN:wilson_vacuum_compression_2026-09-05
<code>vacuum_corner_zero</code>	31	G18; RUN:wilson_vacuum_compression_2026-09-05
<code>vacuum_projection</code>	24	G18; RUN:wilson_vacuum_compression_2026-09-05

W6Residual.lean

Theorem	Line	Formalizes / promotes
<code>factored_residual_dual_bound</code>	205	—
<code>factored_w6_bound</code>	243	—

Theorem	Line	Formalizes / promotes
<code>functional_le_energyDualNorm</code>	41	—
<code>gaussian_energy_coefficient_one</code>	273	—
<code>gaussian_energy_coefficient_three</code>	289	—
<code>gaussian_energy_coefficient_two</code>	280	—
<code>inverse_energy_eq_dual_norm_sq</code>	113	—
<code>inverse_energy_le_dual</code>	81	—
<code>residual_square_completion</code>	48	—
<code>variational_residual_dual_identity</code>	132	—
<code>variational_residual_error_bound</code>	147	—
<code>variational_residual_identity</code>	60	—

WilsonGridAlgebra.lean

Theorem	Line	Formalizes / promotes
<code>wilson_grid_block_energy_identity</code>	20	RESULT:W6_SHARED_EDGE_GAUSSIAN_FLOOR
<code>wilson_grid_block_energy_lower_bound</code>	27	RESULT:W6_SHARED_EDGE_GAUSSIAN_FLOOR
<code>wilson_grid_block_energy_upper_bound</code>	33	RESULT:W6_SHARED_EDGE_GAUSSIAN_FLOOR
<code>wilson_grid_block_energy_zero_iff</code>	39	RESULT:W6_SHARED_EDGE_GAUSSIAN_FLOOR

WilsonSquareForce.lean

Theorem	Line	Formalizes / promotes
<code>wilson_square_finite_radial_synthesis_bound</code>	91	RESULT:W6_SQUARE_SHARP_INVERSE_ENERGY
<code>wilson_square_first_source_saturates</code>	42	RESULT:W6_SQUARE_SHARP_INVERSE_ENERGY
<code>wilson_square_radial_ratio_bound</code>	63	RESULT:W6_SQUARE_SHARP_INVERSE_ENERGY
<code>wilson_square_radial_ratio_difference</code>	50	RESULT:W6_SQUARE_SHARP_INVERSE_ENERGY
<code>wilson_square_radial_ratio_strict</code>	76	RESULT:W6_SQUARE_SHARP_INVERSE_ENERGY
<code>wilson_square_sharp_constant</code>	25	RESULT:W6_SQUARE_SHARP_INVERSE_ENERGY
<code>wilson_square_sharp_constant_pos</code>	34	RESULT:W6_SQUARE_SHARP_INVERSE_ENERGY

G The reading library

260 acquired papers with local full text (99 in the first volume, 161 in the second), grouped by the library’s own categories and ordered by year within each.

This is a *coverage* instrument, not an evidence map. Its purpose is to answer “has this been done?” before an agent spends a week, and its *Connection* column is a curated reading note recording why the paper was acquired. A connection is not an evidence edge and carries no weight in any claim.

The evidence edges — the 54 relations actually verified against a named claim of this program — are the separate and much smaller set in §12. The distance between 260 acquired and 34 read-against-a-claim is the honest measure of how far the external comparison has actually been carried. 260 acquired papers, all with local full text, grouped by the library’s own categories. *Connection* is the curated one-line statement of what the paper bears on in this program; it is a reading note, not an evidence edge. The evidence edges are the separate, smaller, verified set in §12. A paper here has been *obtained*, which is not the same as *read against a claim*.

Hamiltonian and quantum-simulation formulations

Year	Title	arXiv	Connection
2026	Continuum limit of a qubit-regularized SU(3) lattice gauge theory with glueballs	2603.01215	2026 qubit-regularized SU(3) continuum-limit/glueball preprint; direct but too new for citation stability
2026	Minimally Truncated SU(3) Lattice Gauge Theory and String Tension	2601.10065	2026 minimal SU(3) truncation with direct string-tension target; promising but not yet peer reviewed
2025	Accurate ground states of SU(2) lattice gauge theory in 2+1D and 3+1D	2509.12323	variational SU(2) ground states in 2+1D and 3+1D
2025	Asymptotic-freedom and massive glueballs in a qubit-regularized SU(2) gauge theory	2512.11068	peer-reviewed qubit-regularized SU(2) glueball toy model
2025	Dynamics in Hamiltonian Lattice Gauge Theory: Approaching the Continuum Limit with Partitionings of SU(2)	2503.03397	Recent peer-reviewed dynamical SU(2) approach toward the continuum limit
2025	Efficient Truncations of SU(N _c) Lattice Gauge Theory for Quantum Simulation	2503.11888	controlled efficient gauge-field truncations for SU(N _c) Hamiltonians
2025	Emergent Hydrodynamic Mode on SU(2) Plaquette Chains and Quantum Simulation	2502.17551	emergent hydrodynamic mode in SU(2) plaquette-chain Hamiltonian dynamics
2025	Exponential speedup in quantum simulation of Kogut-Susskind Hamiltonian via orbifold lattice	2506.00755	orbifold-lattice route to faster Kogut-Susskind Hamiltonian simulation
2025	Generic Hilbert Space Fragmentation in Kogut-Susskind Lattice Gauge Theories	2502.03533	Hilbert-space fragmentation and structural sectors in Kogut-Susskind Hamiltonians

Year	Title	arXiv	Connection
2025	Improved Honeycomb and Hyperhoneycomb Lattice Hamiltonians for Quantum Simulations of Non-Abelian Gauge Theories	2503.09688	improved non-Abelian Hamiltonian lattices
2025	Loop-string-hadron approach to SU(3) lattice Yang-Mills theory, II: Operator representation for the trivalent vertex	2512.11796	SU(3) loop-string-hadron operator representation on a trivalent vertex
2025	Non-Abelian dynamics on a cube: improving quantum compilation through qudit-based simulations	2506.10945	non-Abelian cube dynamics and qudit-based quantum compilation
2025	Obtaining continuum physics from dynamical simulations of Hamiltonian lattice gauge theories	2506.16559	2025 preprint on recovering continuum physics from Hamiltonian dynamics; early citation traction
2025	Perturbation theory, irrep truncations, and state preparation methods for quantum simulations of SU(3) lattice gauge theory	2509.25865	SU(3) strong-coupling perturbation theory, irrep truncation, and state preparation
2025	Quantum Circuits for SU(3) Lattice Gauge Theory	2503.08866	electric-basis circuits for pure SU(3) Hamiltonian dynamics in two and three dimensions
2024	A Fully Gauge-Fixed SU(2) Hamiltonian for Quantum Simulations	2409.10610	fully gauge-fixed SU(2) Hamiltonian for quantum simulation
2024	An efficient finite-resource formulation of non-Abelian lattice gauge theories beyond one dimension	2409.04441	loop-variable finite-resource non-Abelian lattice gauge theory beyond one dimension
2024	Floquet evolution of the q-deformed $\text{SU}(3)_1$ Yang-Mills theory on a two-leg ladder	2409.20263	Floquet dynamics of q-deformed SU(3) Yang-Mills on a ladder
2024	From square plaquettes to triamond lattices for SU(2) gauge theory	2401.14570	alternative lattice geometries for SU(2) Hamiltonian gauge theory
2024	Gauge Loop-String-Hadron Formulation on General Graphs and Applications to Fully Gauge Fixed Hamiltonian Lattice Gauge Theory	2409.13812	Modern general-graph loop-string-hadron construction with full gauge fixing
2024	Loop-string-hadron approach to SU(3) lattice Yang-Mills theory: I. Hilbert space of a trivalent vertex	2407.19181	SU(3) gauge-invariant loop-string-hadron Hilbert space
2024	Protecting gauge symmetries in the the dynamics of SU(3) lattice gauge theories	2404.12158	gauge-symmetry protection in real-time SU(3) Hamiltonian evolution
2024	Quantum Simulation of SU(3) Lattice Yang Mills Theory at Leading Order in Large N	2402.10265	SU(3) Hamiltonian quantum simulation at large N
2024	Tensor Networks for Lattice Gauge Theories beyond one dimension: a Roadmap	2407.03058	tensor-network roadmap for lattice gauge theories beyond one spatial dimension
2023	q-deformed formulation of Hamiltonian SU(3) Yang-Mills theory	2306.12324	q-deformed Hamiltonian formulation of SU(3) Yang-Mills theory
2023	A new basis for Hamiltonian SU(2) simulations	2307.11829	Fast-cited modern basis for Hamiltonian SU(2), designed for controlled continuum extrapolation

Year	Title	arXiv	Connection
2023	Hamiltonians and gauge-invariant Hilbert space for lattice Yang-Mills-like theories with finite gauge group	2301.12224	explicit gauge-invariant Hilbert-space construction for finite-group Yang-Mills analogues
2023	Quantum Simulation of Lattice QCD with Improved Hamiltonians	2307.05593	improved Hamiltonians for lattice-QCD quantum simulation
2023	Quantum and classical spin network algorithms for q -deformed Kogut-Susskind gauge theories	2304.02527	q -deformed Kogut-Susskind Hamiltonians and finite-resource spin-network algorithms
2023	Quantum simulation of lattice gauge theories via deterministic duality transformations assisted by measurements	2305.12277	measurement-assisted deterministic duality transformations for gauge-invariant simulation
2023	SU(2) Gauge Theory in $2+1$ Dimensions on a Plaquette Chain Obeys the Eigenstate Thermalization Hypothesis	2303.14264	eigenstate thermalization and Wilson-loop matrix elements in a $2+1$ D SU(2) plaquette-chain Hamiltonian
2023	Simple Hamiltonian for Quantum Simulation of Strongly Coupled $2+1$ D SU(2) Lattice Gauge Theory on a Honeycomb Lattice	2307.00045	strong-coupling honeycomb Hamiltonian for $2+1$ D SU(2) Yang-Mills
2023	Spin exchange-enabled quantum simulator for large-scale non-Abelian gauge theories	2305.06373	spin-exchange quantum simulator for scalable non-Abelian gauge dynamics
2023	Strategies for simulating time evolution of Hamiltonian lattice field theories	2312.11637	time-evolution strategies and error analysis for Hamiltonian lattice field theory
2023	String-net formulation of Hamiltonian lattice Yang-Mills theories and quantum many-body scars in a nonabelian gauge theory	2305.05950	Hamiltonian lattice Yang-Mills string-net structure and non-Abelian scars
2022	Gauge Theory Couplings on Anisotropic Lattices	2208.10417	anisotropic SU(N) gauge couplings matched toward the Hamiltonian limit
2022	General quantum algorithms for Hamiltonian simulation with applications to a non-Abelian lattice gauge theory	2212.14030	general non-Abelian Hamiltonian algorithms using gauge-invariant variables
2022	Loop-string-hadron formulation of an SU(3) gauge theory with dynamical quarks	2212.04490	modern SU(3) gauge-invariant loop-string-hadron formulation
2022	Quantum Simulation for High Energy Physics	2204.03381	peer-reviewed high-energy quantum-simulation review
2022	Self-mitigating Trotter circuits for SU(2) lattice gauge theory on a quantum computer	2205.09247	self-mitigating real-time circuits for SU(2) Hamiltonian lattice gauge theory
2021	Exact duality and local dynamics in SU(N) lattice gauge theory	2109.00992	Exact duality and local dynamics for SU(N) Hamiltonian lattice gauge theory
2021	Preparation of the SU(3) Lattice Yang-Mills Vacuum with Variational Quantum Methods	2112.09083	pure SU(3) Hamiltonian vacuum preparation on plaquette chains

Year	Title	arXiv	Connection
2019	Loop, String, and Hadron Dynamics in SU(2) Hamiltonian Lattice Gauge Theories	1912.06133	Highly cited loop-string-hadron formulation and dynamics for SU(2) Hamiltonian LGT
2013	Ultracold Quantum Gases and Lattice Systems: Quantum Simulation of Lattice Gauge Theories	1305.1602	foundational quantum-simulation review for gauge theories
2003	SU(N) Glueball Masses in 2+1 Dimensions	hep-lat/0303016	Hamiltonian/analytic SU(N) glueball masses in 2+1D, a direct alternate spectral calculation
2003	SU(N) Lattice Gauge Theory on a Single Cube	hep-lat/0303022	gauge-invariant SU(N) Hamiltonian theory on a cube
2003	The Large N Glueball Mass Spectrum in 2+1 Dimensions	hep-lat/0303018	Large-N continuation of the Hamiltonian 2+1D glueball calculation
2000	Green's Function Monte Carlo study of SU(3) lattice gauge theory in (3+1)D	hep-lat/0005009	Green-function Monte Carlo SU(3) Hamiltonian study
2000	Towards a Many-Body Treatment of Hamiltonian Lattice SU(N) Gauge Theory	hep-lat/0001028	Many-body/coupled-cluster treatment specifically formulated for Hamiltonian SU(N)
1999	The basis of the physical Hilbert space of lattice gauge theories	hep-lat/9906036	Explicit spin-network basis of the gauge-invariant lattice Hilbert space
1996	The Coupled Cluster Method in Hamiltonian Lattice Field Theory	hep-lat/9603026	coupled-cluster Hamiltonian lattice gauge theory
1992	The Shifted Coupled Cluster Method: A New Approach to Hamiltonian Lattice Gauge Theories	hep-lat/9212025	Gauge-invariant shifted coupled-cluster construction from linked strong-coupling clusters

Glueball spectrum and continuum limits

Year	Title	arXiv	Connection
2026	Learning the generating functional for variance reduction in lattice QCD	2606.15986	learned generating functional for glueball-correlator variance reduction
2026	Update on the computation of the quenched $SU(6)$ Yang-Mills lattice spectrum	2603.13138	current quenched SU(6) Yang-Mills spectrum update
2026	Variance reduction in lattice QCD observables via normalizing flows	2603.02984	normalizing-flow variance reduction demonstrated on SU(3) glueball correlators
2025	Casimir scaling in glueballs in $SU(N)$ and $Sp(2N)$ gauge theories: hints from constituent approaches	2509.09454	very recent Casimir-scaling test for glueballs
2025	Glueball mass spectrum at finite temperature revisited: Constant contribution in glueball correlators in the deconfinement phase	2504.09120	finite-temperature SU(3) glueball correlators on anisotropic lattices

Year	Title	arXiv	Connection
2025	Lattice evidence that scalar glueballs are small	2508.21821	scalar-glueball form factors and radius
2025	Test of a two-level algorithm for the glueball spectrum in $SU(N_c)$ Yang-Mills theory	2501.19312	two-level glueball-spectrum test for $SU(N_c)$
2025	Update on Glueballs	2502.02547	modern peer-reviewed-proceedings review of glueball spectroscopy
2024	Exponential Error Reduction for Glueball Calculations Using a Two-Level Algorithm in Pure Gauge Theory	2406.12656	two-level exponential error reduction with a glueball variational basis
2024	Matching Wilson flow time for calculating the topological charge density and the pseudoscalar glueball mass in quenched lattice QCD	2409.02756	Wilson-flow matching for pseudoscalar glueball mass
2024	The θ -dependence of the Yang-Mills spectrum from analytic continuation	2402.03096	θ dependence of Yang-Mills spectrum
2023	$Sp(2N)$ Lattice Gauge Theories and Extensions of the Standard Model of Particle Physics	2304.01070	review of $Sp(2N)$ lattice spectra and large- N extrapolation
2023	Boundary states and Non-Abelian Casimir effect in lattice Yang-Mills theory	2302.00376	3+1D $SU(3)$ boundary-state and chromometallic Casimir spectrum
2023	Glueballs in $N_f=1$ QCD	2312.00470	Very recent $N_f=1$ glueball spectrum; citation count is still age-limited
2023	Performance of two-level sampling for the glueball spectrum in pure gauge theory	2312.11372	two-level sampling for glueball spectrum
2023	The quenched glueball spectrum from smeared spectral densities	2311.14806	smeared spectral densities for glueballs
2022	Confining strings, axions and glueballs in the planar limit	2205.03642	planar-limit confining strings and glueballs
2022	Glueball spectroscopy in lattice QCD using gradient flow	2211.15176	gradient-flow glueball spectroscopy
2022	Spectrum of Large N Glueballs: Holography vs Lattice	2206.14826	large- N lattice glueball spectrum comparison and synthesis
2022	The glueball spectrum with $N_f=4$ light fermions	2212.02080	Recent $N_f=4$ spectrum for understanding fermion effects; currently lightly cited
2022	Towards glueball masses of large- N $\mathrm{SU}(N)$ pure-gauge theories without topological freezing	2205.06190	large- N glueball masses without topological freezing
2021	$SU(N)$ gauge theories in 3+1 dimensions: glueball spectrum, string tensions and topology	2106.00364	modern continuum and large- N $SU(N)$ spectrum
2020	Color dependence of tensor and scalar glueball masses in Yang-Mills theories	2004.11063	color dependence of scalar and tensor glueballs
2020	Glueballs and Strings in $Sp(2N)$ Yang-Mills theories	2010.15781	comparative $Sp(2N)$ continuum glueball and string spectrum
2020	Prospects for large N gauge theories on the lattice	2001.10859	Modern survey of twisted reduction and large- N lattice prospects

Year	Title	arXiv	Connection
2020	Spectrum of Trace Deformed Yang-Mills Theories	2010.03618	Peer-reviewed spectrum of trace-deformed Yang-Mills, useful as a controlled deformation comparison
2020	The glueball spectrum of SU(3) gauge theory in 3+1 dimension	2007.06422	modern continuum SU(3) spectrum in 3+1 dimensions
2018	The spectrum of 2+1 dimensional Yang-Mills theory on a twisted spatial torus	1807.03481	Spectral study on a twisted spatial torus, directly probing volume reduction and winding sectors
2017	Glueball spectrum from $N_f=2$ lattice QCD study on anisotropic lattices	1702.08174	Anisotropic $N_f=2$ glueball spectroscopy with gluonic operators and mixing diagnostics
2016	SU(N) gauge theories in 2+1 dimensions: glueball spectra and k-string tensions	1609.03873	continuum SU(N) spectrum in 2+1 dimensions
2015	Topology and glueballs in SU(7) Yang-Mills with open boundary conditions	1512.00806	SU(7) open-boundary calculation linking topology sampling to glueball spectroscopy
2013	Introductory lectures to large-N QCD phenomenology and lattice results	1309.3638	Peer-reviewed lectures synthesizing large-N phenomenology and lattice evidence
2012	Recent results in large-N lattice gauge theories	1210.5510	Citation-guided review of then-current large-N lattice results
2012	The string tension from smeared Wilson loops at large N	1206.0049	Highly cited string-tension extraction using smeared Wilson loops at large N
2012	Towards the glueball spectrum from unquenched lattice QCD	1208.1858	Modern high-impact unquenched glueball spectrum and mixing study
2010	Glueball mass measurements from improved staggered fermion simulations	1005.2473	Improved-staggered calculation of glueball masses with dynamical sea quarks
2010	Glueball masses in the large N limit	1007.3879	large-N glueball masses
2005	Glueball Spectrum and Matrix Elements on Anisotropic Lattices	hep-lat/0510074	anisotropic-lattice spectrum and matrix elements
2005	Monte Carlo study of glueball masses in the Hamiltonian limit of SU(3) lattice gauge theory	hep-lat/0503038	Hamiltonian-limit SU(3) glueball masses
2004	Glueball Regge trajectories and the Pomeron – a lattice study –	hep-ph/0409183	High-impact lattice extraction of glueball Regge trajectories and the Pomeron connection
2004	Glueballs and k-strings in SU(N) gauge theories : calculations with improved operators	hep-lat/0404008	improved glueball and k-string operators with large-N scaling
2004	The spectrum of SU(N) gauge theories in finite volume	hep-lat/0412021	Dedicated finite-volume SU(N) spectrum and winding-state contamination analysis
2003	Glueball Regge Trajectories in (2+1) Dimensional Gauge Theories	hep-lat/0306019	Direct 2+1D lattice determination of glueball Regge trajectories

Year	Title	arXiv	Connection
2002	SU(N) gauge theories in 2+1 dimensions – further results	hep-lat/0206027	High-impact follow-up with additional SU(N) 2+1D spectrum/string results
2001	On the glueball spectrum in O(a)-improved lattice QCD	hep-lat/0108022	O(a)-improved unquenched spectrum with finite-volume/torelon mixing analysis
2001	SU(N) gauge theories in four dimensions: exploring the approach to $N = \infty$	hep-lat/0103027	Highly cited 4D SU(N) lattice study establishing the approach to $N=\infty$
2000	Static potentials and glueball masses from QCD simulations with Wilson sea quarks	hep-lat/0003012	Highly cited sea-quark study that explicitly treats glueball and torelon contamination
1999	The glueball spectrum from an anisotropic lattice study	hep-lat/9901004	benchmark glueball spectrum and operator construction
1998	Glueball masses and other physical properties of SU(N) gauge theories in $D=3+1$: a review of lattice results for theorists	hep-th/9812187	Citation-heavy synthesis of 3+1D SU(N) lattice masses and observables
1998	SU(N) gauge theories in 2+1 dimensions	hep-lat/9804008	Citation-heavy continuum SU(N) study in 2+1D, foundational for later spectrum tables
1997	Efficient glueball simulations on anisotropic lattices	hep-lat/9704011	Introduces efficient anisotropic-lattice glueball spectroscopy techniques used by the landmark 1999 spectrum
1993	A comprehensive lattice study of SU(3) glueballs	hep-lat/9304012	Classic high-statistics SU(3) spectrum; a major backward-citation anchor for modern continuum spectra

Flux tubes, strings and torelons

Year	Title	arXiv	Connection
2026	Entanglement Enabled Tomography of Flux Tubes in (2+1)D Yang-Mills Theory	2601.17199	2026 Yang-Mills flux-tube tomography preprint; unusually fast early citation uptake
2026	Imprints of asymptotic freedom on confining strings	2602.15097	2026 PRL connecting asymptotic freedom to measurable confining-string imprints
2025	The spectrum of open confining strings in the large- N_c limit	2506.16342	Latest peer-reviewed open-string spectrum extrapolated toward large N_c
2025	Thermal nature of confining strings	2510.23919	Very recent peer-reviewed analysis of the thermal character of confining strings

Year	Title	arXiv	Connection
2024	Born-Oppenheimer Potentials for $SU(3)$ Gauge Theory	2406.05123	lattice-constrained $SU(3)$ static spectrum connecting gluelumps and strings
2024	Confining Strings and the Worldsheet Axion from the Lattice	2411.03435	Modern lattice-focused synthesis of confining strings and the worldsheet axion
2024	Confining strings in three-dimensional gauge theories beyond the Nambu-Gotō approximation	2407.10678	Recent high-precision study of confining strings beyond Nambu-Goto
2024	Entanglement entropy of a color flux tube in (2+1)D Yang-Mills theory	2410.00112	Recent, well-cited lattice measurement of color-flux-tube entanglement structure
2024	Multiparticle Flux Tube S-matrix Bootstrap	2404.10812	Recent PRL multiparticle extension of the flux-tube S-matrix bootstrap
2023	Confining Strings and Glueballs in $\mathbb{Z}_N$ Gauge Theories	2312.03855	bulk glueball and flux-tube mixing isolated through Z_N comparisons
2023	Eight very excited spectra and one possible axion in $SU(3)$ lattice gauge theory	2303.15152	Eight excited $SU(3)$ flux-tube channels with candidate worldsheet-axion signals
2021	Hybrid static potentials in $SU(3)$ lattice gauge theory at small quark-antiquark separations	2111.00741	$SU(3)$ hybrid static potentials and flux-tube excitations at short source separation
2021	Revisiting the flux tube spectrum of 3d $SU(2)$ lattice gauge theory	2102.06413	Reanalysis of the 3D $SU(2)$ flux-tube spectrum and nonuniversal corrections
2021	Spectrum of very excited Σ_g^+ flux tubes in $SU(3)$ gauge theory	2105.12159	Direct $SU(3)$ extraction of very highly excited Σ_g^+ flux-tube levels
2021	The torelon spectrum and the world-sheet axion	2112.11213	modern $SU(3)$, $SU(5)$, and $SU(6)$ torelon spectrum and worldsheet axion
2019	Flux Tube S-matrix Bootstrap	1906.08098	Highly cited S-matrix bootstrap constraints for flux-tube worldsheet scattering
2018	Spectrum of the open QCD flux tube and its effective string description	1811.11779	Proceedings update/companion on the open QCD flux-tube spectrum
2018	Undressing Confining Flux Tubes with $T\bar{T}$	1808.01339	Modern $T\bar{T}$ interpretation of confining-flux-tube spectra
2017	Spectrum of the open QCD flux tube and its effective string description I: 3d static potential in $SU(N=2,3)$	1705.03828	Detailed open-flux-tube spectrum and effective-string comparison in 3D $SU(2,3)$
2016	Closed flux tubes in $D=2+1$ $SU(N)$ gauge theories: dynamics and effective string description	1602.07634	Detailed closed-flux-tube dynamics and effective-string analysis in 2+1D $SU(N)$
2016	Effective string description of confining flux tubes	1603.06969	modern QCD flux-tube worldsheet theory

Year	Title	arXiv	Connection
2015	Towards a Theory of the QCD String	1511.01908	Influential axionic-string proposal motivated by lattice QCD-string data
2014	A different kind of string	1406.5127	Influential analysis of non-Nambu-Goto worldsheet dynamics and its lattice signatures
2014	Flux Tube Spectra from Approximate Integrability at Low Energies	1404.0037	Highly cited approximate-integrability analysis benchmarked against lattice flux-tube spectra
2013	Evidence for a new particle on the worldsheet of the QCD flux tube	1301.2325	worldsheet dynamics of confining strings
2013	Quantisation of the effective string with TBA	1305.1278	Highly cited TBA quantization framework for the effective confining string
2013	The Effective Theory of Long Strings	1302.6257	effective theory of long strings
2012	Effective String Theory Revisited	1203.1054	effective string theory and finite-volume spectra
2012	SU(N) gauge theories at large N	1210.4997	large-N gauge theories lattice review
2012	The Lorentz-invariant boundary action of the confining string and its universal contribution to the inter-quark potential	1202.1984	Citation-established derivation of the universal Lorentz-invariant boundary correction
2011	Closed flux tubes and their string description in D=2+1 SU(N) gauge theories	1103.5854	closed flux tubes in 3+1 dimensions
2010	Closed flux tubes and their string description in D=3+1 SU(N) gauge theories	1007.4720	closed flux-tube spectrum and string description
2010	Corrections to Nambu-Goto energy levels from the effective string action	1008.2648	Systematic nonuniversal corrections to Nambu-Goto energy levels from the effective string action
2010	The Width of the Color Flux Tube at 2-Loop Order	1006.2252	Two-loop effective-string prediction for the color-flux-tube width
2010	The Width of the Confining String in Yang-Mills Theory	1002.4888	Highly cited lattice/effective-string determination of confining-string broadening in Yang-Mills
2009	Large N and confining flux tubes as strings - a view from the lattice	0912.3339	large-N and confining-flux-tube lattice review
2009	On the effective action of confining strings	0903.1927	Foundational classification of the confining-string effective action and its universal terms
2009	Spectrum of the QCD flux tube in 3d SU(2) lattice gauge theory	0905.4195	Direct lattice spectrum of the 3D SU(2) QCD flux tube
2008	Closed k-strings in SU(N) gauge theories : 2+1 dimensions	0802.1490	Closed k-string spectroscopy across SU(N) gauge groups
2008	On the spectrum of closed k=2 flux tubes in D=2+1 SU(N) gauge theories	0812.0334	Dedicated lattice spectrum of closed k=2 flux tubes in 2+1D SU(N)

Year	Title	arXiv	Connection
2007	The closed string spectrum of $SU(N)$ gauge theories in 2+1 dimensions	0709.0693	Closed-string spectrum in 2+1D $SU(N)$, an early direct Nambu-Goto comparison
2004	String excitation energies in $SU(N)$ gauge theories beyond the free-string approximation	hep-th/0406205	universal effective-string corrections
2004	Universal Subleading Spectrum of Effective String Theory	hep-th/0411017	Highly cited derivation of the universal subleading closed-string spectrum
2003	Excitations of the static quark-antiquark system in several gauge theories	hep-lat/0312019	Early multi-gauge-theory comparison of static-source excitation spectra
2002	Fine Structure of the QCD String Spectrum	hep-lat/0207004	Landmark high-citation lattice resolution of the fine structure of the QCD-string excitation spectrum
2001	Confining strings in $SU(N)$ gauge theories	hep-lat/0107007	Foundational numerical $SU(N)$ confining-string and k-string study

Rigorous lattice Yang–Mills

Year	Title	arXiv	Connection
2026	Confinement as Decoding: Higher Form Codes and Lattice Yang-Mills Theory	2608.02452	higher-form-code route to confinement and transfer-matrix mass
2026	Direct and Indirect Loop Equations in Lattice Yang-Mills Theory	2601.04316	modern direct and indirect lattice loop equations
2026	On the leading order term of the lattice Yang-Mills free energy		modern rigorous infinite-volume lattice Yang-Mills free-energy asymptotics
2026	The heat-kernel master field on $\mathbb{Z}^d$ at strong coupling	2606.28945	strong-coupling heat-kernel master field, $1/N$ expansion, area law, and master-loop equation
2025	$U(N)$ lattice Yang-Mills in the 't Hooft regime	2510.22788	new uniform strong-coupling $U(N)$ results
2025	Bootstrapping $SU(3)$ Lattice Yang-Mills Theory	2502.14421	$SU(3)$ lattice Yang-Mills bootstrap
2025	Correlation decay for $U(1)$ lattice Higgs theory: the case of small mass	2509.19176	constructive cluster expansion and correlation decay for small-mass $U(1)$ lattice Higgs theory
2025	Current expansion and couplings for Ising lattice gauge theory	2502.19942	random-current expansion and switching lemma for lattice gauge theory
2025	Dynamical approach to area law for lattice Yang-Mills	2509.04688	dynamical mass-gap condition implying Wilson area law
2025	Expanded regimes of area law for lattice Yang-Mills theories	2505.16585	expanded strong-coupling area-law regimes at large N

Year	Title	arXiv	Connection
2025	Surface sums in two-dimensional large- N lattice Yang-Mills: Cancellations and explicit computations for general loops	2508.13827	large- N two-dimensional lattice-Yang-Mills surface-sum cancellations
2025	Villain action in lattice gauge theory		modern rigorous Wilson/Manton-to-Villain limit for compact gauge groups
2024	Bootstrap for Finite N Lattice Yang-Mills Theory	2404.16925	finite- N lattice Yang-Mills bootstrap
2024	Langevin dynamics of lattice Yang-Mills-Higgs and applications	2401.13299	strong-coupling mass gap and exponential ergodicity with Higgs matter
2024	Makeenko-Migdal equations for 2D Yang-Mills: from lattice to continuum	2412.15422	rigorous convergence of two-dimensional lattice Makeenko-Migdal equations to continuum area derivatives
2024	Surface sums for lattice Yang-Mills in the large- N limit	2411.11676	large- N surface sums with signed-Catalan weights
2023	Geometric Derivation of the Finite N Master Loop Equation	2309.07399	geometric derivation of the finite- N lattice master loop equation
2023	Invariant measure and universality of the 2D Yang-Mills Langevin dynamic	2302.12160	invariant measure and universality for Yang-Mills Langevin dynamics
2023	Random surfaces and lattice Yang-Mills	2307.06790	random-surface expansion for lattice Yang-Mills
2022	A new derivation of the finite N master loop equation for lattice Yang-Mills	2202.00880	finite- N master loop equation for $SU(N)$, $U(N)$, and $SO(N)$
2022	A stochastic analysis approach to lattice Yang-Mills at strong coupling	2204.12737	strong-coupling mass gap and exponential clustering
2022	Bootstrap for Lattice Yang-Mills theory	2203.11360	large- N lattice Yang-Mills bootstrap from loop equations and positivity
2022	Correlation decay for finite lattice gauge theories at weak coupling	2202.10375	exponential correlation decay for finite lattice gauge theories at weak coupling
2022	Large N limit of the Yang-Mills measure on compact surfaces II: Makeenko-Migdal equations and planar master field	2201.05886	modern compact-surface large- N master field and Makeenko-Migdal theory
2022	Stochastic quantisation of Yang-Mills-Higgs in 3D	2201.03487	stochastic quantization and continuum control for three-dimensional Yang-Mills-Higgs
2021	A state space for 3D Euclidean Yang-Mills theories	2111.12813	rigorous state space for three-dimensional Euclidean Yang-Mills
2021	Decay of correlations in finite Abelian lattice gauge theories	2104.03752	rigorous exponential decay of correlations in finite Abelian lattice gauge theories

Year	Title	arXiv	Connection
2021	Wilson Loop Expectations for Non-Abelian Gauge Fields Coupled to a Higgs Boson at Low and High Disorder	2111.07540	non-Abelian Wilson-loop asymptotics with Higgs matter via vortex and polymer expansions
2021	Wilson lines in the Abelian lattice Higgs model	2111.06620	rigorous Wilson-line asymptotics in the Abelian lattice Higgs model
2020	A probabilistic mechanism for quark confinement	2006.16229	probabilistic confinement mechanism
2020	Langevin dynamic for the 2D Yang-Mills measure	2006.04987	rigorous Yang-Mills Langevin dynamics and Wilson observables
2020	Wilson loop expectations in lattice gauge theories with finite gauge groups	2001.05627	rigorous weak-coupling Wilson-loop estimates for finite gauge groups
2018	Wilson loops in Ising lattice gauge theory	1811.09770	rigorous weak-coupling Wilson-loop asymptotics in four-dimensional lattice gauge theory
2018	Yang-Mills for probabilists	1803.01950	rigorous/probabilistic lattice Yang-Mills review
2016	Loop Equations and bootstrap methods in the lattice	1612.08140	positivity bootstrap for lattice loop equations
2016	The $1/N$ expansion for $SO(N)$ lattice gauge theory at strong coupling	1604.04777	all-orders genus-graded $1/N$ surface expansion for lattice gauge theory
2016	Three proofs of the Makeenko-Migdal equation for Yang-Mills theory on the plane	1601.06283	three rigorous derivations of the planar Makeenko-Migdal equation
2016	Wilson loop expectations in $SU(N)$ lattice gauge theory	1610.03821	Wilson-loop expectations in $SU(N)$ lattice gauge theory
2015	Rigorous solution of strongly coupled $SO(N)$ lattice gauge theory in the large N limit	1502.07719	rigorous convergent large- N strong-coupling Wilson-loop surface expansion
2011	The master field on the plane	1112.2452	planar large- N master field characterized by Makeenko-Migdal equations

Spectral extraction and operator overlap

Year	Title	arXiv	Connection
2026	Charmonium-Glueball spectroscopy with improved hadron creation operators	2603.20178	modern derivative-based glueball operators and overlap analysis
2024	Bayesian solution to the inverse problem and its relation to Backus-Gilbert methods	2409.04413	Modern Bayesian interpretation and extension of Backus-Gilbert inverse methods
2024	Block Lanczos algorithm for lattice QCD spectroscopy and matrix elements	2412.04444	Rapidly cited block-Lanczos spectroscopy/matrix-element method published in 2025
2024	Lanczos, the transfer matrix, and the signal-to-noise problem	2406.20009	transfer-matrix Lanczos extraction with two-sided error bounds

Year	Title	arXiv	Connection
2024	Spectral densities from Euclidean lattice correlators via the Mellin transform	2407.04141	Recent Mellin-transform framework with controlled lattice spectral-density reconstruction
2023	Beyond Generalized Eigenvalues in Lattice Quantum Field Theory	2309.05111	Modern proposal going beyond the standard generalized-eigenvalue analysis
2023	Teaching to extract spectral densities from lattice correlators to a broad audience of learning-machines	2307.00808	Peer-reviewed learning-machine benchmark for spectral-density extraction
2022	Yang-Mills glueball masses from spectral reconstruction	2212.01113	independent Yang-Mills spectral reconstruction of glueball masses
2021	Källén-Lehmann Spectral Representation of the Scalar SU(2) Glueball	2103.11846	Kallen-Lehmann reconstruction of the scalar SU(2) glueball
2020	Reconstruction of smeared spectral function from Euclidean correlation functions	2001.11779	Widely used reconstruction method for smeared spectral functions from Euclidean data
2019	On the extraction of spectral densities from lattice correlators	1903.06476	Highly cited regularized method for extracting smeared spectral densities from Euclidean correlators
2013	Extended hadron and two-hadron operators of definite momentum for spectrum calculations in lattice QCD	1303.6816	Highly cited group-theoretic construction of extended momentum-projected operator bases for lattice spectra
2010	Properties and uses of the Wilson flow in lattice QCD	1006.4518	Wilson flow and smoothed gauge observables
2009	A novel quark-field creation operator construction for hadronic physics in lattice QCD	0905.2160	distillation and optimized hadron operators
2009	On the generalized eigenvalue method for energies and matrix elements in lattice field theory	0902.1265	generalized eigenvalue method with error control
2008	Efficient use of the Generalized Eigenvalue Problem	0808.1017	Practical companion on efficient GEVP use, distinct from the 2009 asymptotic-error paper already collected
2003	Analytic Smearing of SU(3) Link Variables in Lattice QCD	hep-lat/0311018	analytic stout-link smearing
2002	Locality and Statistical Error Reduction on Correlation Functions	hep-lat/0209145	local multilevel error reduction for glueball correlation functions
2001	Locality and exponential error reduction in numerical lattice gauge theory	hep-lat/0108014	multilevel exponential error reduction

Weingarten and Haar integration

Year	Title	arXiv	Connection
2026	Universal dualities for Wilson loops in lattice Yang-Mills	2604.16252	finite-N Wilson-loop dualities and Weingarten state sums
2025	Weingarten Calculus with Virtual Isometries	2510.21186	new virtual-isometry extension
2022	Expectation values of polynomials and moments on general compact Lie groups	2203.11607	Weingarten maps and polynomial moments for general compact Lie groups including $SU(N)$
2022	Moment Methods on compact groups: Weingarten calculus and its applications	2207.08418	compact-group moment methods
2021	The Weingarten Calculus	2109.14890	modern Weingarten-calculus review with lattice-gauge application
2021	Weingarten Calculus	2101.00921	comprehensive Weingarten-calculus survey
2018	Matrix Group Integrals, Surfaces, and Mapping Class Groups I: $SU(n)$	1802.04862	Haar unitary integrals encoded by surfaces and mapping-class groups
2018	$SU(N)$ polynomial integrals and some applications	1812.06069	explicit $SU(N)$ polynomial integrals for lattice-gauge applications
2017	Weingarten calculus via orthogonality relations: new applications	1701.04493	orthogonality-recursion method
2016	Revisiting $SU(N)$ integrals	1611.00236	explicit invariant integration formulas for $SU(N)$
2013	Weingarten calculus for matrix ensembles associated with compact symmetric spaces	1301.5401	Weingarten calculus for compact symmetric spaces
2009	On some properties of orthogonal Weingarten functions	0903.5143	structural and asymptotic properties of orthogonal Weingarten functions
2008	Complete homogeneous symmetric polynomials in Jucys-Murphy elements and the Weingarten function	0811.3595	Jucys-Murphy and symmetric-function formula for the unitary Weingarten function
2006	Asymptotics of unitary and orthogonal matrix integrals	math/0608193	asymptotics of unitary and orthogonal matrix integrals used in surface weights
2005	Integration over compact quantum groups	math/0511253	Weingarten integration extended to compact quantum groups
2004	Integration with respect to the Haar measure on unitary, orthogonal and symplectic group	math-ph/0402073	foundational Haar-unitary integration formula
2002	Moments and Cumulants of Polynomial random variables on unitary groups, the Itzykson-Zuber integral and free probability	math-ph/0205010	unitary matrix moments, cumulants, and Itzykson-Zuber asymptotics

Effective Hamiltonians and strong coupling

Year	Title	arXiv	Connection
2025	The Spectral Renormalization Flow Based on the Smooth Feshbach–Schur Map: The Introduction of the Semi-Group Property	2508.07643	modern smooth Feshbach-Schur spectral flow with a semigroup property
2024	The geometry of the Hermitian matrix space and the Schrieffer–Wolff transformation	2407.10478	modern geometric treatment of Schrieffer-Wolff
2022	A mathematical framework for quantum Hamiltonian simulation and duality	2208.11941	precise mathematical framework for Hamiltonian simulation and duality
2021	The Feshbach-Schur map and perturbation theory	2105.02058	explicit Feshbach-Schur perturbation estimates
2020	Hamiltonian simulation in the low-energy subspace	2006.02660	explicit leakage and error bounds for low-energy-subspace Hamiltonian simulation
2017	Efficient optimization of perturbative gadgets	1709.02705	optimized perturbative-gadget error bounds at arbitrary order
2017	Universal Quantum Hamiltonians	1701.05182	rigorous low-energy Hamiltonian-simulation framework and universal Hamiltonians
2014	Perturbative gadgets without strong interactions	1408.5881	convergence and error control for perturbative gadgets without strong interactions
2011	Schrieffer-Wolff transformation for quantum many-body systems	1105.0675	rigorous Schrieffer-Wolff transformation and error bounds
2010	Strong-coupling expansion and effective hamiltonians	1005.2495	strong-coupling effective-Hamiltonian expansion review
2008	Perturbative Gadgets at Arbitrary Orders	0802.1874	arbitrary-order perturbative gadgets with explicit effective interactions
2008	Simulation of Many-Body Hamiltonians using Perturbation Theory with Bounded-Strength Interactions	0803.2686	bounded-strength many-body simulation by perturbative effective Hamiltonians
2007	On the Smooth Feshbach-Schur Map	0704.3244	smooth Feshbach-Schur map
2004	Higher order effective low-energy theories	cond-mat/0407255	fourth-order low-energy expansion comparing canonical and projection-resolvent methods
2004	The Complexity of the Local Hamiltonian Problem	quant-ph/0406180	rigorous low-energy projection gadgets for encoded Hamiltonians

Flat bands and compact localized states

Year	Title	arXiv	Connection
2026	New class of exactly flat topological bands - compact localised states protected by local graph topology	2607.22831	exact topological flat bands protected by local graph-index structure

Year	Title	arXiv	Connection
2023	Flat bands of periodic graphs	2304.06465	mathematical classification of flat bands on periodic graphs
2020	Fragile topology in line-graph lattices with two, three, or four gapped flat bands	2010.11953	line-graph cycle-space flat bands and fragile topology
2020	Singular flat bands	2012.04279	singular-flat-band classification, topology, and compact-state completeness
2019	Line-Graph Lattices: Euclidean and Non-Euclidean Flat Bands, and Implementations in Circuit Quantum Electrodynamics	1902.02794	line-graph lattices and homological flat bands
2018	Artificial flat band systems: from lattice models to experiments	1801.09378	highly cited flat-band systems review
2018	Classification of flat bands according to the band-crossing singularity of Bloch wave functions	1808.05926	classification of singular and nonsingular flat bands
2017	Compact localized states and flat bands from local symmetry partitioning	1709.07806	local-symmetry construction of compact localized states and flat bands
2016	Compact localized states and flatband generators in one dimension	1610.02970	classification and generation of one-dimensional compact localized states
2015	Strongly correlated flat-band systems: The route from Heisenberg spins to Hubbard electrons	1502.02729	strongly correlated flat-band review
2008	Band touching from real space topology in frustrated hopping models	0803.3628	real-space topology and singular band touching
2005	On the structure of eigenfunctions corresponding to embedded eigenvalues of locally perturbed periodic graph operators	math-ph/0511084	compactly supported eigenfunctions for perturbed periodic graph operators
1998	Aharonov-Bohm cages in two-dimensional structures	cond-mat/9806068	Aharonov-Bohm cages and exact compact localization by flux interference
1997	From Nagaoka's ferromagnetism to flat-band ferromagnetism and beyond: An introduction to ferromagnetism in the Hubbard model	cond-mat/9712219	rigorous flat-band ferromagnetism review and theorem synthesis

Strong-coupling expansions

Year	Title	arXiv	Connection
2010	Centre symmetric 3d effective actions for thermal SU(N) Yang-Mills from strong coupling series	1010.0951	strong-coupling effective theory review/application
2008	Strong coupling expansion for finite temperature Yang-Mills theory in the confined phase	0805.1163	strong-coupling thermodynamics and equation of state
2007	Strong coupling expansion for Yang-Mills theory at finite temperature	0710.0512	finite-temperature strong-coupling expansion

Year	Title	arXiv	Connection
1985	Effective transfer matrix for low-lying glueball states in lattice gauge theory		effective transfer-matrix eigenvalues and low-lying glueball dispersion
1983	Strong coupling and mean field methods in lattice gauge theories		canonical review of character, graph, cluster, correlation, and spectral strong-coupling machinery

H The page-level extraction index

777 extracted pages across 28 works, indexed by topic and line position. The workflow this supports is in §3: search by paper and topic, follow the locator into the reading copy, verify the equation in the pinned PDF.

The index records locators rather than republishing source text. A topic hit may be a citation rather than a result, or an extraction error; candidate theorem labels may sit inside citations or carry OCR faults. Five pages extracted fewer than eighty characters and are flagged unreliable in the source index — their contents must not be inferred from the text. Exact equations from legacy scans must be checked in the PDF, not in the extraction.

777 extracted pages across 28 works, indexed by topic and line. A row says how many pages of that work are searchable and which topics occur in it; the index records *locators*, not text, so the reading copy under `literature/inbox/<ID>/extracted/` is where the words are. A hit may be a citation or an extraction error: the index is a search aid and certifies nothing.

Work	Pages	Labelled	Topics present
BFS_1980_II	85	52	gauge_geometry (89), constructive (86), uniform_limits (78), reconstruction (11), ground_state (3)
MMM_2019_CURVATURE	80	0	curvature (93), gauge_geometry (82), ground_state (45), spectral_gap (18), constructive (2), uniform_limits (2)
WITTEN_1994_FOUR_MANIFOLD	67	0	gauge_geometry (57), spectral_gap (37), ground_state (35), constructive (5), curvature (1), uniform_limits (1)
MRS_1993	60	11	gauge_geometry (171), uniform_limits (141), constructive (63), reconstruction (5), curvature (1)
BRS_1976	58	0	gauge_geometry (46), constructive (20), ground_state (2)
SW_1994	45	0	gauge_geometry (67), ground_state (21), constructive (12), spectral_gap (3)
IMBRIE_1981_I	44	20	ground_state (26), constructive (17), uniform_limits (4), reconstruction (3), spectral_gap (2), curvature (1)
MONDAL_2023_GEOMETRIC	42	10	curvature (179), gauge_geometry (135), spectral_gap (42), ground_state (40), constructive (28), uniform_limits (5)
BBIJ_1984	39	16	gauge_geometry (30), spectral_gap (21), uniform_limits (11), constructive (10), ground_state (3), reconstruction (3)
IMBRIE_1981_II	39	27	ground_state (13), uniform_limits (13), constructive (12), reconstruction (4), spectral_gap (2)

Work	Pages	Labelled	Topics present
OS_1973	30	19	reconstruction (68), uniform_limits (9), gauge_geometry (2), spectral_gap (2), ground_state (1), constructive (1)
OS_1975	26	15	reconstruction (49), uniform_limits (4), gauge_geometry (2), ground_state (1)
FROHLICH_1982	25	0	constructive (11), reconstruction (4), uniform_limits (3), ground_state (1)
MALDACENA_1998	22	0	gauge_geometry (11), curvature (2), ground_state (2), constructive (2)
JAFFE_2000_CQFT	16	0	constructive (33), ground_state (20), reconstruction (17), gauge_geometry (15), spectral_gap (10), curvature (2)
JW_2006	14	0	gauge_geometry (66), spectral_gap (25), constructive (24), reconstruction (22), ground_state (9), uniform_limits (6)
THOOF_1974	13	0	gauge_geometry (24), spectral_gap (1)
OS_SENEOR_1976	13	9	reconstruction (11), uniform_limits (11), ground_state (2), constructive (2)
SIMON_1983_DISCRETE	12	2	spectral_gap (20), ground_state (2), gauge_geometry (1), uniform_limits (1)
MR_1976_CRITICAL	10	3	reconstruction (6), uniform_limits (5), constructive (4), ground_state (1), spectral_gap (1)
CREUTZ_1980	8	0	gauge_geometry (39), uniform_limits (22), constructive (22), ground_state (1), reconstruction (1)
GJ_1968_I	7	5	uniform_limits (18), ground_state (8), reconstruction (2), gauge_geometry (1), constructive (1), spectral_gap (1)
YANG_MILLS_1954	5	0	gauge_geometry (31)
GJ_1974_SINGLE_PHASE	4	2	gauge_geometry (6), ground_state (3), constructive (3), uniform_limits (2), reconstruction (1)
GW_1973	4	0	constructive (26), gauge_geometry (26), ground_state (3), uniform_limits (1)
POLITZER_1973	4	0	constructive (13), gauge_geometry (6), ground_state (4), uniform_limits (1)
WEINBERG_1967	3	0	gauge_geometry (10), constructive (7), ground_state (5)
WILSON_1977_LECTURES	2	0	gauge_geometry (8), constructive (2), uniform_limits (1)

Searchable topics, with total line hits: **gauge_geometry** (925), **constructive** (406), **uniform_limits** (340), **curvature** (284), **ground_state** (251), **reconstruction** (207), **spectral_gap** (185).

I Symbols and join keys

28 registered symbols. Each carries a canonical name, the spellings the source corpus uses, the names the code uses, the exact values, and the claims it appears in.

This registry is what makes a quantity findable across a manuscript written in one notation, code written in another, and a claim recorded in a third. It is also where the corpus records *forbidden* names — spellings that regenerate a dissolved contradiction every time someone reads them, such as writing a single “ m_4 ” for two differently anchored coordinates (§4) — and names

coined inside this program that the source corpus never uses, which must not be searched for in the archive as though they were.

The join keys of this corpus are exact values and spellings, not concepts: no natural-language query retrieves 109151/249696. The registry below is what makes a value findable across a document written in one notation, code written in another, and a claim recorded in a third.

A_shp values: 5/48, 0.1041666666666728

Sealed fourth-order shape coefficient 5/48. Both disputed kernels agree on it, which is what "sealed core" means.

corpus spellings: 5/48, A_{old} in code: A_SHP_3, A_SHP_3_NUM claims: R7, G14

beta_3^bal values: 15644916262153/34416487661400

The balanced (epsilon-free) continuation of the all-rank shape formula at $N = 3$: $P17(9)/(3 R20(9))$, equal to the pinned structured B_N expression there. Differs from the historical β_3 by exactly 25/64 (the recorded C10 shift);

corpus spellings: B_3^{bal} , q_N^{bal} in code: beta_formula claims: C2, C10

q_a, e_2, e_3

Elementary symmetric functions of $a_i = 4 \sin^2(k_i/2)$. Every fourth-order shape coefficient is a symmetric function of these – that is U2.

corpus spellings: q_a , e_2 , e_3 claims: U2, G14

c_4^adj values: -53/768, -31/1536

The cube-completion channel between two ADJACENT faces of a unit cube, $-106/(N(N^2-1)^3)$ at every rank: the primitive simple-loop law's -88 plus ten multi-loop orderings at -18 (ADR 0024). In the kernel's own (0,2) basis it is the -53/768 the rotation record carries;

in code: CUBE_COMPLETION_ADJACENT_4, CUBE_COMPLETION_ADJACENT_HISTORICAL_4, cube_completion_adjacent claims: C2, G3, G14

c_{2,prim}^tet values: -1/3

Tetrahedral primitive local Haar-resolvent coefficient, the $r = 2$ row of the corpus's primitive completion law $c_{\{r, \mathrm{prim}\}}$. Asserted in the corpus with no artifact or reference SHA anywhere;

corpus spellings: $-8/[N(N^2-1)]$, tetrahedral completion, $c_{\{r, \mathrm{prim}\}}$ in code: TETRA_COMPLETION_2_SU3, c_prim_printed claims: C15, G5

C_shp values: -211835444920651/4405310420659200, -0.04808638318135875, -0.020213328886166577

Off-axis fourth-order shape coefficient. Resolved 2026-09-04 (ADR 0024): $C_{\mathrm{historical}} + 25/1024$.

corpus spellings: C_shape, C_{old} , C_{new} , $C^{\{4\}}$ in code: C_SHP_HISTORICAL, C_SHP_NEW_NUM claims: C2, R5, G3

d_corner(N) values: 2580244782961/398756546697600, -56022878647/4153714028100

The corner-cluster cumulant that decided C2, as a function of N (ADR 0025): (23, 30) rational functions in both sectors with denominator $(N^2-1)^3(4N^2-9)^3(4N^2-1)(9N^2-25)(2N^2-1)^3(3N^2-1)(4N^2-5)(4N^2-2N-5)(4N^2+2N-5)$. At $N = 3$, in the kernel's (0,2) basis, +2580244782961/398756546697600 and -56022878647/4153714028100.

in code: CORNER_DRESSING_ODD_N, CORNER_DRESSING_EVEN_N claims: G14, C2, G3

Delta_C values: 0.027873054295192174

Off-axis shift $C_{\mathrm{new}} - C_{\mathrm{old}}$. Gamma-point data cannot constrain it, because the kernel it multiplies vanishes at Gamma.

corpus spellings: `Delta_C`, `\Delta_C` in code: `DELTA_C_NUM` claims: C2, G3

Delta_Gamma values: 2.0827701250956417, 2.0827701250956414

Scalar shift $m_{\text{Gamma}}^{(4)} - q_{\text{band}}^{(4)}$. Fixed by Gamma-point data.

corpus spellings: `\Delta_Gamma` in code: `DELTA_GAMMA_NUM`, `DELTA_GAMMA_AS_PRINTED_NUM` claims: C1, R6

ell_N values: -11/306

All-rank C-even second-order hopping. Printed in the corpus and gated in a notebook as `factor(Wmix + Wlike + 1/CF)`, unregistered here until 2026-08-28.

corpus spellings: `ell_N`, `-2*N*(3*N**2 - 5)` in code: `even_hopping`, `T_PLUS_2` claims: C13, R2

h_4^side values: -2861009/84387303000

Isotropic pentagonal-prism fourth-order side coefficient, $A_+ - A_-$. Separate geometry;

corpus spellings: `h4_side`, `h_4^{\mathrm{side}}` in code: `H4_SIDE` claims: R20, R21, G4

lambda_X, lambda_M, lambda_R

Generalized Hodge pencil eigenvalues. $\lambda_R = 2 \lambda_M - \lambda_X$ is the blind holdout used to test an extraction without fitting to it.

corpus spellings: `lambda_R`, `\lambda_R` claims: G3

m_6

Exact sixth-order coefficient. Belongs to the branch whose fourth-order scalar is $q_{\text{band}}^{(4)}$;

corpus spellings: `m_6` claims: R15, G9

m_Gamma^(4) values: -0.7751458630189173

Vacuum-subtracted physical Gamma-point fourth-order coefficient. Blind match to Hamer's a_4 through $m_n = 2^{**}(n-1) a_n$.

corpus spellings: `m_Gamma^{(4)}` in code: `M_GAMMA_4_NUM` claims: C1, R4

Phi_C

The crosswalk kernel $4 e_2(k) / q_a(k)$. $\Phi_C(0) = 0$, which is why the Gamma-point scalar places no constraint on `Delta_C` – the structural reason C2 cannot be closed by re-anchoring.

corpus spellings: `\frac{4e_2}{q_a}`, `4e_2/q` in code: `phi_c` claims: C2, G3

pi~ values: -205125054864713/12642703205932800

The coplanar shared-link fourth-order amplitude with the two-hop shadow removed: $\pi_{\sim} = \pi + 2u$. In the Hodge form of the kernel it multiplies S_{sq} itself, and $C_{\text{shp}} = -5/96 - (\rho + \pi_{\sim})/2$ exactly, with no u .

in code: `PI_REDUCED` claims: C2, G3, G14

q_band^(4) values: -20721577909065127111/7250590288602460800, -2.857915988114559

Fourth-order rest scalar in the band-kernel anchoring. NOT a competing estimate of $m_{\text{Gamma}}^{(4)}$ – a differently anchored coordinate.

corpus spellings: `q_{\mathrm{band}}^{(4)}`, `q_{\mathrm{old}}^{(4)}`, `q_old` in code: `Q_BAND_4` claims: C1, R4

rho_assembled values: -588708011765248393/14501180577204921600, -13035490122347/550663802582400

The cluster-assembled rotation amplitude read in the kernel's own basis, the x-then-z traversal of the (0,2) face in which all 24 cross-plane records are ρ times the L_{down} incidence: minus the 2026-09-02 run's `RHO_CLUSTER`. It differs from `RHO_ORBIT` by exactly 25/512, the adjacent-face cube completion, and gives $C_{\text{shp}} = C_{\text{historical}} + 25/1024$ (ADR 0024).

in code: `RHO_ASSEMBLED, C_SHP_ASSEMBLED` claims: C2, G3

w_R

The resolvent weight of one shared-link fusion channel, $-(d_R/N^2)/(C_F + C_R/2)$, summing to A_N over $F(x)$ F_{bar} and B_N over $F(x)$ F . The numerator d_R/N^2 is the manuscripts' "isotropy" sentence: asserted there, and derived here from the order-2 Weingarten values – the six nonshared links collapse to two independent Haar matrices, and two degree-(2,2) moments (N^2 and N) settle both families.

corpus spellings: `w_R, W_mix, Wlike` in code: `channel_weight, channel_data, shared_link_weights` claims: G24, R2

sigma~

values: -1249434547013129251/483372685906830720

The on-site fourth-order amplitude with the two-hop shadow removed: $\text{sigma}\sim = \text{sigma} - 12u$. It is the constant term of the Hodge form and enters the carrier band through c_0 alone.

in code: `SIGMA_REDUCED` claims: C2

sigma(u)

values: -61/408, -22/153, -137767222189182735950309/2009803206414863779920000

Native string tension through fifth order. Odd physical coefficients carry $(-1)^n$ relative to the raw unit-insertion magnitudes.

corpus spellings: `\sigma_n, sigma_5` in code: `SIGMA_0, SIGMA_2, SIGMA_3, SIGMA_4, SIGMA_5, SIGMA_5_RAW` claims: C5, R14, G7

d_single(N)

values: -385/1997568, -121/249696

The fourth-order dressing of a shared-link pair by a plaquette touching one face only, as a function of N (ADR 0025): C-odd $2N^3(N^2-4)(10N^2-13)/((N^2-1)^3(4N^2-9)^2(2N^2-1)^2)$, C-even $-\ell_N^2/(2C_F)$. Reconstructed from the third engine at $N = 3..70$ and verified on held-out ranks (ADR 0025), then computed outright over the field $Q(N)$ (ADR 0029): the same rational function, derived.

in code: `SINGLE_CONTACT_DRESSING_ODD_N, SINGLE_CONTACT_DRESSING_EVEN_N` claims: G14, C2

t_N

values: 5/612

All-rank second-order hopping, $B_N - A_N$. $t_2 = 0$, $t_3 = 5/612$.

corpus spellings: `t_N, t_-` in code: `T_MINUS_2, hopping` claims: R2

tau_4

values: -2861009/16877460600

Pentagonal cap hopping coefficient, h_4^{side} times 5 by exact D_5 covariance. The symbol is $2\tau_4 \cos k$.

corpus spellings: `\tau_4, tau_4` in code: `TAU_4` claims: R21, G4

u_2hop(N)

values: 360421351/40327601932800, 948253471/40327601932800

The two-hop chain cumulant, the one dynamical input of the Hodge form (ADR 0019), as a function of N (ADR 0025): (21, 28) rational functions in both sectors with denominator $2(N^2-1)^3(4N^2-9)^3(9N^2-16)(2N^2-3)(2N^2-1)^3(3N^2-2)(4N^2-N-4)(4N^2+N-4)$; the C-odd numerator carries $(N^2-4)^2$.

in code: `TWO_HOP_WEIGHT_ODD_N, TWO_HOP_WEIGHT_EVEN_N` claims: G14, C2

u_even

values: 948253471/40327601932800

The C-even fourth-order chain amplitude: the cumulant $W(P,Q,R) - W(P,R)$ on the $P \rightarrow R$ element in the C-even sector, for Q sharing one link with each of the link-disjoint P and R . The same rational on all three chain types (coplanar-coplanar, coplanar-perpendicular, perpendicular-perpendicular), with no incidence sign because the C-even shared-link hop -11/306 carries none.

in code: `U_CHAIN_C_EVEN` claims: G3, G14, C2

W_4

values: 0.48061786909826, 0.9265867378213348

Fourth-order bandwidth. Enters the near-Gamma crossover constant only through $\text{sqrt}(W_4)$, which is why the G11 criterion survives C2.

corpus spellings: `W_4` in code: `W4_HISTORICAL_NUM, W4_NEW_NUM` claims: C2, G11

x (degree-3 record quantum) values: 360421351/40327601932800

The single amplitude quantizing every degree-3 channel coefficient of the historical 189-record kernel: each block's q_{e_2} and q_{e_3} amplitude is an integer multiple of x , itself a raw record weight, and the tier collapse is $(+1,+1,-2)x$ and $(-3,+6,-3)x$ exactly (suite "off-axis channel ledger"). Coined by the maintainer's off-axis ledger (`WORK_SINCE_2026-08`);

in code: `X_QUANTUM` claims: C2, G14

J Documents and their standing

248 source documents with a registered alias and a standing.

Standing is the field that matters most when reading the archive, because a superseded document is not marked in its own text: it reads exactly like a current one, states its conclusions with the same confidence, and was correct when written. The register is the only thing that distinguishes them.

The archive's governing instruction is that nothing is erased, which makes this register load-bearing rather than administrative. Preserved material includes failed attempts, superseded drafts and duplicates, all deliberately — deleting a refuted claim destroys the evidence that it was tried, which is the same principle behind the dead-route register in §17.

248 source documents carry a registered alias and a *standing*. Standing is the single most important field for an agent reading the archive: it says whether a document is current, superseded, or preserved only as evidence of a stage. A superseded document is not deleted and reads exactly like a current one.

Standing: repo (159)

Alias	Path
ARCHIVE_DERIVATION_CHAIN	paper/research_notes/ARCHIVE_DERIVATION_CHAIN_20260911.md
ARCHIVE_FUNCTIONAL	docs/derivations/archive-functional-inequalities.md
ARCHIVE_REVIEW_SOURCES	runs/archive_derivations_2026-09-11/README.md
ARCHIVE_SOURCE_TRANSPORT	docs/derivations/archive-source-transport.md
CONDITIONAL_GRADIENT_REPAIR	paper/research_notes/G19_CONDITIONAL_GRADIENT_REPAIR_20260905.md
CONDITIONAL_QUANTUM_PATH_COVARIANCE	paper/research_notes/G19_CONDITIONAL_QUANTUM_PATH_COVARIANCE_20260906.md
COUPLED_OSCILLATOR_METHOD_REVIEW	paper/research_notes/G19_COUPLED_OSCILLATOR_METHOD_REVIEW_20260905.md
CUBIC_GROUND_TRANSFER	paper/research_notes/G19_CUBIC_GROUND_TRANSFER_20260906.md
DYNAMIC_CONDITIONAL_CUBIC_ENERGY	paper/research_notes/G19_DYNAMIC_FIBER_COVARIANCE_AND_CUBIC_ENERGY_20260906.md
EARLIER_PROOF_RECOVERY	docs/derivations/earlier-proof-recovery.md
EARLIER_PROOF_RECOVERY_RESULT	paper/research_notes/EARLIER_PROOF_RECOVERY_20260911.md
EARLIER_RECOVERY_SOURCE_01	runs/earlier_proof_recovery_2026-09-11/sources/05_affine_laplacian_law_analytic_proof.md

Alias	Path
EARLIER_RECOVERY_SOURCE_02	runs/earlier_proof_recovery_2026-09-11/sources/WORKHOUSE_CARRIER_TO_PARTICLE_PROOF_DOSSIER.md
EARLIER_RECOVERY_SOURCE_03	runs/earlier_proof_recovery_2026-09-11/sources/LYAPUNOV_08_transversality_rank3_and_binomial_tail_drift.md
EARLIER_RECOVERY_SOURCE_04	runs/earlier_proof_recovery_2026-09-11/sources/ENGINE_FLUX_singular_geometry_derivations.py
EARLIER_RECOVERY_SOURCE_05	runs/earlier_proof_recovery_2026-09-11/sources/LYAPUNOV_09_appendix_disjoint_staple_coordinates_d4.md
EARLIER_RECOVERY_SOURCE_06	runs/earlier_proof_recovery_2026-09-11/sources/NOTE_FLUX_singular_geometry_derivations_2026-08-28.md
EARLIER_RECOVERY_SOURCE_07	runs/earlier_proof_recovery_2026-09-11/sources/VSU_Convex_Poisson_WellPosedness.md
EARLIER_RECOVERY_SOURCE_08	runs/earlier_proof_recovery_2026-09-11/sources/b6_shell_block_krylov_reduction_certificate.json
EARLIER_RECOVERY_SOURCE_09	runs/earlier_proof_recovery_2026-09-11/sources/01_conditional_spectral_floor_monotonicity.md
EARLIER_RECOVERY_SOURCE_10	runs/earlier_proof_recovery_2026-09-11/sources/FINITE_ORDER_NESTED_QUOTIENT_SPECTRAL_REDUCTION_THEOREM_FULL_DERIVATION_2026-08-28.md
EARLIER_RECOVERY_SOURCE_11	runs/earlier_proof_recovery_2026-09-11/sources/LATTICE_QCD_phase_isolation_principle.md
EARLIER_RECOVERY_SOURCE_12	runs/earlier_proof_recovery_2026-09-11/sources/01_determinant_reduction_theorem.md
EARLIER_RECOVERY_SOURCE_13	runs/earlier_proof_recovery_2026-09-11/sources/cleanroom_b6_symbolic_certificate.json
EARLIER_RECOVERY_SOURCE_14	runs/earlier_proof_recovery_2026-09-11/sources/B6_SHELL_BLOCK_KRYLOV_REDUCTION_2026-08-28.md
EARLIER_RECOVERY_SOURCE_15	runs/earlier_proof_recovery_2026-09-11/sources/NOTE_G17_SU3_COVARIANCE_ATTACK_2026-08-30.md
EARLIER_RECOVERY_SOURCE_16	runs/earlier_proof_recovery_2026-09-11/sources/NOTE_G19_balaban_blocking_reflection_positivity_2026-08-30.md
EARLIER_RECOVERY_SOURCE_17	runs/earlier_proof_recovery_2026-09-11/sources/verify_balaban_corner_reflection_exact.py
FESHBACH_RESOLVENT_COMPARISON	paper/research_notes/G22_FESHBACH_RESOLVENT_COMPARISON_20260909.md
FLAT_HOLONOMY_SOURCES_AND_RANK_REPAIR	paper/research_notes/G19_FLAT_HOLONOMY_SOURCES_AND_RANK_REPAIR_20260907.md
FOLDER_VALIDATION_PBH_WILSON_FOCUSED_2026_09_08_XML	docs/validation/pbh-wilson-focused-2026-09-08.xml
FOLDER_VALIDATION_PBH_WILSON_GRAPH_2026_09_08_XML	docs/validation/pbh-wilson-graph-2026-09-08.xml
FOLDER_VALIDATION_WILSON_BACKGROUND_GRAPH_XML	docs/validation/wilson-background-graph.xml
FOLDER_VALIDATION_WILSON_BACKGROUND_INVARIANTS_XML	docs/validation/wilson-background-invariants.xml
FOLDER_VALIDATION_WILSON_BACKGROUND_LEDGER_XML	docs/validation/wilson-background-ledger.xml

Alias	Path
FOLDER_VALIDATION_WILSON_BLOCK_FRESHNESS_XML	docs/validation/wilson-block-freshness.xml
FOLDER_VALIDATION_WILSON_BLOCK_GRAPH_XML	docs/validation/wilson-block-graph.xml
FOLDER_VALIDATION_WILSON_BLOCK_INVARIANTS_XML	docs/validation/wilson-block-invariants.xml
FOLDER_VALIDATION_WILSON_BLOCK_LEDGER_XML	docs/validation/wilson-block-ledger.xml
FOLDER_VALIDATION_WILSON_MARKED_AXIOMS_2026_09_08_TXT	docs/validation/wilson-marked-axioms-2026-09-08.txt
FOLDER_VALIDATION_WILSON_MARKED_TESTS_2026_09_08_XML	docs/validation/wilson-marked-tests-2026-09-08.xml
FOLDER_VALIDATION_WILSON_SC17_GRAPH_XML	docs/validation/wilson-sc17-graph.xml
FOLDER_VALIDATION_WILSON_SC17_INVARIANTS_XML	docs/validation/wilson-sc17-invariants.xml
FOLDER_VALIDATION_WILSON_SC17_LEAN_AXIOMS_TXT	docs/validation/wilson-sc17-lean-axioms.txt
FOLDER_VALIDATION_WILSON_SC17_LEAN_BUILD_TXT	docs/validation/wilson-sc17-lean-build.txt
FOLDER_VALIDATION_WILSON_SC17_LEDGER_XML	docs/validation/wilson-sc17-ledger.xml
FOLDER_VALIDATION_WILSON_SELECTED_FOCUSED_2026_09_08_XML	docs/validation/wilson-selected-focused-2026-09-08.xml
FOLDER_VALIDATION_WILSON_SELECTED_GRAPH_2026_09_08_XML	docs/validation/wilson-selected-graph-2026-09-08.xml
FOLDER_VALIDATION_WILSON_SHELL_AXIOMS_2026_09_08_TXT	docs/validation/wilson-shell-axioms-2026-09-08.txt
FOLDER_VALIDATION_WILSON_SHELL_HOOK_RETRY_2026_09_08_XML	docs/validation/wilson-shell-hook-retry-2026-09-08.xml
FOLDER_VALIDATION_WILSON_SHELL_LEDGER_RETRY_2026_09_08_XML	docs/validation/wilson-shell-ledger-retry-2026-09-08.xml
FOLDER_VALIDATION_WILSON_SHELL_TESTS_2026_09_08_XML	docs/validation/wilson-shell-tests-2026-09-08.xml
FOLDER_VALIDATION_WILSON_SPATIAL_AXIOMS_2026_09_08_TXT	docs/validation/wilson-spatial-axioms-2026-09-08.txt
FOLDER_VALIDATION_WILSON_SPATIAL_FOCUSED_2026_09_08_XML	docs/validation/wilson-spatial-focused-2026-09-08.xml
FOLDER_VALIDATION_WILSON_SPATIAL_FOCUSED_RETRY_2026_09_08_XML	docs/validation/wilson-spatial-focused-retry-2026-09-08.xml
FOLDER_VALIDATION_WILSON_SPATIAL_GRAPH_2026_09_08_XML	docs/validation/wilson-spatial-graph-2026-09-08.xml
FOLDER_VALIDATION_WILSON_SPATIAL_LEAN_BUILD_2026_09_08_TXT	docs/validation/wilson-spatial-lean-build-2026-09-08.txt
FOLDER_VALIDATION_WILSON_VACUUM_FOCUSED_XML	docs/validation/wilson-vacuum-focused.xml

Alias	Path
FOLDER_VALIDATION_WILSON_VACUUM_GRAPH_XML	docs/validation/wilson-vacuum-graph.xml
FOLDER_VALIDATION_WILSON_VACUUM_LEAN_2026_09_09_TXT	docs/validation/wilson-vacuum-lean-2026-09-09.txt
FOLDER_VALIDATION_WILSON_VACUUM_LEDGER_XML	docs/validation/wilson-vacuum-ledger.xml
FOLDER_VALIDATION_WILSON_WEIGHTED_FOCUSED_XML	docs/validation/wilson-weighted-focused.xml
FOLDER_VALIDATION_WILSON_WEIGHTED_GRAPH_XML	docs/validation/wilson-weighted-graph.xml
FOLDER_VALIDATION_WILSON_WEIGHTED_LEDGER_XML	docs/validation/wilson-weighted-ledger.xml
FOLDER_VALIDATION_YANGMILLS_DERIVATION_TESTS_2026_09_08_XML	docs/validation/yangmills-derivation-tests-2026-09-08.xml
FOLDER_VALIDATION_YANGMILLS_DERIVATION_UTF8_RERUN_2026_09_08_XML	docs/validation/yangmills-derivation-utf8-rerun-2026-09-08.xml
FORM_SCHUR_SCALE_COMPARISON	paper/research_notes/G19_FORM_SCHUR_SCALE_COMPARISON_20260905.md
G14_UNIVERSAL_CELLULAR_HODGE_TETRAHEDRAL	paper/research_notes/G14_UNIVERSAL_CELLULAR_HODGE_TETRAHEDRAL_20260911.md
G17_CORRECTED_ALPHA_DEPENDENT_KP_STABILITY	paper/research_notes/G17_CORRECTED_ALPHA_DEPENDENT_KP_STABILITY_20260910.md
G19_CAUCHY_DRIFT_REPAIR	paper/research_notes/G19_CAUCHY_DRIFT_REPAIR_20260909.md
G19_CONTINUUM_BRIDGE	paper/research_notes/G19_CONTINUUM_BRIDGE_INSERT.tex
G19_DIMENSION_FIVE_IRRELEVANCE	paper/research_notes/G19_DIMENSION_FIVE_IRRELEVANCE_LEMMA_20260910.md
G19_PROJECTIVE_MARTINGALE_REPAIR	paper/research_notes/G19_PROJECTIVE_MARTINGALE_REPAIR_20260909.md
G19_ROUGH_GAUGE_COERCIVITY	paper/research_notes/G19_ROUGH_GAUGE_COERCIVITY_AND_CURVATURE_20260910.md
G9_SIXTH_ORDER_COMBINED	paper/research_notes/G9_SIXTH_ORDER_COMBINED_20260911.md
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GAUSSIAN_FULL_FAST_GREEN	paper/research_notes/G19_FULL_GAUSSIAN_FAST_GREEN_20260906.md
GAUSSIAN_OS_OBSERVABILITY	paper/research_notes/G19_GAUSSIAN_OS_HISTORY_OBSERVABILITY_20260905.md
GAUSSIAN_PATH_ENDPOINT_BASELINE	paper/research_notes/G19_GAUSSIAN_PATH_ENDPOINT_BASELINE_20260906.md
GAUSSIAN_QUANTUM_FAST_SOURCES	paper/research_notes/G19_GAUSSIAN_QUANTUM_FAST_SOURCES_20260905.md
GLUEBALL_REVERSE_TARGET_DATA	paper/research_notes/G19_GLUEBALL_REVERSE_TARGET_DATA_20260905.md
GROUND_MARGINAL_SCHUR_SCORE	paper/research_notes/G19_GROUND_MARGINAL_SCHUR_SCORE_20260905.md
HODGE_FESHBACH_CHANNEL	paper/research_notes/G14_HODGE_FESHBACH_CHANNEL_20260908.md
HODGE_POLYMER_SUPPORTED	docs/derivations/hodge-polymer-supported-statements.md
LITERAL_ENDPOINT_COMPLETE_WINDOW	paper/research_notes/G19_LITERAL_ENDPOINT_COMPLETE_WINDOW_20260905.md
LOCAL_CLASS_WICK_RESULTS	paper/research_notes/LOCAL_CLASS_WICK_SPECTRUM_20260911.md
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LOCAL_COVARIANT_AVERAGED_PATH_SOURCES	paper/research_notes/G19_LOCAL_COVARIANT_AVERAGED_PATH_SOURCES_20260905.md
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MASTER paper	paper/master_paper_2026-08-28.tex

Alias	Path
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MOVING_TIME_SPECTRAL_GAP	docs/derivations/moving-time-spectral-gap.md
ODD_ORDER_CENTRE_PARITY	paper/research_notes/ODD_ORDER_CENTRE_PARITY_20260911.md
OS_HISTORY_BLOCKING	paper/research_notes/G19_OS_BLOCKING_AND_REVERSE_MASS_MATCHING_20260905.md
OS_KERNEL_GAP	docs/derivations/os-kernel-moving-time-gap.md
OS_KERNEL_GAP_RESULT	paper/research_notes/G19_KERNEL_CLOSURE_ATTEMPT_20260911.md
PAPER_FLATBAND	paper/homological_flat_bands_2026-08-28.pdf
PUBLICATION rev5	paper/workhouse_publication_edition_rev5_2026-08-30.tex
PUBLICATION v2	paper/workhouse_publication_edition_v2_2026-08-30.tex
REVIEW_SPECTRAL_BUDGETS	docs/derivations/review-derived-spectral-budgets.md
REVIEW_SPECTRAL_BUDGET_RESULTS	paper/research_notes/PILLARS_SPECTRAL_BUDGET_RESULTS_20260911.md
SEPTEMBER10_UPSTREAM_SCALARS	docs/derivations/september10-upstream-scalar-inventory.md
SOURCE_CURRENT_SPECTRAL_BRIDGES	docs/derivations/source-currents-and-spectral-totality.md
THEORY_CURRENT_BRIDGES	paper/research_notes/THEORY_CURRENT_BRIDGES_20260910.md
THEORY_GEOMETRY_RICCATI_BRIDGES	docs/derivations/theory-geometry-and-riccati-bridges.md
TRUE_GROUND_CENTER_SCORE_	paper/research_notes/G19_TRUE_GROUND_CENTER_SCORE_OBSTRUCTION_20260905.md
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URZUA_COUPLED_OSCILLATORS_2019	literature/fulltext/quantumrep-01-00009.pdf
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W6_CONDITIONAL_SCORE_TAIL_CONTROL	docs/derivations/w6-conditional-score-tail-control.md
W6_CONDITIONAL_TRANSPORT_	docs/derivations/w6-conditional-transport-obstruction.md
OBSTRUCTION	
W6_GROUND_JETS_TRANSPORT_BUDGET	docs/derivations/w6-ground-jets-and-transport-budget.md
W6_M10_REPAIR_RESULTS	paper/research_notes/W6_M10_REPAIR_RESULTS_20260911.md
W6_R10_REPAIRED_RESULTS	paper/research_notes/W6_R10_REPAIRED_IDENTITIES_20260911.md
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W6_SOURCE_GENERATOR_SCORE_FRAME	docs/derivations/w6-source-generator-score-frame.md
W6_SYNCHRONIZED_M10_DOMINATION	docs/derivations/w6-synchronized-m10-domination.md
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FLOOR	
WILSON_CREATOR_LIMIT	paper/research_notes/G18_WILSON_CREATOR_THERMODYNAMIC_LIMIT_20260905.md
WILSON_CREATOR_PARENT	paper/research_notes/G18_WILSON_CREATOR_PARENT_AND_SPECTRAL_FLOW_20260905.md
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Alias	Path
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WILSON_FIBER_FAST_GAP	paper/research_notes/G19_WILSON_PHYSICAL_FIBER_FAST_GAP_20260905.md
WILSON_FINITE_CELL_GAP	paper/research_notes/G19_WILSON_FINITE_CELL_GAP_AND_BOUNDARY_FORM_20260905.md
WILSON_FIXED_ORDER_CHART	paper/research_notes/G18_VACUUM_CHART_RECURSION_20260905.md
WILSON_G19_REVIEW	docs/validation/wilson-g19-corpus-reconciliation.md
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WILSON_GROUND_BUNDLE_RELATIVE_FORM	paper/research_notes/G19_WILSON_GROUND_BUNDLE_RELATIVE_FORM_20260905.md
WILSON_HARMONIC_BOUNDARY	paper/research_notes/G19_WILSON_HARMONIC_BOUNDARY_COMPARISON_20260905.md
WILSON_HARMONIC_CUBIC_OBSTRUCTION	paper/research_notes/G19_WILSON_HARMONIC_CUBIC_OBSTRUCTION_20260906.md
WILSON_INFINITY_PHYSICAL_BAND	paper/research_notes/G18_WILSON_INFINITY_VOLUME_PHYSICAL_BAND_20260905.md
WILSON_KINETIC_WINDOW	paper/research_notes/G19_UNIFORM_WILSON_WINDOW_20260904.md
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WILSON_SC17_TIME	docs/derivations/wilson-sc17-physical-time-limit.md
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WILSON_STRIP_BO	paper/research_notes/G19_WILSON_STRIP_BO_AND_TWO_STRIP_SPLITTING_20260905.md
WILSON_THREE_DIMENSIONAL_HARMONIC_BOUNDARY	paper/research_notes/G19_WILSON_THREE_DIMENSIONAL_HARMONIC_BOUNDARY_20260905.md
WILSON_TRANSFER_RESOLVENT	paper/research_notes/G19_TRANSFER_RESOLVENT_UNIFORM_BOUNDS_20260905.md
WILSON_TRUE_BLOCK_CHECK	docs/derivations/wilson-true-blocks-independent-check.md
WILSON_TWO_SQUARE_PHYSICAL_SHELLS	paper/research_notes/G19_WILSON_TWO_SQUARE_PHYSICAL_SHELLS_20260905.md
WILSON_VACUUM_COMPRESSION	paper/research_notes/G18_VACUUM_COMPRESSION_BOUND_20260905.md
WILSON_WEIGHTED_ACTIVITIES	paper/research_notes/G18_WILSON_WEIGHTED_ACTIVITY_BOUND_20260905.md
YM_FLAT_DIRECTIONS	docs/derivations/yangmills-simon-flat-directions.md
YM_GROUND_STATE	docs/derivations/yangmills-weighted-curvature.md
YM_RECONSTRUCTION	docs/derivations/yangmills-reconstruction.md

Standing: current (74)

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GUIDE	theory/GLUEBALL_SOURCE_CONSOLIDATION_GUIDE_2026-08-20_v4_3.md
HODGE_FESHBACH	docs/research/september_feshbach_integration.md

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SEPT_DOC_G19_GAUSSIAN_PATH_ENDPOINT_BASELINE_20260906	docs/derivations/wilson-spatial-inputs/G19_GAUSSIAN_PATH_ENDPOINT_BASELINE_20260906.md
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SEPT_DOC_PBH_WILSON_2026_09_08	docs/validation/pbh-wilson-2026-09-08.json
SEPT_DOC_README	docs/derivations/wilson-shell-inputs/README.md
SEPT_DOC_README_8ba09c37	docs/derivations/wilson-spatial-inputs/README.md
SEPT_DOC_SPATIAL_CONTINUUM_PASSAGE_2026_09_08	docs/research/spatial-continuum-passage-2026-09-08.md
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SEPT_DOC_WILSON_BACKGROUND_INDEPENDENT_REVIEW	docs/validation/wilson-background-independent-review.md

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SEPT_DOC_WILSON_G19_CORPUS_REVIEW_MANIFEST	docs/validation/wilson-g19-corpus-review-manifest.json
SEPT_DOC_WILSON_G19_FIBERS_REVIEW	docs/validation/wilson-g19-fibers-review.md
SEPT_DOC_WILSON_G19_GAUSSIAN_REVIEW	docs/validation/wilson-g19-gaussian-review.md
SEPT_DOC_WILSON_G19_MARTINGALE_REPAIR	docs/validation/wilson-g19-martingale-repair.md
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SEPT_DOC_WILSON_SC17_2026_09_09	docs/validation/wilson-sc17-2026-09-09.json
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SEPT_DOC_WILSON_SELECTED_2026_09_08	docs/validation/wilson-selected-2026-09-08.json
SEPT_DOC_WILSON_SHELL_2026_09_08	docs/validation/wilson-shell-2026-09-08.json
SEPT_DOC_WILSON_SHELL_AXIOMS	docs/validation/wilson-shell-axioms.lean
SEPT_DOC_WILSON_SPATIAL_2026_09_08	docs/validation/wilson-spatial-2026-09-08.json
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SEPT_DOC_WILSON_TRUE_BLOCKS_CHECK_2026_09_09	docs/validation/wilson-true-blocks-check-2026-09-09.json
SEPT_DOC_WILSON_VACUUM_2026_09_09	docs/validation/wilson-vacuum-2026-09-09.json
SEPT_DOC_WILSON_VACUUM_LEDGER_SNAPSHOT	docs/validation/wilson-vacuum-ledger-snapshot.yaml
SEPT_DOC_WILSON_VACUUM_THEOREMS_SNAPSHOT	docs/validation/wilson-vacuum-theorems-snapshot.yaml
SEPT_DOC_WILSON_WEIGHTED_2026_09_08	docs/validation/wilson-weighted-2026-09-08.json
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SEPT_DOC_YANGMILLS_FORMAL_BRIDGES	docs/derivations/yangmills-formal-bridges.md
SEPT_DOC_YANGMILLS_GPU_AUDIT_2026_09_09	docs/validation/yangmills-gpu-audit-2026-09-09.json
SEPT_DOC_YANGMILLS_SOURCES_2026_09_08	docs/validation/yangmills-sources-2026-09-08.json

Alias	Path
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WILSON_ACTIVITY_INPUT	docs/derivations/wilson-shell-inputs/G18_WILSON_WEIGHTED_ACTIVITY_BOUND_20260905.md
WILSON_BAND_INPUT	docs/derivations/wilson-shell-inputs/G18_WILSON_INFINITY_VOLUME_PHYSICAL_BAND_20260905.md
WILSON_CHART_INPUT	docs/derivations/wilson-shell-inputs/G18_WILSON_CARDINALITY_UNITARY_CHART_20260905.md
WILSON_SELECTED	docs/derivations/wilson-selected-inverse-wall.md
WILSON_SHELL	docs/derivations/wilson-marked-shell-transport.md
WILSON_SPATIAL	docs/derivations/wilson-spatial-schur-excess.md
WILSON_SPATIAL_FLAT_INPUT	docs/derivations/wilson-spatial-inputs/G19_FLAT_HOLONOMY_SOURCES_AND_RANK_REPAIR_20260907.md
WILSON_SPATIAL_SCHUR_INPUT	docs/derivations/wilson-spatial-inputs/G19_FORM_SCHUR_SCALE_COMPARISON_20260905.md
WILSON_TRUE_BLOCK	docs/derivations/wilson-true-vacuum-block-estimates.md
WILSON_VACUUM_ASSEMBLY	docs/derivations/wilson-vacuum-aligned-assembly.md
WILSON_WEIGHTED_REPAIR	docs/derivations/wilson-weighted-repair-and-rotor-gap.md
YM_GPU_AUDIT	docs/derivations/yangmills-gpu-resolution-audit.md
v4.3	theory/MASTER_THEORY_UNIFIED_2026-08-20_v4_3.md

Standing: corpus (8)

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PAPER	corpus-import/papers/flat_band/PAPER_FLUX_glueball_flat_band_v1_1.tex
THM_FLUX	corpus-import/theory/notes/THM_FLUX_hodge_cellular_circuit_mobility_theorem.md
TRACK_A_RECEIVED_G17	runs/track_a_formalization_2026-09-09/sources/paper/research_notes/G17_HAMILTONIAN_OSTERWALDER_SEILER_TRANSCRIPTION_20260910.md
TRACK_A_RECEIVED_SC17	runs/track_a_formalization_2026-09-09/sources/docs/derivations/wilson-sc17-spatial-closure.md
TRACK_A_RECEIVED_W6	runs/track_a_formalization_2026-09-09/sources/paper/research_notes/G19_W6_CUBIC_RESIDUAL_BOUNDS_20260910.md
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Standing: unrecorded (3)

Alias	Path
CANON	
MOB	
PUB edition	

Standing: quarantined (3)

Alias	Path
SEPT_DOC_YANGMILLS_RESOLVENT_LOCALIZATION_AND_DIRICHLET_GAP	docs/derivations/yangmills-resolvent-localization-and-dirichlet-gap.md
YM_BALABAN_MULTISCALE	docs/derivations/yangmills-continuum-balaban-multiscale-proof.md
quarantined master v3	corpus-import/corpus/MASTER_THEORY_UNIFIED_2026-08-20_v3.md

Standing: superseded (1)

Alias	Path
MASTER_THEORY	theory/superseded/MASTER_THEORY.md

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